Probabilistic Methods

Lecture Notes

A clear route from counting and
events to random variables and data.

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Chapter 1

Introduction

These notes are an introduction to probabilistic thinking: the mathematical art of reasoning under uncertainty. Probability is not presented here as a collection of ready-made formulas, but as a language for describing experiments, incomplete information, variability, randomness, and data.

The purpose of the course is to teach how to build probabilistic models carefully. Before one computes a probability, expectation, variance, confidence interval, or approximation, one should understand what is random, what is being observed, what assumptions are being made, and what mathematical object represents the situation. In this sense, probability is not only a computational tool, but also a discipline of precise modelling.

The course is intended for readers who want to understand why probabilistic methods work, not only how to apply them. The emphasis is therefore placed on interpretation, model construction, and the connection between formal definitions and concrete situations. Calculations appear throughout the notes, but they are treated as consequences of a well-defined model rather than as isolated procedures.

1.1 How to Use These Notes

These notes are meant to be read before problem classes. Their role is not only to present the material, but also to show how mathematical argumentation is built: how one starts from a situation, identifies the relevant objects, formulates assumptions, chooses notation, develops a line of reasoning, and only then performs computations. A problem class is therefore not the place where the basic language of the topic is introduced for the first time; it is the place where that language is used, tested, discussed, and refined through problems.

Students are expected to read the relevant chapter in advance and to treat the text as a model of reasoning. The important question is not only "What is the answer?" but also "How is the solution argued?" While reading, students should pay attention to the order of the argument, the meaning of each definition, the reason for each assumption, and the way examples are connected to general concepts. The aim is to learn how to construct a probabilistic explanation, not merely how to recognize a formula that seems to fit a task.

Problem lists will accompany the chapters and will be used during classes. They are meant to be solved after the corresponding text has been read. The chapter gives the language, concepts, and patterns of argumentation needed to approach the problems responsibly. Without this preparation, a solution may look complete on paper while still being mathematically empty: it may contain formulas, calculations, or generated text, but no clear understanding of what model is being used or why the reasoning is valid.

Artificial intelligence tools may be useful for practice, checking calculations, generating additional examples, or asking for alternative explanations. Students are encouraged to use such tools responsibly when they support learning. However, AI-generated text cannot replace the student's own reading and understanding. A student must be able to read, question, correct, and explain any solution they present. If one cannot explain the experiment, the elementary outcomes, the assumptions, the chosen model, and the steps of the argument, then the work has not yet been done.

These notes also help students learn what to ask for. To use problem lists, textbooks, or AI tools well, one must first know what matters: which object is being modelled, which assumptions are decisive, which definitions are involved, and what kind of justification is expected. The PDF is therefore an essential part of the course: it teaches the material, but it also teaches how to read mathematics, how to demand proper explanations, and how to recognize whether an answer contains real reasoning.

The expected preparation for classes is conceptual as well as computational. After reading a chapter, a student should be able to summarize its main ideas, explain the definitions in their own words, reconstruct the key examples, identify the assumptions in a probabilistic model, follow and reproduce the main arguments, and answer questions that test understanding of the text. The notes are deliberately written with full explanations and detailed argumentation so that this preparation is possible.

1.2 Structure of the Course

These lecture notes are organized as a gradual construction of probabilistic thinking. The aim is not to begin with formulas and then search for examples, but to develop the mathematical language needed to describe uncertainty in a precise way. We start from the problem of counting and representing possible outcomes, then move toward events, probability, random variables, probability laws, empirical data, sampling, limit theorems, and statistical inference.

The course begins with combinatorics because many probabilistic models depend on the ability to count possible outcomes correctly. At this stage probability itself is not yet needed. What is needed is the more elementary skill of recognizing what kind of object an outcome is: a sequence, a selection, an arrangement, a choice with or without repetition, or an object whose elements may or may not be distinguishable. This first step builds the counting language that later allows us to describe sample spaces and favorable outcomes with precision.

The second stage has a special conceptual role. It is devoted to discovering the formal language of probability theory. The purpose is to show how a mathematical theory can be built from careful thinking, analysis, and abstraction. We examine how statements about a random situation select sets of outcomes, how these sets become events, how logical operations correspond to set operations, and why a numerical theory of probability should be placed on such a structure. This part is not primarily computational; it is meant to teach how probabilistic language is formed.

After this conceptual preparation, the notes turn to sample spaces and elementary probability. Several canonical examples are developed from the beginning: one first understands the experiment, decides what should count as an elementary outcome, constructs the sample space, defines events, and only then assigns probabilities. Different representations of the same structure, such as lists, tables, trees, and symbolic notation, are used to show that a sample space is an abstract mathematical model rather than a collection of observed data.

The next step is conditional probability. Here the central idea is that information changes the space in which we reason. Conditional probability is introduced by asking what it means to restrict attention to the cases in which some event has already occurred. From this point of view, the formula for conditional probability becomes natural, and it leads to independence, dependence, partitions, the law of total probability, and Bayes' theorem. Bayes' theorem is treated as a central tool for reversing conditional reasoning and for understanding situations where intuition is easily misled.

The course then introduces random variables through the discrete case. A random variable appears as a function that extracts a numerical quantity from an elementary outcome: the number of successes, the number of errors, the number of arrivals, a payoff, or another measurable feature of the experiment. This leads to the support of a discrete random variable, its probability mass function, its cumulative distribution function, expectation, variance, moments, and interpretation. Classical discrete laws are presented as models arising from specific mechanisms, not as isolated formulas.

The continuous case is then developed as a parallel but conceptually more demanding part of the theory. For a continuous random variable, probability is no longer concentrated at individual points, so probabilities of intervals become the fundamental objects. This leads to density functions, cumulative distribution functions, integrals, continuous expectations, variances, moments, quantiles, and the interpretation of probability as area under a density. Important continuous models, including the uniform, exponential, normal, gamma, and beta distributions, are introduced with attention to their meaning and use.

Once theoretical probability laws have been introduced, the course moves to empirical distributions and descriptive statistics. The same ideas now appear in data: frequency tables, histograms, empirical cumulative distribution functions, means, medians, variances, standard deviations, quartiles, interquartile ranges, and outliers. This stage is meant to clarify the difference between an idealized probability law and a finite dataset, and also to show how empirical summaries mirror the theoretical quantities defined earlier.

The following stage adds another level of randomness by studying sampling and distributions of statistics. A sample mean, sample proportion, sample variance, median, or other statistic is not merely a number; it is itself a random variable because it depends on the random sample. This perspective prepares the ground for understanding sampling distributions and for seeing why repeated sampling creates a new layer of probabilistic structure.

The final stage of the present course outline is devoted to limit theorems and statistical inference. Once empirical summaries and sampling distributions are understood, we can ask what regularities appear as the sample size grows or as experiments are repeated many times. This leads to the law of large numbers, the central limit theorem, normal approximations, standard errors, confidence intervals, and the basic logic of statistical inference. The goal is to show how probability theory makes inference from data possible: it allows us to estimate unknown quantities, measure uncertainty, and justify conclusions mathematically.

Chapter 2

Elements of Combinatorics

2.1 Counting begins with elementary outcomes

Combinatorics is often presented as a list of formulas. This is misleading. Before a formula can be used, we must know what one elementary outcome is. A physical situation does not determine a mathematical space of possibilities by itself. The space of possibilities depends on what is recorded.

Suppose that three balls are drawn from a box. If the balls are labelled and the order of drawing is recorded, then an outcome is a sequence of labelled objects. If only the selected labelled balls are recorded, then an outcome is a set. If only the colors are recorded, then labelled balls of the same color may become indistinguishable. If only the number of red balls is recorded, then the outcome is a numerical summary of the draw rather than the full draw itself.

Thus the first question in a counting problem is not

Which formula should be used?

The first question is

What is one outcome?

Definition

An **elementary outcome** is a complete description of one possible result of a situation, according to the recording rule chosen for the model.

The phrase "according to the recording rule" is essential. The same physical situation may produce different mathematical spaces of possibilities if different information is recorded.

Example

A box contains three balls labelled R_1, R_2, B_1. Two balls are drawn without replacement. If order and labels are recorded, then examples of outcomes are

$$(R_1, B_1), \qquad (B_1, R_1), \qquad (R_1, R_2).$$

If only the set of labelled balls is recorded, then examples of outcomes are

$$\{R_1, B_1\}, \qquad \{R_1, R_2\}.$$

If only the sequence of colors is recorded, then examples of outcomes are

$$(R, B), \qquad (B, R), \qquad (R, R).$$

If only the number of red balls is recorded, then the possible recorded values are

$$0, 1, 2.$$

These are not four different answers to the same counting question. They are four different mathematical models of the same physical situation.

Remark

A counting error often begins before any arithmetic is done. It begins when the outcome type is chosen incorrectly.

2.2 Recording rules and spaces of possibilities

A recording rule tells us which differences between physical results are mathematically visible. Some differences are recorded; others are ignored. Once the recording rule is fixed, the space of possibilities is the set of all possible recorded outcomes.

Definition

A **recording rule** is a decision about what information is included in the mathematical outcome and what information is ignored.

For finite situations, the construction usually follows this pattern:

$$\text{physical situation} \longrightarrow \text{recording rule} \longrightarrow \text{elementary outcome} \longrightarrow \text{space of possibilities.}$$

The following table shows common outcome types.

Recording rule	Outcome type	Typical notation
Order is recorded	sequence	$(a_1, a_2, \ldots, a_k)$
Only membership is recorded	set	$\{a_1, a_2, \ldots, a_k\}$
All objects are arranged in a row	linear arrangement	$(a_{\sigma(1)}, \ldots, a_{\sigma(n)})$
Objects are arranged on a circle	circular arrangement	relative order
Only numbers of types are recorded	count vector	$(n_1, n_2, \ldots, n_r)$
Only a numerical summary is recorded	summary value	x

2.3 Representing the space of possibilities

Counting is easier when the space of possibilities is represented in a form that matches the structure of the problem. A representation is not decoration. It is part of the reasoning, because it makes visible which outcomes are different, which descriptions are equivalent, and which construction steps are being used.

Different situations call for different representations:

- a list is useful when the space is small and every outcome can be written explicitly;
- a table is useful when two choices are made and rows and columns organize the possibilities;
- a tree is useful when an outcome is built through successive stages;
- a set notation is useful when only membership matters;
- a tuple notation is useful when positions or order matter;
- a count vector is useful when only the numbers of objects of each type matter.

Example

If two letters are chosen from $\{A, B, C\}$ and order is recorded with repetition allowed, the space can be represented as a table:

	A	B	C
A	(A, A)	(A, B)	(A, C)
B	(B, A)	(B, B)	(B, C)
C	(C, A)	(C, B)	(C, C)

The table shows two things at once: there are 3 choices for the first position and 3 choices for the second position, and the ordered pairs (A, B) and (B, A) are different outcomes.

Example

If two letters are chosen from $\{A, B, C\}$ and only the selected set is recorded, then

$$\{A, B\} = \{B, A\}.$$

A table of ordered pairs would still be possible as an auxiliary representation, but it would overdescribe the outcomes. The actual outcomes are sets:

$$\{A, B\}, \qquad \{A, C\}, \qquad \{B, C\}.$$

The change of representation reflects a change of model.

A useful representation should answer the question: what differences are visible in this model? If the representation hides an important difference, the count may be too small. If it records irrelevant differences, the count may be too large.

Example

A committee of three people is chosen from ten people. If the committee is recorded only as a group, then

$$\{A, B, C\} = \{C, A, B\}.$$

The order of listing the names is irrelevant. The outcome is a set.
If the same three people are assigned the roles of chair, secretary, and treasurer, then

$$(A, B, C) \neq (C, A, B).$$

The outcome is now an ordered assignment of roles.

The two situations may look similar in everyday language: in both cases three people are selected. But mathematically they are different because the recording rules are different.

2.4 Diagnostic questions before counting

Before using a formula, answer the following questions.

1. What is one elementary outcome?
2. Is the outcome a sequence, a set, an arrangement, or a count vector?
3. Does order matter?

4. Are all objects used, or only some of them?

5. Can the same object be used more than once?

6. Are the objects distinguishable or indistinguishable?

7. Are we first counting labelled descriptions and then identifying descriptions that represent the same outcome?

These questions are more important than the names of formulas. The names are only labels attached to structures that have already been recognized.

Remark

Do not begin a problem by asking which formula you remember. Begin by asking what kind of outcome is being constructed.

2.5 The multiplication principle as a construction rule

Many counting problems can be solved by describing how an outcome is constructed step by step.

Theorem

Multiplication principle. Suppose a construction is performed in k successive stages. If there are m_1 choices for the first stage, m_2 choices for the second stage, and so on up to m_k choices for the k-th stage, then the number of possible complete constructions is

$$m_1 m_2 \cdots m_k.$$

The multiplication principle is not only a formula. It is a way of reading an outcome as a construction process.

Example

A four-digit PIN code is formed from the digits $0, 1, \ldots, 9$, and digits may repeat. One outcome is a sequence such as

$$(0, 7, 7, 3).$$

There are 10 choices for each position, hence

$$10 \cdot 10 \cdot 10 \cdot 10 = 10^4$$

possible PIN codes.
The reason is not that the formula n^k was memorized. The reason is that a PIN code is a sequence of four independent choices from a set of ten symbols.

Example

A three-letter code is formed from the letters A, B, C, D, and letters may not repeat. One outcome is an ordered triple, for example

$$(A, D, B).$$

There are 4 choices for the first position, 3 for the second, and 2 for the third. Therefore the number of codes is

$$4 \cdot 3 \cdot 2 = 24.$$

When the same number of choices is available at every stage, powers appear. When the number of choices decreases, falling products appear. When order is later ignored, division appears. Most elementary combinatorial formulas can be understood from these three ideas.

2.6 Sequences with repetition

A sequence with repetition is an ordered list in which each position is filled separately from the same set of symbols.

Definition

Let S be a set with n elements. A **sequence of length k with repetition** over S is an ordered k-tuple

$$(a_1, a_2, \ldots, a_k),$$

where each $a_i \in S$. The same element may occur many times.

The word "ordered" means that positions matter. The first position, the second position, and the third position are different places in the description. The phrase "with repetition" means that choosing one symbol at one position does not remove it from later positions.

Now we derive the count. To construct one sequence,

- choose a_1: there are n choices;
- choose a_2: there are again n choices, because repetition is allowed;
- continue in the same way up to a_k: there are n choices at each position.

By the multiplication principle, the number of complete sequences is

$$\underbrace{n \cdot n \cdots n}_{k \text{ factors}} = n^k.$$

Thus the formula n^k is not a separate rule to memorize. It is the multiplication principle applied to k ordered positions, each with the same n choices.

This model is used when:

- order is recorded,
- each position is chosen separately,
- repetition is allowed,
- the outcome is a full sequence.

Typical examples include PIN codes, repeated coin tosses, repeated die rolls, strings over an alphabet, and repeated selections with replacement.

Example

The space of possibilities for three tosses of a coin is

$$\{H, T\}^3.$$

It has

$$2^3 = 8$$

elementary outcomes. Each outcome is a sequence, for example

$$(H, T, H).$$

The outcomes (H, T, H) and (T, H, H) are different because the order of positions is recorded.
A tree representation shows the same count visually. At the first toss there are 2 branches, from each of these there are 2 branches for the second toss, and from each of those there are 2 branches for the third toss. The tree has

$$2 \cdot 2 \cdot 2 = 8$$

terminal branches, one for each complete sequence.

2.7 Ordered selections without repetition

Sometimes only some objects are chosen, but the order or position of the chosen objects matters.

Definition

An **ordered selection without repetition** of length k from n distinct objects is an ordered list of k different objects chosen from the n available objects.

To derive the count, construct the ordered list position by position. There are n choices for the first position. Once the first object has been used, it cannot be used again, so there are $n-1$ choices for the second position. After two objects have been used, there are $n-2$ choices for the third position. Continuing in this way, the k-th position has

$$n - (k-1) = n - k + 1$$

choices. Therefore the number of ordered selections without repetition is

$$P(n, k) = n(n-1)(n-2)\cdots(n-k+1).$$

This product can also be written using factorials. Since

$$n! = n(n-1)\cdots(n-k+1)(n-k)(n-k-1)\cdots 1,$$

dividing by $(n-k)!$ removes the unused tail:

$$P(n, k) = \frac{n!}{(n-k)!}.$$

This model is used when:

- only some objects are selected,
- order or position matters,
- no object may be used twice.

Example

Gold, silver, and bronze medals are assigned among 12 runners. An outcome is not a set of three runners. It is an ordered triple

$$(\text{gold}, \text{silver}, \text{bronze}).$$

There are 12 choices for gold, then 11 choices for silver, then 10 choices for bronze. Hence the number of possible assignments is

$$12 \cdot 11 \cdot 10 = P(12, 3).$$

In some European terminology, ordered selections are called variations. In these notes we use the name ordered selections without repetition, or k-permutations, because it describes directly what the outcome is.

2.8 Arrangements of all objects

A permutation is an arrangement of all objects in a row.

Definition

A **permutation** of n distinct objects is a linear arrangement of all n objects.

This is the special case of an ordered selection without repetition in which all objects are used. Thus $k = n$, and

$$P(n, n) = n(n-1)(n-2)\cdots 2 \cdot 1 = n!.$$

So there are

$$n!$$

permutations of n distinct objects.

Example

If 7 students are arranged in a line, then one outcome is a complete ordered list of all students. The number of arrangements is

$$7!.$$

This is not because 7! is a name attached to the word "arrangement". It is because there are 7 choices for the first position, 6 for the second, 5 for the third, and so on until all positions are filled.

Permutations are used when:

- all objects are used,
- order matters,
- the objects are distinguishable,
- the arrangement is linear.

2.9 Circular arrangements

A circular arrangement is different from a linear arrangement because rotating the whole arrangement does not create a new relative seating order.

Example

Suppose four people A, B, C, D sit around a round table. The linear lists

$$(A, B, C, D), \qquad (B, C, D, A), \qquad (C, D, A, B), \qquad (D, A, B, C)$$

represent the same circular arrangement if only relative positions matter.

For n distinct objects placed around a circle, with rotations identified, the number of circular arrangements is

$$(n-1)!.$$

One way to see this is to fix one object as a reference point. This does not reduce any real choice; it only chooses one representative from each group of rotated descriptions. Once the reference object is fixed, the remaining $n-1$ objects can be placed around it in a linear order:

$$(n-1)(n-2)\cdots 2\cdot 1 = (n-1)!.$$

Remark

A circular arrangement requires a modelling decision. If seats are numbered or if there is a distinguished position, then rotations may become different outcomes and the linear count may be appropriate.

2.10 Unordered selections: combinations

A combination is an unordered selection.

Definition

A **combination** of size k from an n-element set is a subset with k elements.

The number of combinations is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

This formula can be derived from ordered selections. There are

$$P(n, k) = \frac{n!}{(n-k)!}$$

ordered selections of length k. But each unordered selection of k objects can be ordered in $k!$ ways. Therefore

$$\binom{n}{k} = \frac{P(n, k)}{k!} = \frac{n!}{k!(n-k)!}.$$

Combinations are used when:

- only membership is recorded,
- order is ignored,

- each object may be chosen at most once,
- the outcome is a set.

Example

A five-card hand from a standard deck is a set of five cards, not a sequence of five draws, if the final hand is all that is recorded. The number of possible hands is

$$\binom{52}{5}.$$

If the order of drawing is recorded, the correct count is instead

$$P(52,5) = 52 \cdot 51 \cdot 50 \cdot 49 \cdot 48.$$

These are different sample spaces.

2.11 Overcounting and equivalent descriptions

Many combinatorial formulas arise from a two-step idea:

1. Count descriptions that are easy to construct.
2. Divide by the number of descriptions representing the same outcome.

This is the idea of overcounting.

Definition

Overcounting occurs when the same intended outcome is counted more than once because it has several different descriptions.

Overcounting is not a mistake if it is controlled. It becomes a method when we know exactly how many times each outcome has been counted.

Example

Suppose we want to choose a committee of three people from n people. The order of names in the committee does not matter.
It is easier first to count ordered selections:

$$n(n-1)(n-2).$$

But each committee of three people has been counted 3! times, because the same three people can be listed in 3! different orders. Therefore the number of committees is

$$\frac{n(n-1)(n-2)}{3!}.$$

The essential idea is not division itself. The essential idea is identifying which descriptions should be treated as the same outcome.

2.12 Distinguishable and indistinguishable objects

Objects are distinguishable if the model treats them as separate individuals. Objects are indistinguishable if only their type matters.

Example

The objects

$$R_1, R_2, R_3$$

are three distinguishable red balls. The symbols

$$R, R, R$$

represent three indistinguishable red balls of the same type.

A physical object may be distinguishable in one model and indistinguishable in another. This depends on the recording rule.

Example

An urn contains red and blue balls. If each ball has a label, and labels are recorded, then two red balls are different outcomes. If only colors are recorded, then the red balls are treated as indistinguishable.

This distinction is crucial in probability. If the physical random mechanism selects labelled balls uniformly, but the sample space records only colors, then the color outcomes may not be equally likely.

2.13 Permutations with repeated elements

When some objects are indistinguishable, a linear arrangement may have several labelled descriptions that represent the same visible arrangement.

Suppose we arrange n objects, where there are n_1 objects of type 1, n_2 objects of type 2, and so on, with

$$n_1 + n_2 + \cdots + n_r = n.$$

The number of distinct visible arrangements is

$$\frac{n!}{n_1! n_2! \cdots n_r!}.$$

Example

The word BANANA contains six letters: three A's, two N's, and one B. If all six positions are arranged, then the number of distinct words is

$$\frac{6!}{3!2!1!}.$$

The division appears because permuting the three A's among themselves does not change the visible word, and neither does permuting the two N's.

This formula should be read as an overcounting correction:

$$\text{labelled arrangements} \quad \longrightarrow \quad \text{visible arrangements.}$$

2.14 Combinations with repetition

Sometimes we choose k objects from n types, and the same type may be chosen several times. The order of the chosen objects is ignored. The outcome is then not a sequence of choices, but a vector of counts.

Definition

A **combination with repetition** of size k from n types is a choice of non-negative integers

$$(x_1, x_2, \ldots, x_n)$$

such that

$$x_1 + x_2 + \cdots + x_n = k.$$

Here x_i records how many objects of type i are chosen.

The important modelling decision is that the positions of the chosen objects are not recorded. For example, choosing two objects of type 1, none of type 2, and three of type 3 is recorded as

$$(2, 0, 3),$$

not as an ordered sequence of five choices.

The stars and bars representation

We derive the formula by representing each count vector visually. Write k stars to represent the chosen objects and insert $n-1$ bars to divide the stars into n groups. The number of stars before the first bar is x_1, the number of stars between the first and second bar is x_2, and so on, until the number of stars after the last bar is x_n.

For example, if $n = 3$ and $k = 5$, the count vector

$$(2, 0, 3)$$

is represented by

$$* * \,|\,|\, * * * .$$

There are two stars before the first bar, no stars between the two bars, and three stars after the second bar. This represents

$$x_1 = 2, \qquad x_2 = 0, \qquad x_3 = 3.$$

Conversely, every arrangement of k stars and $n-1$ bars determines exactly one count vector. For instance,

$$|\, * * * \,|\, * *$$

represents

$$(0, 3, 2).$$

This is a one-to-one correspondence:

$$\text{count vectors } (x_1, \ldots, x_n) \text{ with } x_1 + \cdots + x_n = k \longleftrightarrow \text{arrangements of } k \text{ stars and } n-1 \text{ bars.}$$

Therefore the counting problem becomes a simpler arrangement problem. We have

$$k + (n-1) = n + k - 1$$

total positions. To form an arrangement, we only need to decide which k of these positions contain stars; the remaining $n-1$ positions will contain bars. Hence the number of arrangements is

$$\binom{n+k-1}{k}.$$

Equivalently, we may choose the $n-1$ positions of the bars, giving

$$\binom{n+k-1}{n-1}.$$

Thus

$$\binom{n+k-1}{k} = \binom{n+k-1}{n-1}.$$

Theorem

The number of combinations with repetition of size k from n types is

$$\binom{n+k-1}{k} = \binom{n+k-1}{n-1}.$$

Example

How many ways are there to choose 5 donuts from 3 flavors if only the numbers of each flavor matter?
An outcome is a count vector

$$(x_1, x_2, x_3), \qquad x_1 + x_2 + x_3 = 5.$$

The stars and bars representation uses 5 stars and 2 bars, so there are 7 positions in total. We may choose the 5 positions of the stars:

$$\binom{7}{5},$$

or the 2 positions of the bars:

$$\binom{7}{2}.$$

Thus the number of possible choices is

$$\binom{3+5-1}{5} = \binom{7}{5} = \binom{7}{2}.$$

Remark

Combinations with repetition do not describe ordered repeated choices. If order matters, the model is a sequence with repetition and the count is n^k. If order is ignored and only counts of types matter, the model is combinations with repetition.

2.15 Recognizing the correct counting model

The following diagnostic table summarizes the main finite counting structures.

Outcome type	Order?	Repetition?	Count
sequence with repetition	yes	yes	n^k
ordered selection	yes	no	$P(n,k) = \dfrac{n!}{(n-k)!}$
permutation	yes	no, all used	$n!$
circular arrangement	relative order	no, all used	$(n-1)!$
combination	no	no	$\binom{n}{k}$
combination with repetition	no	yes	$\binom{n+k-1}{k}$
multiset permutation	positions yes	identical objects	$\dfrac{n!}{n_1! \cdots n_r!}$

The table should be used only after the outcome type has been identified. It is a summary, not a substitute for modelling.

Remark

A formula is a compressed argument. If the argument does not match the situation, the formula gives the wrong count even when the arithmetic is correct.

2.16 Common traps

Trap 1: confusing a sequence with a set

The ordered outcomes

$$(A, B, C) \quad \text{and} \quad (C, B, A)$$

are different sequences but the same set:

$$\{A, B, C\} = \{C, B, A\}.$$

Use sequences when order is recorded. Use sets when only membership is recorded.

Trap 2: treating indistinguishable objects as distinguishable

The letters in BANANA are not six distinct letters. The three A's are indistinguishable, and the two N's are indistinguishable. Therefore 6! overcounts the number of visible arrangements.

Trap 3: using combinations when roles matter

A committee is a set. A medal assignment is an ordered assignment. A group of three athletes and a gold-silver-bronze result are not the same kind of outcome.

Trap 4: using a representation that records too much

Sometimes an auxiliary representation is more detailed than the intended outcome. For example, ordered lists may be useful for deriving the number of committees, but the committee itself is not an ordered list. If the extra descriptions are not corrected for, the count is too large.

Trap 5: using a representation that records too little

Sometimes a representation hides distinctions that the model requires. If positions, roles, labels, or order are part of the outcome, then a representation that ignores them gives too small a count. A good representation must preserve exactly the distinctions that the recording rule declares visible.

2.17 What combinatorics teaches here

The role of this chapter is not to memorize more formulas. Its role is to teach how to build and count finite spaces of possibilities. A formula is useful only after the underlying structure has been recognized.

The general workflow is

$$\text{situation} \longrightarrow \text{recording rule} \longrightarrow \text{elementary outcome} \longrightarrow \text{space of possibilities} \longrightarrow \text{count.}$$

Each arrow in this workflow represents a mathematical decision. The same everyday description may lead to different counts if the recording rule changes. The same physical objects may be treated as distinguishable or indistinguishable depending on what is visible in the model. The same collection may be represented as a list, a set, an arrangement, or a count vector.

Good counting therefore has two parts. First, one must construct the correct model. Second, one must justify the count by a construction argument, an overcounting correction, or a representation such as a table, a tree, or a stars-and-bars diagram.

2.18 Final checklist

When solving a counting problem, write down the following before doing arithmetic:

1. What is the physical situation?
2. What information is recorded?
3. What is one elementary outcome?
4. What is the space of possibilities?
5. How can this space be represented: as a list, table, tree, set, tuple, arrangement, or count vector?
6. Are outcomes sequences, sets, arrangements, or count vectors?
7. Does the representation record exactly the distinctions required by the model?
8. Which construction, overcounting correction, or bijection gives the count?
9. What does the final number mean in the original situation?

This checklist is the main lesson of the chapter. The formulas are useful, but they are useful only when the underlying structure has been discovered first and the count has been justified.

Chapter 3

Discovering Formalism

3.1 Why formalism has to be discovered

The previous chapter taught a first lesson: before one counts, one must know what is being counted. The present chapter begins from a closely related but deeper observation. Before one assigns any numerical value to uncertainty, one must know what kind of object a statement about a situation selects.

Mathematics often begins when ordinary language becomes too elastic. In ordinary language we can say things such as

$$\text{the result is favourable,} \qquad \text{at least one die shows 6,}$$

$$\text{the first toss is heads,} \qquad \text{it rains on Wednesday or Friday.}$$

These statements are meaningful, but they are not yet mathematical objects. They have grammar, but not yet structure. To reason with them reliably, we must translate them into something that can be compared, combined, negated, represented, and eventually measured.

The central discovery of this chapter is that a statement about a situation acts as a test on elementary outcomes. An elementary outcome either satisfies the statement or it does not. Therefore a statement selects a set of elementary outcomes. This selected set is what probability theory calls an event.

Remark

The formal language is not introduced to make the subject look more abstract. It is introduced because without it we cannot reliably say what is being discussed.

At this stage, we are not primarily interested in numerical answers. We are interested in the birth of a language. We want to see how the following chain of ideas appears naturally:

$$\text{situation} \longrightarrow \text{elementary outcomes} \longrightarrow \text{statements about outcomes} \longrightarrow \text{sets of outcomes} \longrightarrow \text{operations on set}$$

Only after this language exists can numerical probability be introduced responsibly.

3.2 A first universe of discourse

Every formal discussion needs a universe in which its statements are interpreted. In our setting, that universe is the set of all elementary outcomes under consideration.

Definition

The **sample space** Ω is the set of elementary outcomes that are possible under the recording rule chosen for the situation.

In this chapter, Ω should be read as a universe of discourse. It tells us what the word "outcome" means in the current model. A statement is not evaluated in the air. It is evaluated relative to Ω.

Example

Consider two coin tosses, with order recorded. The sample space is

$$\Omega = \{HH, HT, TH, TT\}.$$

The statement "exactly one head occurs" is true for HT and TH, but false for HH and TT. Thus the statement selects the set

$$\{HT, TH\}.$$

The selected set is the mathematical object corresponding to the statement.

The same phrase may select a different set if the sample space changes. For example, if only the number of heads is recorded, then the sample space is

$$\Omega' = \{0, 1, 2\}.$$

In that model, the statement "exactly one head occurs" selects

$$\{1\}.$$

The ordinary sentence is similar, but the formal object is different because the underlying space is different.

Remark

Formalism forces us to keep track of the level at which we are speaking. Are we describing the full ordered outcome, a set of visible outcomes, or only a numerical summary? The answer changes the mathematical object.

This is the methodological point of the chapter. We do not begin with symbols because symbols are fashionable. We begin with a question: what must be preserved so that reasoning remains honest? The symbols appear as a disciplined way to preserve exactly that structure.

3.3 Statements become events

Let Ω be fixed. A statement about the situation becomes mathematical when we can say exactly for which elementary outcomes it is true. The set of all such outcomes is called an event.

Definition

An **event** is a subset of the sample space Ω. If $A \subseteq \Omega$, then A consists of precisely those elementary outcomes for which a certain statement is true.

This definition may look simple, but it is a major conceptual move. It tells us that statements and sets can play the same role once Ω has been fixed:

$$\text{statement about the situation} \quad \longleftrightarrow \quad \text{subset of } \Omega.$$

The statement gives meaning. The subset gives structure.

Example

Suppose two dice are rolled and order is recorded. Then

$$\Omega = \{(i,j) : i,j \in \{1,2,3,4,5,6\}\}.$$

The statement

the sum is 7

selects the event

$$A = \{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}.$$

The statement

the first die is greater than the second

selects the event

$$B = \{(i,j) \in \Omega : i > j\}.$$

The notation $B = \{(i,j) \in \Omega : i > j\}$ is not a shortcut for avoiding thought. It records the rule by which membership in the event is decided.

3.4 Representing events as regions

A set of outcomes can be written as a list, but lists are not always the best representation. When Ω has a visible structure, an event should often be represented as a region inside that structure. This is not merely a visual aid. It is a way of seeing relations.

For two dice, the sample space is naturally represented as a 6×6 table:

	1	2	3	4	5	6
1	.	.	.	.	.	.
2	.	.	.	.	.	.
3	.	.	.	.	.	.
4	.	.	.	.	.	.
5	.	.	.	.	.	.
6	.	.	.	.	.	.

Each cell represents one ordered pair (i,j). The row records the first die, and the column records the second die.

The event $B = \{(i,j) : i > j\}$ occupies the cells below the diagonal:

	1	2	3	4	5	6
1	.	.	.	.	.	.
2	X	.	.	.	.	.
3	X	X	.	.	.	.
4	X	X	X	.	.	.
5	X	X	X	X	.	.
6	X	X	X	X	X	.

The diagonal itself corresponds to $i = j$. The region above the diagonal corresponds to $i < j$. The geometry of the table reveals the structure of the statement.

Remark

A representation is good when it preserves the distinctions relevant to the statement and makes relations visible. A poor representation may hide the structure that the problem is trying to teach.

3.5 Reading a region as a statement

The translation works in both directions. A verbal statement selects a set, but a marked set also asks to be interpreted as a statement.

Example

In the two-dice table, suppose the marked cells are exactly the diagonal:

	1	2	3	4	5	6
1	X	.	.	.	.	.
2	.	X	.	.	.	.
3	.	.	X	.	.	.
4	.	.	.	X	.	.
5	.	.	.	.	X	.
6	.	.	.	.	.	X

This region can be described in several equivalent ways:

the two dice show the same number,

the first result equals the second result,

$$\{(i, j) \in \Omega : i = j\}.$$

The ability to move between these descriptions is part of understanding the formalism.

Students often think that formal notation is a translation from language into symbols. That is only half of the work. One must also be able to read symbols and diagrams back into language. A set without interpretation is mute; a sentence without a set is vague. Mathematics becomes powerful when both sides illuminate each other.

3.6 Logical operations become set operations

Once statements have become events, logical operations on statements become set operations on events. This is one of the most important structural discoveries in elementary probability theory. The words

and, or, not

are not informal decorations. They determine precise operations on subsets of Ω.

Let $A \subseteq \Omega$ and $B \subseteq \Omega$ be events.

Language	Set operation	Meaning
A and B	$A \cap B$	outcomes satisfying both statements
A or B	$A \cup B$	outcomes satisfying at least one statement
not A	$A^c = \Omega \setminus A$	outcomes not satisfying A
A but not B	$A \setminus B$	outcomes satisfying A and not B

The table is not a dictionary of symbols to memorize. It expresses a deeper fact: logical structure and set structure are the same structure viewed in two languages.

Definition

For events $A, B \subseteq \Omega$:

$$A \cap B = \{\omega \in \Omega : \omega \in A \text{ and } \omega \in B\},$$

$$A \cup B = \{\omega \in \Omega : \omega \in A \text{ or } \omega \in B\},$$

$$A^c = \{\omega \in \Omega : \omega \notin A\}.$$

3.7 Compound statements in a two-dice universe

Consider again two dice, with

$$\Omega = \{(i, j) : i, j \in \{1, 2, 3, 4, 5, 6\}\}.$$

Define three events:

$$A = \{(i, j) : i + j = 7\},$$

$$B = \{(i, j) : i > j\},$$

$$C = \{(i, j) : i = 6 \text{ or } j = 6\}.$$

Now each compound statement can be built systematically.

The statement

the sum is 7 or at least one die shows 6

corresponds to

$$A \cup C.$$

The statement

the sum is 7 and at least one die shows 6

corresponds to

$$A \cap C.$$

The statement

the sum is 7 but the first die is not greater than the second

corresponds to

$$A \cap B^c.$$

The statement

at least one die shows 6 but the sum is not 7

corresponds to

$$C \setminus A = C \cap A^c.$$

This is the beginning of an algebra of events. We are not merely naming sets. We are learning to calculate with statements.

Remark

The purpose of the notation is not to replace thought. The purpose is to make the structure of thought inspectable. If a student writes $A \cap B^c$, they should also be able to say what it means in words and mark the corresponding region in the sample space.

3.8 Why "or" needs care

In everyday speech, the word "or" is sometimes ambiguous. In mathematics, the union $A \cup B$ means inclusive or: an outcome belongs to $A \cup B$ if it belongs to A, or to B, or to both.

This matters because the overlap is not automatically empty. If A and B overlap, then the union is not obtained by placing two disjoint blocks side by side. The intersection must be seen.

Example

Let

$$A = \{(i,j) : i + j = 7\}, \qquad C = \{(i,j) : i = 6 \text{ or } j = 6\}.$$

The outcome $(1,6)$ belongs to both A and C. The outcome $(6,1)$ also belongs to both. Thus

$$A \cap C = \{(1,6),(6,1)\}.$$

The phrase "sum is 7 or at least one die shows 6" includes these outcomes once, not twice. The union is a set of outcomes, not a list of reasons.

This is a methodological lesson. A sentence may have several reasons for being true, but an elementary outcome either belongs to the event or it does not. Formalism prevents double counting at the level of meaning before any numerical computation begins.

3.9 Complementation and the boundary of the universe

The complement A^c is always taken relative to the current sample space Ω. This is why Ω must be fixed before complements are meaningful.

Example

If two dice are rolled and

$$A = \{(i,j) : i + j = 7\},$$

then

$$A^c = \{(i,j) \in \Omega : i + j \neq 7\}.$$

The complement is not an abstract collection of all things in the world for which the sum is not 7. It is the collection of outcomes inside the two-dice sample space for which the sum is not 7.

The complement teaches a philosophical lesson hidden inside a technical definition: negation is always negation within a universe. To say "not A" mathematically, one must first say what counts as a possible case.

3.10 From events to observed frequencies

So far, an event is only a set of outcomes. This is already a substantial achievement: we can now state clearly what a sentence means. But probability theory eventually wants to do something more. It wants to attach numbers to events.

The first honest source of such numbers is observation. If an experiment is repeated many times, and we record how often each outcome or event occurs, then we obtain observed frequencies. These frequencies are not yet theoretical probabilities. They are empirical measurements. Nevertheless, they reveal the shape that a future theory of probability should have.

Suppose a die is rolled 1000 times. The elementary outcomes are

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

Assume the recorded numbers are

$$n(\{1\}) = 168, \quad n(\{2\}) = 154, \quad n(\{3\}) = 181,$$

$$n(\{4\}) = 167, \quad n(\{5\}) = 160, \quad n(\{6\}) = 170.$$

For an event $A \subseteq \Omega$, define its observed frequency by

$$f(A) = \frac{n(A)}{1000},$$

where $n(A)$ is the number of recorded trials whose outcome belongs to A.

Example

Let

$$A = \{2, 4, 6\}.$$

Then A occurs whenever the recorded die result is 2, 4, or 6. Therefore

$$n(A) = n(\{2\}) + n(\{4\}) + n(\{6\}) = 154 + 167 + 170 = 491,$$

and hence

$$f(A) = \frac{491}{1000} = 0.491.$$

The addition is justified because the elementary outcomes 2, 4, and 6 are disjoint possibilities. A single roll cannot be both 2 and 4.

This example shows a transition. We did not begin with a formula for probability. We began with a set A, observed how many recorded outcomes fell inside that set, and divided by the total number of trials.

3.11 What frequencies teach us

Observed frequencies have structural properties. These properties are not arbitrary. They come from the way sets of outcomes are counted in repeated observations.

First, frequencies are non-negative:

$$f(A) \geq 0.$$

They cannot be negative because counts cannot be negative.

Second, frequencies are at most 1:

$$f(A) \leq 1.$$

The event A cannot occur more often than the total number of trials.

Third, the whole sample space occurs on every trial:

$$f(\Omega) = 1.$$

Every recorded outcome belongs to Ω by construction.

Fourth, the impossible event never occurs:

$$f(\varnothing) = 0.$$

No recorded outcome belongs to the empty set.

The most important structural property concerns disjoint events. If $A \cap B = \varnothing$, then no trial can contribute to both A and B. Therefore

$$n(A \cup B) = n(A) + n(B),$$

and after division by the total number of trials,

$$f(A \cup B) = f(A) + f(B).$$

Theorem

For observed frequencies on a fixed finite sample space:

$$f(\varnothing) = 0, \qquad f(\Omega) = 1,$$

$$0 \leq f(A) \leq 1,$$

and if $A \cap B = \varnothing$, then

$$f(A \cup B) = f(A) + f(B).$$

These are not yet axioms. They are observations about how empirical frequencies behave. The axioms of probability will later abstract exactly this pattern.

3.12 When simple addition fails

If two events are not disjoint, simple addition counts the overlap twice. This is not a numerical accident; it is a structural fact about sets.

Let

$$M = \{1, 2, 3\}, \qquad N = \{3, 4, 5\}.$$

Then

$$M \cap N = \{3\}.$$

If we add $n(M)$ and $n(N)$, the trials with outcome 3 are counted once as part of M and once as part of N. Therefore

$$n(M \cup N) = n(M) + n(N) - n(M \cap N).$$

At the frequency level,

$$f(M \cup N) = f(M) + f(N) - f(M \cap N).$$

Remark

The correction term is not a trick. It is the price of overlap. Whenever an outcome has two reasons to be included, addition counts it twice unless the overlap is subtracted once.

The philosophy remains the same: the numerical rule is not primary. The structure of the event comes first; the numerical rule follows from that structure.

3.13 From observed frequency to probability

Observed frequencies are tied to a particular record of trials. If the same experiment is repeated again, the frequencies may change slightly. Yet the structure suggested by frequencies is stable:

events receive numbers between 0 and 1, the whole sample space receives 1, the empty event receives 0, and disjoint unions receive sums.

This suggests a new mathematical object. Instead of recording what happened in one finite collection of trials, we introduce a function that assigns a number to each event in a way that preserves the structural rules discovered above.

Definition

A **probability assignment** on a finite sample space Ω is a function

$$P : \mathcal{P}(\Omega) \to [0, 1]$$

that assigns a number $P(A)$ to each event $A \subseteq \Omega$.

The notation $\mathcal{P}(\Omega)$ denotes the power set of Ω, that is, the set of all subsets of Ω. Since events are subsets of Ω, the domain of P is the collection of events.

But not every function from $\mathcal{P}(\Omega)$ to $[0, 1]$ deserves to be called probability. The function must respect the structure of events.

Theorem

In a finite setting, a probability assignment should satisfy:

1. **Non-negativity:** for every event A,
$$P(A) \geq 0.$$
2. **Normalization:**
$$P(\Omega) = 1.$$
3. **Additivity for disjoint events:** if $A \cap B = \varnothing$, then
$$P(A \cup B) = P(A) + P(B).$$

These properties are not decorative. They are the minimum conditions needed for the numerical language to remain faithful to the set language.

3.14 Consequences of the axioms in the finite case

Once the axioms are accepted, several familiar rules become consequences rather than assumptions.

First, the impossible event has probability zero. Since

$$\Omega = \Omega \cup \varnothing$$

and the union is disjoint, additivity gives

$$P(\Omega) = P(\Omega) + P(\varnothing).$$

Subtracting $P(\Omega)$ from both sides gives

$$P(\varnothing) = 0.$$

Second, complements add up to one. For any event A, the events A and A^c are disjoint and

$$A \cup A^c = \Omega.$$

Therefore

$$P(A) + P(A^c) = P(\Omega) = 1,$$

so

$$P(A^c) = 1 - P(A).$$

Third, if $A \subseteq B$, then $P(A) \leq P(B)$. Indeed, B can be decomposed as

$$B = A \cup (B \setminus A),$$

where the two parts are disjoint. Hence

$$P(B) = P(A) + P(B \setminus A).$$

Since probabilities are non-negative,

$$P(B \setminus A) \geq 0,$$

and therefore

$$P(B) \geq P(A).$$

Finally, for arbitrary events A and B, the union formula follows from splitting the union into disjoint pieces:

$$A \cup B = A \cup (B \setminus A).$$

Thus

$$P(A \cup B) = P(A) + P(B \setminus A).$$

But B itself decomposes as

$$B = (B \setminus A) \cup (A \cap B),$$

so

$$P(B) = P(B \setminus A) + P(A \cap B).$$

Therefore

$$P(B \setminus A) = P(B) - P(A \cap B),$$

and hence

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Remark

The formulas are not separate facts to memorize. They are consequences of preserving the structure of events. Once the event algebra is understood, the probability rules become almost inevitable.

3.15 The axiomatic point of view

The modern theory of probability is axiomatic. This does not mean that it ignores intuition. It means that it takes the stable part of intuition and turns it into a structure that can be used reliably.

In finite spaces, the essential ideas are already visible:

- events are subsets of a sample space;
- probability assigns numbers to events;
- the whole space has probability 1;

- disjoint alternatives add.

For general probability spaces, the same philosophy is retained, but one must be more careful. Instead of assigning probabilities to all subsets of Ω, one usually assigns probabilities to a chosen family of events $\mathcal{F}$. This family must be closed under the operations needed for logical reasoning: complements, countable unions, and therefore also countable intersections. Such a family is called a σ-algebra.

Definition

A **probability space** is a triple

$$(\Omega, \mathcal{F}, P),$$

where Ω is a sample space, $\mathcal{F}$ is a σ-algebra of events, and $P : \mathcal{F} \to [0, 1]$ is a probability measure satisfying:

1. $P(A) \geq 0$ for all $A \in \mathcal{F}$,
2. $P(\Omega) = 1$,
3. if $A_1, A_2, A_3, \ldots$ are pairwise disjoint events, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

The third condition is countable additivity. In finite examples it reduces to the additivity rules already seen. In infinite settings it becomes a deep structural requirement: it says that probability remains compatible with decompositions into countably many disjoint alternatives.

This is the philosophical endpoint of the chapter. We began with informal statements about outcomes. We discovered that statements select sets. We discovered that logical relations between statements become set operations. We observed that frequencies assign numbers to events in a structured way. Finally, we abstracted the stable part of that structure into axioms.

The beauty of the formalism is that it is not imposed from outside. It is discovered by asking, step by step, what must be preserved if reasoning under uncertainty is to become mathematics.

Chapter 4

Sample Spaces and Elementary Probability

This chapter develops several canonical examples of sample spaces and elementary probability from the ground up. The emphasis is on understanding the experiment, deciding what should count as an elementary outcome, constructing the sample space, defining events, and assigning probabilities only after the mathematical model has been made explicit.

Different representations of the same probabilistic structure are used throughout the chapter: lists, tables, trees, diagrams, and symbolic notation. The goal is to show that a sample space is an abstract object produced by analysis and modelling, not a collection of observed data.

4.1 From an experiment to a sample space

The previous chapter introduced the formal language of probability: a sample space, events as subsets of the sample space, operations on events, and a probability assignment. We now use that language in concrete models. The main point of the present chapter is methodological:

a sample space is not found inside the experiment; it is constructed.

A physical or conceptual experiment may be described in ordinary language, but ordinary language does not by itself say what the elementary outcomes are. We must decide what information is recorded, what distinctions are visible in the model, and what distinctions are deliberately ignored.

Definition

A **probabilistic model** begins with a decision about what is recorded. A **sample space** Ω is the set of all elementary outcomes under that recording rule. A **probability assignment** then gives probabilities to events, that is, to subsets of Ω.

Thus the construction has the following order:

$$\text{experiment} \longrightarrow \text{recording rule} \longrightarrow \text{elementary outcome} \longrightarrow \Omega \longrightarrow \text{events} \longrightarrow P.$$

The order matters. If we skip the recording rule, then we do not really know what one elementary outcome is. If we do not know what one elementary outcome is, then we do not know what the sample space is. If we do not know the sample space, then symbols such as $A \subseteq \Omega$ and $P(A)$ have no precise meaning.

The same physical situation may have different sample spaces

Consider the instruction:

toss a coin twice.

This instruction describes a physical procedure, but it does not yet fully describe the mathematical model. Several recording rules are possible.

If we record the full ordered sequence of outcomes, then one elementary outcome is a sequence such as HT. The sample space is

$$\Omega_1 = \{HH, HT, TH, TT\}.$$

In this model $HT \neq TH$, because the first position and the second position are different pieces of information.

If instead we record only the number of heads, then one elementary outcome is a number. The sample space is

$$\Omega_2 = \{0, 1, 2\}.$$

In this model the sequences HT and TH are not separate elementary outcomes. They both correspond to the same recorded value, namely 1.

If we record only whether at least one head appeared, then the sample space is

$$\Omega_3 = \{\text{yes}, \text{no}\}.$$

This is another legitimate model, but it answers a different question.

Remark

The phrase "the experiment is the same" can be misleading. The physical action may be the same, but the mathematical sample space depends on what is recorded. Probability theory does not work directly with the physical action; it works with the mathematical description produced from that action.

Elementary outcomes are model-dependent

An elementary outcome is not necessarily the smallest physical object involved. It is the smallest complete result according to the model.

Example

Suppose a ball is drawn from an urn containing three red balls and two blue balls. If the balls are labelled

$$R_1, R_2, R_3, B_1, B_2,$$

and the label is recorded, then the sample space is

$$\Omega = \{R_1, R_2, R_3, B_1, B_2\}.$$

Here the elementary outcomes are individual labelled balls.
If only the colour is recorded, then the sample space is instead

$$\Omega' = \{R, B\}.$$

Here the elementary outcomes are colours, not labelled balls. The two sample spaces describe different levels of information.

The second model is not wrong. It may be exactly the model we need if the question is only about colour. But it must be treated honestly: changing the recording rule changes the sample space and may also change how probabilities are assigned.

The sample space is only half of the model

After Ω has been constructed, we still have to assign probabilities. The sample space tells us which elementary outcomes exist in the model. It does not automatically say how likely they are.

Example

For a single coin toss we may choose

$$\Omega = \{H, T\}.$$

If the coin is modelled as symmetric, then we assign

$$P(H) = P(T) = \frac{1}{2}.$$

If the coin is modelled as biased toward heads, then we may assign

$$P(H) = p, \qquad P(T) = 1 - p, \qquad p \neq \frac{1}{2}.$$

The sample space is the same in both models, but the probability assignment is different.

This distinction is fundamental. A list of possible outcomes is not yet a probability model. A probability model requires both:

$$\text{possible outcomes} \quad \text{and} \quad \text{probabilities of events.}$$

Uniform and non-uniform assignments

In a finite sample space

$$\Omega = \{\omega_1, \omega_2, \ldots, \omega_n\},$$

a probability assignment is determined by the probabilities of the elementary outcomes:

$$P(\{\omega_1\}) = p_1, \quad P(\{\omega_2\}) = p_2, \quad \ldots, \quad P(\{\omega_n\}) = p_n.$$

These numbers must satisfy

$$p_i \geq 0 \quad \text{for every } i,$$

and

$$p_1 + p_2 + \cdots + p_n = 1.$$

Then the probability of an event is obtained by adding the probabilities of the elementary outcomes contained in that event:

$$P(A) = \sum_{\omega_i \in A} p_i.$$

Definition

A finite probability model is **uniform** if all elementary outcomes have the same probability. If $|\Omega| = n$, this means

$$P(\{\omega\}) = \frac{1}{n}$$

for every $\omega \in \Omega$.

Only in a uniform finite model do we obtain the classical formula

$$P(A) = \frac{|A|}{|\Omega|}.$$

This formula is not the definition of probability in every finite model. It is a consequence of the special assumption that all elementary outcomes are equally likely.

Remark

A common error is to count how many named outcomes appear in a sample space and then assume that all names have equal probability. Equal naming is not the same as equal probability. Equal probability must come from the mechanism of the model or from an explicit modelling assumption.

Events are questions asked of the sample space

Once Ω is fixed, an event is a subset of Ω. Conceptually, an event answers a question about the elementary outcome.

Example

If a die is rolled and

$$\Omega = \{1, 2, 3, 4, 5, 6\},$$

then the question

$$\text{is the result even?}$$

selects the event

$$A = \{2, 4, 6\}.$$

The question

$$\text{is the result at least 5?}$$

selects the event

$$B = \{5, 6\}.$$

The compound question

$$\text{is the result even and at least 5?}$$

selects the event

$$A \cap B = \{6\}.$$

This is why the sample space must be fixed before events are discussed. The same sentence can select different sets in different spaces.

A first complete modelling pattern

When solving elementary probability problems, we shall repeatedly use the following pattern.

1. Describe the physical or conceptual experiment.
2. State what information is recorded.
3. Identify one elementary outcome.
4. Construct the sample space Ω.
5. Describe the event or events of interest as subsets of Ω.
6. Decide whether elementary outcomes are equally likely.
7. Assign probabilities to elementary outcomes or directly to events.
8. Compute the required probability and interpret it in the original situation.

This pattern may look slow. That is intentional. At the beginning, speed is less important than correctness. Most wrong solutions to elementary probability problems do not fail because of difficult arithmetic. They fail because the sample space, the elementary outcomes, or the probability assignment were never made explicit.

Remark

The goal of this chapter is not to collect examples. The goal is to learn a method: construct the model first, calculate only after the model is clear.

4.2 Coin models

Coin tossing is the simplest family of probability models, but it is not trivial. It already contains several ideas that will appear throughout the course: a sample space, a recording rule, a probability assignment, repeated experiments, product spaces, events described by conditions, and the difference between a particular sequence and a numerical summary of that sequence.

The physical object called a coin does not determine a probability model by itself. We must say what is recorded and what assumptions are made about the mechanism. A coin may be treated as symmetric, or it may be treated as biased. A single toss may be recorded, or a whole sequence of tosses may be recorded. Each of these choices changes the model or the probabilities inside the model.

One toss of a coin

Suppose first that a coin is tossed once and that we record which side lands up. The elementary outcome is one of two symbols:

$$H \quad \text{or} \quad T,$$

where H denotes heads and T denotes tails. The sample space is

$$\Omega = \{H, T\}.$$

At this point we have only described the possible outcomes. We have not yet assigned probabilities.

If the coin is modelled as symmetric, then the two elementary outcomes are given equal probability:

$$P(H) = P(T) = \frac{1}{2}.$$

This is a modelling assumption based on symmetry. It is not a consequence of the fact that the set Ω has two elements. The fact that $|\Omega| = 2$ tells us how many elementary outcomes there are. It does not by itself tell us how likely they are.

If the coin is biased, we may instead write

$$P(H) = p, \qquad P(T) = 1 - p,$$

where

$$0 \le p \le 1.$$

The same sample space $\{H, T\}$ now carries a different probability assignment. When $p = 1/2$, the model is symmetric. When $p \neq 1/2$, one side is more likely than the other.

> **Remark**
>
> The symbols H and T are not probabilities. They are elementary outcomes. The numbers assigned to them come from an additional assumption about the coin and the tossing mechanism.

Events in the one-toss model

In the one-toss model there are only four events:

$$\varnothing, \qquad \{H\}, \qquad \{T\}, \qquad \Omega.$$

Their probabilities are determined by the probabilities of H and T:

$$P(\varnothing) = 0, \qquad P(\Omega) = 1,$$

$$P(\{H\}) = p, \qquad P(\{T\}) = 1 - p.$$

For a symmetric coin this becomes

$$P(\{H\}) = P(\{T\}) = \frac{1}{2}.$$

Even this small example illustrates the general finite rule. If an event is a set of elementary outcomes, then its probability is obtained by adding the probabilities of the elementary outcomes inside it.

Two tosses: sequences as elementary outcomes

Now suppose the coin is tossed twice and the complete order of results is recorded. An elementary outcome is no longer a single symbol. It is an ordered sequence of length two. The sample space is

$$\Omega = \{HH, HT, TH, TT\}.$$

The outcomes HT and TH are different because the first position and the second position are different. The outcome HT means heads first and tails second. The outcome TH means tails first and heads second.

If the coin is symmetric and the two tosses are modelled as independent repetitions of the same mechanism, then every sequence has probability

$$\frac{1}{4}.$$

Thus

$$P(HH) = P(HT) = P(TH) = P(TT) = \frac{1}{4}.$$

If the coin has $P(H) = p$ and $P(T) = 1 - p$, then the probabilities of the four sequences are

$$P(HH) = p^2, \qquad P(HT) = p(1-p),$$

$$P(TH) = (1-p)p, \qquad P(TT) = (1-p)^2.$$

The middle two values are equal because multiplication of numbers is commutative:

$$p(1-p) = (1-p)p.$$

But the sequences HT and TH remain different elementary outcomes.

Example

Let $p = 0.7$. Then

$$P(HH) = 0.49, \qquad P(HT) = 0.21, \qquad P(TH) = 0.21, \qquad P(TT) = 0.09.$$

These probabilities add to one:

$$0.49 + 0.21 + 0.21 + 0.09 = 1.$$

The sample space has four outcomes, but the model is not uniform.

The event "exactly one head"

The event

$$A = \{\text{exactly one head occurs}\}$$

contains two elementary outcomes:

$$A = \{HT, TH\}.$$

Therefore, for a symmetric coin,

$$P(A) = P(HT) + P(TH) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

For a biased coin with $P(H) = p$,

$$P(A) = p(1-p) + (1-p)p = 2p(1-p).$$

Notice the difference between an elementary outcome and an event. The sequence HT is one elementary outcome. The phrase "exactly one head" describes an event containing two elementary outcomes. Probability problems often ask for the probability of an event, not of a single elementary outcome.

Three tosses and the structure of a product space

For three tosses, with order recorded, the sample space is

$$\Omega = \{H, T\}^3.$$

This notation means the set of all ordered triples whose entries are H or T. Explicitly,

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}.$$

There are

$$2^3 = 8$$

elementary outcomes, because each of the three positions has two possible symbols.

For a symmetric coin, each sequence has probability

$$\frac{1}{8}.$$

For a biased coin with $P(H) = p$, the probability of a sequence is obtained by multiplying the probabilities of its symbols. For example,

$$P(HTH) = p(1-p)p = p^2(1-p),$$

and

$$P(TTH) = (1-p)(1-p)p = (1-p)^2 p.$$

The exponent records how many times a symbol appears in the sequence. A sequence with k heads and $3 - k$ tails has probability

$$p^k(1-p)^{3-k}.$$

The general model of n tosses

For n tosses, with order recorded, the sample space is

$$\Omega = \{H, T\}^n.$$

An elementary outcome is a sequence

$$\omega = (a_1, a_2, \ldots, a_n),$$

where each a_i is either H or T. There are

$$|\Omega| = 2^n$$

such sequences.

If the coin is symmetric and each toss is modelled as an independent repetition, then the model is uniform:

$$P(\omega) = \frac{1}{2^n}$$

for every $\omega \in \Omega$.

If $P(H) = p$ and $P(T) = 1 - p$, then a sequence with k heads and $n - k$ tails has probability

$$p^k(1-p)^{n-k}.$$

This formula is not yet the probability of "exactly k heads". It is the probability of one particular sequence with k heads.

Example

For five tosses, the sequence

$$HHTTH$$

has three heads and two tails. Its probability in the biased model is

$$p^3(1-p)^2.$$

The sequence

$$HTHTH$$

also has three heads and two tails, so it has the same probability

$$p^3(1-p)^2.$$

But the two sequences are different elementary outcomes.

Exactly k heads in n tosses

Now consider the event

$$A_k = \{\omega \in \{H, T\}^n : \omega \text{ contains exactly } k \text{ heads}\}.$$

This event contains all sequences of length n in which exactly k positions are occupied by H, and the remaining $n - k$ positions are occupied by T.

To count these sequences, we do not choose the heads themselves as physical objects. We choose the positions at which heads occur. There are n positions, and we choose k of them to contain H. Once those positions are chosen, the remaining positions must contain T. Hence

$$|A_k| = \binom{n}{k}.$$

For a symmetric coin, the whole space is uniform, so

$$P(A_k) = \frac{|A_k|}{|\Omega|} = \frac{\binom{n}{k}}{2^n}.$$

For a biased coin, the space is generally not uniform. However, every sequence in A_k has the same probability $p^k(1-p)^{n-k}$. Since there are $\binom{n}{k}$ such sequences, additivity gives

$$P(A_k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Theorem

If a coin with $P(H) = p$ is tossed n times independently, then the probability of exactly k heads is

$$\binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

This formula has three parts:

- $\binom{n}{k}$ counts the positions of the heads;
- p^k is the contribution of the k heads;
- $(1-p)^{n-k}$ is the contribution of the $n-k$ tails.

What this model teaches

The coin model teaches several lessons that will recur in more complicated examples.

First, the sample space depends on what is recorded. One toss gives $\{H, T\}$; n ordered tosses give $\{H, T\}^n$; recording only the number of heads gives $\{0, 1, \ldots, n\}$.

Second, the probability assignment is separate from the sample space. The same space $\{H, T\}$ can carry the symmetric assignment $1/2, 1/2$ or the biased assignment $p, 1-p$.

Third, an event may contain many elementary outcomes. The event "exactly k heads" is not one sequence; it is a set of sequences.

Fourth, combinatorics enters probability when we count how many elementary outcomes satisfy a condition. The binomial coefficient appears because we count positions of successes, not because it is a formula to be memorized in isolation.

Remark

The coin model is simple enough to be completely transparent, but rich enough to show the entire modelling workflow: choose the recording rule, construct Ω, assign probabilities, define events, count favourable outcomes, and interpret the result.

4.3 Dice models

Dice models are the natural next step after coin models. A die has more than two visible faces, and this makes it easier to see the difference between a uniform sample space and a non-uniform probability assignment. Dice also provide one of the most important examples in elementary probability: the model of two dice, where elementary outcomes are ordered pairs but many questions are about sums, comparisons, or other functions of those pairs.

As before, we do not begin with a formula. We begin with the experiment and the recording rule.

One roll of one die

Suppose a die is rolled once and the number on the upper face is recorded. The sample space is

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

An elementary outcome is one number from this set.

If the die is modelled as fair, then all six elementary outcomes have the same probability:

$$P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}.$$

This is the uniform model on a six-element sample space.

If the die is not assumed to be fair, then we may write

$$P(i) = p_i, \qquad i = 1, 2, 3, 4, 5, 6,$$

where

$$p_i \geq 0$$

for every i, and

$$p_1 + p_2 + p_3 + p_4 + p_5 + p_6 = 1.$$

The sample space is still $\{1, 2, 3, 4, 5, 6\}$, but the probability assignment is no longer necessarily uniform.

Remark

The sentence "a die has six possible results" does not imply that each result has probability $1/6$. The value $1/6$ comes from the additional modelling assumption that the die and the rolling mechanism are symmetric enough to treat the six faces equally.

Events for one die

Events are subsets of Ω. For example, the event that the result is even is

$$E = \{2, 4, 6\}.$$

The event that the result is at least 5 is

$$A = \{5, 6\}.$$

The event that the result is prime is

$$B = \{2, 3, 5\}.$$

In the fair model,

$$P(E) = \frac{|E|}{|\Omega|} = \frac{3}{6} = \frac{1}{2},$$

$$P(A) = \frac{2}{6} = \frac{1}{3},$$

and

$$P(B) = \frac{3}{6} = \frac{1}{2}.$$

In a non-uniform model, we cannot use only the number of favourable outcomes. Instead, we must add the probabilities of the elementary outcomes inside the event:

$$P(E) = p_2 + p_4 + p_6,$$

$$P(A) = p_5 + p_6,$$

$$P(B) = p_2 + p_3 + p_5.$$

Example

Suppose

$$(p_1, p_2, p_3, p_4, p_5, p_6) = (0.10, 0.10, 0.15, 0.15, 0.20, 0.30).$$

Then the event $E = \{2, 4, 6\}$ has probability

$$P(E) = 0.10 + 0.15 + 0.30 = 0.55.$$

Even though E contains three of the six possible faces, its probability is not $3/6$, because the elementary outcomes are not equally likely.

Two dice as ordered pairs

Now suppose two dice are rolled and the number on each die is recorded. If the dice are distinguished, for example by colour, then the outcome is an ordered pair

$$(i, j),$$

where i is the result on the first die and j is the result on the second die. The sample space is

$$\Omega = \{1, 2, 3, 4, 5, 6\} \times \{1, 2, 3, 4, 5, 6\}.$$

Equivalently,

$$\Omega = \{(i, j) : i, j \in \{1, 2, 3, 4, 5, 6\}\}.$$

It has

$$|\Omega| = 6 \cdot 6 = 36$$

elementary outcomes.

If both dice are fair and the rolls are modelled as independent, then every ordered pair has probability

$$\frac{1}{36}.$$

Thus the model is uniform on the set of ordered pairs.

Remark

The outcomes $(2, 5)$ and $(5, 2)$ are different in this model. They represent different results on the first and second die. They may lead to the same sum, but they are not the same elementary outcome.

The event that the sum is seven

Consider the event

$$S_7 = \{\text{the sum of the two dice is } 7\}.$$

In the ordered-pair model this event is the set

$$S_7 = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}.$$

It contains six elementary outcomes. Since the fair two-dice model is uniform,

$$P(S_7) = \frac{|S_7|}{|\Omega|} = \frac{6}{36} = \frac{1}{6}.$$

The count 6 comes from a structural fact. To obtain sum 7, once the first coordinate i is chosen, the second coordinate must be $7 - i$. The possible first coordinates are $1, 2, 3, 4, 5, 6$, and each gives one valid ordered pair.

Example

The event that the sum is 4 is

$$S_4 = \{(1,3),(2,2),(3,1)\}.$$

Hence

$$P(S_4) = \frac{3}{36} = \frac{1}{12}$$

in the fair two-dice model.

The events S_4 and S_7 are both described by a sum, but they contain different numbers of ordered pairs. Therefore they have different probabilities in the fair two-dice model.

Representing two dice by a table

A useful representation of the two-dice sample space is a 6×6 table:

	1	2	3	4	5	6
1	$(1,1)$	$(1,2)$	$(1,3)$	$(1,4)$	$(1,5)$	$(1,6)$
2	$(2,1)$	$(2,2)$	$(2,3)$	$(2,4)$	$(2,5)$	$(2,6)$
3	$(3,1)$	$(3,2)$	$(3,3)$	$(3,4)$	$(3,5)$	$(3,6)$
4	$(4,1)$	$(4,2)$	$(4,3)$	$(4,4)$	$(4,5)$	$(4,6)$
5	$(5,1)$	$(5,2)$	$(5,3)$	$(5,4)$	$(5,5)$	$(5,6)$
6	$(6,1)$	$(6,2)$	$(6,3)$	$(6,4)$	$(6,5)$	$(6,6)$

Rows record the first die and columns record the second die. This table is not just a picture. It makes the elementary outcomes visible and allows events to be seen as regions.

For example, the event

$$A = \{(i,j) : i = j\}$$

that both dice show the same number is the diagonal of the table:

$$A = \{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6)\}.$$

Thus

$$P(A) = \frac{6}{36} = \frac{1}{6}.$$

The event

$$B = \{(i,j) : i > j\}$$

that the first die shows a larger number than the second die is the region below the diagonal. It contains

$$1 + 2 + 3 + 4 + 5 = 15$$

elementary outcomes, so

$$P(B) = \frac{15}{36} = \frac{5}{12}.$$

The trap of recording only the sum

There is another possible recording rule. Instead of recording the ordered pair, we might record only the sum. Then the sample space becomes

$$\Omega' = \{2,3,4,5,6,7,8,9,10,11,12\}.$$

This is a legitimate sample space if the recorded outcome is only the sum. However, it is not a uniform sample space when the sum comes from rolling two fair dice.

The value 7 can occur in six ordered ways:

$$(1,6),(2,5),(3,4),(4,3),(5,2),(6,1).$$

The value 2 can occur in only one ordered way:

$$(1,1).$$

Therefore

$$P(7)=\frac{6}{36}, \qquad P(2)=\frac{1}{36}.$$

They are not equal.

Remark

The set $\{2,3,\dots,12\}$ has eleven elements, but the sums of two fair dice are not equally likely. A smaller or more aggregated sample space may be useful, but its probabilities must be inherited from the underlying mechanism, not assigned uniformly without justification.

The complete distribution of the sum is

s	2	3	4	5	6	7	8	9	10	11	12
# pairs	1	2	3	4	5	6	5	4	3	2	1

Hence

$$P(S=s)=\frac{\#\{(i,j):i+j=s\}}{36}.$$

For example,

$$P(S=10)=\frac{3}{36}=\frac{1}{12}.$$

Non-uniform dice and product assignments

If one die has probabilities

$$P(1)=p_1,\dots,P(6)=p_6$$

and another die has probabilities

$$Q(1)=q_1,\dots,Q(6)=q_6,$$

then, under the assumption that the two rolls are independent, the probability of the ordered pair (i,j) is

$$P((i,j))=p_iq_j.$$

The event that the sum is 7 then has probability

$$P(S_7)=p_1q_6+p_2q_5+p_3q_4+p_4q_3+p_5q_2+p_6q_1.$$

This expression is longer than $6/36$, but it says exactly the same kind of thing: add the probabilities of the elementary outcomes that belong to the event.

Repeated rolls of one die

For n rolls of one die, with order recorded, the sample space is

$$\Omega=\{1,2,3,4,5,6\}^n.$$

An elementary outcome is a sequence

$$(a_1,a_2,\dots,a_n),$$

where each $a_i \in \{1, 2, 3, 4, 5, 6\}$. The number of elementary outcomes is

$$6^n.$$

If the die is fair and the rolls are independent, then each sequence has probability

$$\frac{1}{6^n}.$$

If the die has probabilities $p_1, \ldots, p_6$, then

$$P(a_1, a_2, \ldots, a_n) = p_{a_1} p_{a_2} \cdots p_{a_n}.$$

Example

For three rolls of a non-uniform die, the sequence $(2, 6, 2)$ has probability

$$p_2 p_6 p_2 = p_2^2 p_6.$$

The event "exactly two sixes in three rolls" contains the sequences

$$(6, 6, x), \quad (6, x, 6), \quad (x, 6, 6),$$

where x is not 6. In the fair model its probability is

$$\binom{3}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right).$$

The binomial coefficient counts the positions occupied by the sixes.

What dice models teach

Dice models teach three important lessons.

First, a finite sample space may be uniform or non-uniform. The same set $\{1, 2, 3, 4, 5, 6\}$ can describe a fair die or a biased die.

Second, when an experiment has several stages, an elementary outcome is often a sequence or ordered tuple. For two dice, the natural elementary outcomes are ordered pairs.

Third, an aggregated value such as a sum is not automatically uniform. The sum of two fair dice has values in $\{2, 3, \ldots, 12\}$, but these values have different probabilities because they correspond to different numbers of ordered pairs.

Remark

The safest modelling habit is to construct the detailed sample space first and only then decide whether a coarser description, such as the sum, is appropriate. Aggregation is useful, but it must preserve the probabilities induced by the underlying model.

4.4 Card models

Card models are useful because the phrase "draw cards" is incomplete. It does not say whether we record one card or several cards, whether order matters, whether cards are replaced, or whether the final object is a hand, a sequence, or only a summary such as the number of aces. Each of these decisions changes the sample space.

In this section let D denote a standard deck of 52 distinct cards. The cards are distinct because a card is determined by both rank and suit. For example, the ace of hearts and the ace of spades are different cards.

Drawing one card

Suppose one card is drawn from a well-shuffled deck and the exact card is recorded. The sample space is

$$\Omega = D.$$

It has

$$|\Omega| = 52$$

elementary outcomes. If the deck is well shuffled and no card is favoured, the uniform model assigns

$$P(c) = \frac{1}{52}$$

for every card $c \in D$.

Events are subsets of the deck. For example, the event that the card is an ace is

$$A = \{\text{ace of hearts}, \text{ace of diamonds}, \text{ace of clubs}, \text{ace of spades}\}.$$

Therefore

$$P(A) = \frac{4}{52} = \frac{1}{13}.$$

The event that the card is a heart contains 13 cards, so

$$P(\text{heart}) = \frac{13}{52} = \frac{1}{4}.$$

The event that the card is a face card contains the jacks, queens, and kings in four suits, hence 12 cards, so

$$P(\text{face card}) = \frac{12}{52} = \frac{3}{13}.$$

Remark

The sample space contains individual cards, not ranks. If we record the exact card, then there are 52 elementary outcomes. The event "ace" is not one elementary outcome; it is a set of four elementary outcomes.

A five-card hand as a set

Now suppose five cards are dealt and only the final hand is recorded. The order in which the cards arrived is ignored. Then an elementary outcome is a five-card subset of the deck:

$$S \subseteq D, \qquad |S| = 5.$$

The sample space is

$$\Omega = \{S \subseteq D : |S| = 5\}.$$

Its size is

$$|\Omega| = \binom{52}{5}.$$

This count appears because a hand is a set. We choose which five cards belong to the hand, and the order of listing those cards does not create a new hand. Thus

$$\{A\heartsuit, K\clubsuit, 7\diamondsuit, 3\spadesuit, 2\heartsuit\}$$

is the same hand no matter in which order we write its elements.

If the deck is well shuffled and every five-card hand is equally likely, then the uniform formula applies:

$$P(E) = \frac{|E|}{\binom{52}{5}}$$

for any event E consisting of five-card hands.

Example

Let E be the event that the hand contains exactly one ace. To construct such a hand, choose one of the four aces and choose the remaining four cards from the 48 non-aces. Hence

$$|E| = \binom{4}{1}\binom{48}{4}.$$

Therefore

$$P(E) = \frac{\binom{4}{1}\binom{48}{4}}{\binom{52}{5}}.$$

This is a probability of an event, not of a single hand. The event contains many hands.

Five cards drawn in order

Suppose instead that five cards are drawn one after another and the order is recorded. Then an elementary outcome is a sequence

$$(c_1, c_2, c_3, c_4, c_5),$$

where all five cards are distinct. The sample space is

$$\Omega_{\text{seq}} = \{(c_1, c_2, c_3, c_4, c_5) : c_i \in D,\ c_i \neq c_j \text{ for } i \neq j\}.$$

Its size is

$$52 \cdot 51 \cdot 50 \cdot 49 \cdot 48.$$

The first position has 52 choices, the second has 51 choices because one card has already been used, and so on.

This sample space is different from the five-card hand space. A hand ignores order; a sequence records order. The same five cards can appear in

$$5!$$

different orders. Therefore

$$52 \cdot 51 \cdot 50 \cdot 49 \cdot 48 = 5!\binom{52}{5}.$$

Remark

The phrase "draw five cards" is not enough to determine the sample space. If order is ignored, the elementary outcome is a set. If order is recorded, the elementary outcome is a sequence. Both models are legitimate, but they answer different questions.

The same event in two sample spaces

Consider the statement

the five cards contain exactly one ace.

In the hand model, the event is a set of five-card subsets. In the sequence model, the event is a set of ordered five-tuples. The verbal statement is the same, but the formal event lives in a different sample space.

In the hand model we already counted

$$\binom{4}{1}\binom{48}{4}$$

favourable hands.

In the ordered model, we may count favourable sequences directly. First choose which position contains the ace: 5 choices. Choose the ace: 4 choices. Then fill the remaining four ordered positions with non-aces without repetition:

$$48 \cdot 47 \cdot 46 \cdot 45$$

choices. Thus the number of favourable sequences is

$$5 \cdot 4 \cdot 48 \cdot 47 \cdot 46 \cdot 45.$$

The total number of ordered sequences is

$$52 \cdot 51 \cdot 50 \cdot 49 \cdot 48.$$

So the probability in the ordered model is

$$\frac{5 \cdot 4 \cdot 48 \cdot 47 \cdot 46 \cdot 45}{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}.$$

This value is equal to

$$\frac{\binom{4}{1}\binom{48}{4}}{\binom{52}{5}},$$

because both models describe the same random dealing mechanism at different levels of detail.

The equality is not magic. Each unordered hand corresponds to exactly 5! ordered sequences. The factor 5! appears in both the favourable count and the total count, so it cancels.

Drawing with replacement

A different model appears if each drawn card is returned to the deck before the next draw. Suppose five draws are made, the exact card is recorded each time, and replacement occurs after every draw. Then the same card may appear several times in the recorded sequence. The sample space is

$$\Omega_{\text{rep}} = D^5.$$

Its size is

$$|\Omega_{\text{rep}}| = 52^5.$$

If the deck is well shuffled before each draw, every sequence in D^5 has probability

$$\frac{1}{52^5}.$$

This is not the same as drawing five cards without replacement. For example, under replacement the sequence

$$(A\heartsuit, A\heartsuit, 7\clubsuit, 2\diamondsuit, K\spadesuit)$$

is possible. Without replacement it is impossible, because the same physical card cannot be drawn twice.

Example

In five draws with replacement, what is the probability that all five cards are aces?

Each draw has probability $4/52 = 1/13$ of producing an ace. Since replacement makes the same probability available at each draw, the probability is

$$\left(\frac{4}{52}\right)^5 = \left(\frac{1}{13}\right)^5.$$

Equivalently, there are 4^5 ace-only sequences and 52^5 total sequences, so

$$P(\text{all aces}) = \frac{4^5}{52^5}.$$

Without replacement, the probability that five drawn cards are all aces is zero, because the deck contains only four aces. The change in the physical procedure changes the sample space and therefore changes the probabilities.

At least one ace

Card models are also useful for showing the complement method. In a five-card hand, let

$$A = \{\text{the hand contains at least one ace}\}.$$

Counting hands with at least one ace directly would require cases: exactly one ace, exactly two aces, exactly three aces, exactly four aces. It is often simpler to count the complement.

The complement is

$$A^c = \{\text{the hand contains no aces}\}.$$

There are 48 non-aces, so

$$|A^c| = \binom{48}{5}.$$

Therefore

$$P(A^c) = \frac{\binom{48}{5}}{\binom{52}{5}},$$

and

$$P(A) = 1 - P(A^c) = 1 - \frac{\binom{48}{5}}{\binom{52}{5}}.$$

The complement method is not a trick. It is event algebra: instead of counting all hands that satisfy the event, we count the hands that fail it and subtract from the whole space.

What card models teach

Card models emphasize that the same everyday phrase can hide several different mathematical structures.

If one card is drawn and the exact card is recorded, then $\Omega = D$. If five cards are recorded as a hand, then Ω is a set of five-element subsets. If five cards are recorded in order without replacement, then Ω is a set of sequences with no repeated card. If cards are drawn with replacement, then $\Omega = D^5$, and repeated cards become possible.

Remark

Before solving a card problem, always identify whether the elementary outcome is a single card, a set of cards, an ordered sequence of cards, or a summary of the cards. Most card-counting errors are errors about the sample space, not errors about arithmetic.

4.5 Urn models

Urn models are a standard way to study how probability spaces are constructed. They are especially useful because one physical setup can be described at several levels. We may record the exact labelled object selected from the urn, or we may record only a visible type such as colour. We may sample with replacement or without replacement. We may record order or ignore order. Each choice leads to a different sample space.

One selection from labelled objects

Suppose an urn contains five labelled objects:

$$R_1, R_2, R_3, B_1, B_2,$$

where the letter records type and the subscript records the label. If one object is selected and its label is recorded, then

$$\Omega = \{R_1, R_2, R_3, B_1, B_2\}.$$

If the selection mechanism treats the five labelled objects equally, then each has probability $1/5$.

The event that the selected object is of type R is

$$A = \{R_1, R_2, R_3\}.$$

Therefore

$$P(A) = \frac{3}{5}.$$

The event that the selected object is of type B is

$$A^c = \{B_1, B_2\},$$

so

$$P(A^c) = \frac{2}{5}.$$

Remark

In this model, R_1, R_2, and R_3 are different elementary outcomes. They have the same type, but the labels are recorded. The event "type R" is therefore a set of elementary outcomes, not a single elementary outcome.

Recording only type

Now use the same physical urn, but record only the type of the selected object. Then the sample space is

$$\Omega' = \{R, B\}.$$

This sample space has two elements, but it is not automatically uniform. The probability of type R is inherited from the three labelled objects of type R, and the probability of type B is inherited from the two labelled objects of type B:

$$P(R) = \frac{3}{5}, \qquad P(B) = \frac{2}{5}.$$

This is the central lesson of the urn model: when several detailed elementary outcomes are grouped into one recorded outcome, equal probabilities at the detailed level may become unequal probabilities at the aggregated level.

Example

Writing

$$P(R) = P(B) = \frac{1}{2}$$

would describe a different mechanism: one that first chooses a type uniformly. It would not describe uniform selection from an urn containing three labelled objects of type R and two labelled objects of type B.

Two selections with replacement

Suppose one object is selected, recorded, returned to the urn, and then another selection is made. If labels and order are recorded, then one elementary outcome is an ordered pair

$$(x_1, x_2),$$

where each coordinate belongs to

$$\{R_1, R_2, R_3, B_1, B_2\}.$$

Because the first selected object is returned, the same label may appear twice. Thus

$$\Omega = \{R_1, R_2, R_3, B_1, B_2\}^2, \qquad |\Omega| = 25.$$

The event that both selected objects have type R is

$$A = \{(R_i, R_j) : i, j \in \{1, 2, 3\}\}.$$

There are $3 \cdot 3 = 9$ such ordered pairs, so

$$P(A) = \frac{9}{25}.$$

Equivalently,

$$P(A) = \frac{3}{5} \cdot \frac{3}{5}.$$

Two selections without replacement

Now suppose the first selected object is not returned before the second selection. If labels and order are recorded, then

$$\Omega = \{(x_1, x_2) : x_1, x_2 \in \{R_1, R_2, R_3, B_1, B_2\},\ x_1 \neq x_2\}.$$

There are

$$5 \cdot 4 = 20$$

elementary outcomes.

The event that both selected objects have type R is

$$A = \{(R_i, R_j) : i, j \in \{1, 2, 3\},\ i \neq j\}.$$

There are $3 \cdot 2 = 6$ such ordered pairs, hence

$$P(A) = \frac{6}{20} = \frac{3}{10}.$$

The probability is smaller than in the replacement model because, after one object of type R has been selected, fewer such objects remain available.

Example

The event "one object of each type" has two ordered forms:

$$(R_i, B_j) \quad \text{or} \quad (B_j, R_i).$$

There are $3 \cdot 2 = 6$ outcomes of the first form and $2 \cdot 3 = 6$ outcomes of the second form. Therefore

$$P(\text{one of each type}) = \frac{12}{20} = \frac{3}{5}.$$

Order recorded or ignored

If two objects are selected without replacement and order is ignored, then an elementary outcome is a two-element subset of the five labelled objects:

$$\Omega = \{S \subseteq \{R_1, R_2, R_3, B_1, B_2\} : |S| = 2\}.$$

Thus

$$|\Omega| = \binom{5}{2} = 10.$$

The event that both selected objects have type R has

$$\binom{3}{2} = 3$$

favourable subsets, so

$$P(\text{both type } R) = \frac{\binom{3}{2}}{\binom{5}{2}} = \frac{3}{10}.$$

This agrees with the ordered calculation because each unordered pair corresponds to exactly two ordered pairs.

Only the number of type R objects is recorded

Sometimes we ignore labels and order and record only a count. In two selections with replacement, the possible recorded values for the number of type R objects are

$$\Omega' = \{0, 1, 2\}.$$

The probabilities are not uniform. Since $P(R) = 3/5$ and $P(B) = 2/5$,

$$P(0) = \left(\frac{2}{5}\right)^2 = \frac{4}{25},$$

$$P(1) = 2 \cdot \frac{3}{5} \cdot \frac{2}{5} = \frac{12}{25},$$

$$P(2) = \left(\frac{3}{5}\right)^2 = \frac{9}{25}.$$

The middle value has a factor 2 because exactly one type R object can appear in two orders: first or second.

General formula with replacement

Suppose the urn contains r labelled objects of type R and b labelled objects of type B. If one object is selected uniformly and only type is recorded, then

$$P(R) = \frac{r}{r+b}, \qquad P(B) = \frac{b}{r+b}.$$

If n selections are made with replacement and types are recorded in order, then

$$\Omega = \{R, B\}^n.$$

The probability of exactly k recorded symbols of type R is

$$\binom{n}{k} \left(\frac{r}{r+b}\right)^k \left(\frac{b}{r+b}\right)^{n-k}.$$

The binomial coefficient counts the positions occupied by type R.

General formula without replacement

If n objects are selected without replacement and order is ignored, then the total number of possible labelled selections is

$$\binom{r+b}{n}.$$

To obtain exactly k objects of type R, choose k from the r objects of type R, and choose $n-k$ from the b objects of type B. Thus

$$P(\text{exactly } k \text{ of type } R) = \frac{\binom{r}{k}\binom{b}{n-k}}{\binom{r+b}{n}}.$$

Example

If $r = 7$, $b = 5$, and $n = 4$, then the probability of selecting exactly two objects of type R, without replacement and ignoring order, is

$$\frac{\binom{7}{2}\binom{5}{2}}{\binom{12}{4}}.$$

What urn models teach

Urn models show that probability depends on both the mechanism and the recorded information. A detailed uniform model may become non-uniform after aggregation. Replacement keeps the composition of the urn fixed from selection to selection; without replacement, the composition changes. Recording order gives sequences; ignoring order gives sets or count summaries.

Remark

The phrase "select from an urn" is not yet a probability model. A complete model must say what is recorded, whether replacement occurs, whether order is recorded, and whether labelled objects or only their types are treated as elementary outcomes.

4.6 Waiting for the first success

All examples so far have used finite sample spaces. We now return to the coin model, but change the experiment. Instead of deciding in advance how many tosses will be made, we toss until a certain event happens. This produces a sample space that is infinite but still discrete.

This example is important because it shows that a sample space does not have to be finite. It also shows why a probability model must describe not only the possible outcomes, but also the rule that generates them.

The experiment

Suppose a coin is tossed repeatedly until the first head appears. We stop as soon as the first head occurs. Assume

$$P(H) = p, \qquad P(T) = 1 - p, \qquad 0 < p \leq 1.$$

Here heads will be interpreted as success and tails as failure.

The physical experiment is not a fixed number of tosses. It may stop after one toss, or after two tosses, or after three tosses, and so on. There is no largest possible stopping time.

Recording the whole sequence

One natural recording rule is to record the whole sequence observed up to and including the first head. Then the possible elementary outcomes are

$$H, \quad TH, \quad TTH, \quad TTTH, \quad \ldots$$

The sample space is

$$\Omega = \{H, TH, TTH, TTTH, \ldots\}.$$

Equivalently,

$$\Omega = \{T^{k-1}H : k = 1, 2, 3, \ldots\},$$

where $T^{k-1}H$ means $k-1$ tails followed by one head.

For example:

$$k = 1 \quad \Rightarrow \quad H,$$
$$k = 2 \quad \Rightarrow \quad TH,$$
$$k = 5 \quad \Rightarrow \quad TTTTH.$$

This sample space is countably infinite. Its elements can be listed in a sequence, but the list never ends.

Recording the stopping time

Another natural recording rule is to record only the number of the toss on which the first head appears. Let

$$N = \text{the number of tosses needed to obtain the first head.}$$

Then the sample space is

$$\Omega' = \{1, 2, 3, \ldots\}.$$

The outcome $N = 1$ corresponds to the sequence H. The outcome $N = 2$ corresponds to TH. The outcome $N = k$ corresponds to

$$\underbrace{TT \cdots T}_{k-1 \text{ tails}} H.$$

The two descriptions are equivalent for this experiment. Recording the full stopped sequence and recording the stopping time contain the same information, because once the stopping time is known, the sequence must consist of failures before the last toss and a success at the last toss.

Remark

The sample space $\{1, 2, 3, \ldots\}$ is not the set of all possible numbers in the world. It is the set of possible stopping times for this particular experiment.

Deriving the probability formula

We now derive the probability that the first head appears on toss k. The condition

$$N = k$$

means that the first $k-1$ tosses are tails and the k-th toss is heads. Thus the corresponding sequence is

$$\underbrace{TT\cdots T}_{k-1 \text{ times}}H.$$

The probability of one tail is $1-p$, and the probability of one head is p. Therefore

$$P(N = k) = \underbrace{(1-p)(1-p)\cdots(1-p)}_{k-1 \text{ factors}}\,p.$$

Hence

$$P(N = k) = (1-p)^{k-1}p, \qquad k = 1, 2, 3, \ldots.$$

Theorem

If independent tosses are repeated until the first success, and each toss has success probability p, then the probability that the first success occurs on trial k is

$$P(N = k) = (1-p)^{k-1}p, \qquad k = 1, 2, 3, \ldots.$$

This is called the geometric model. In this chapter we do not need the name as much as the construction. The formula appears because the outcome $N = k$ requires exactly $k-1$ failures followed by one success.

Examples

Example

Suppose the coin is fair, so $p = 1/2$. Then

$$P(N = 1) = \frac{1}{2},$$

$$P(N = 2) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4},$$

$$P(N = 3) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8},$$

and in general

$$P(N = k) = \left(\frac{1}{2}\right)^k.$$

For a fair coin, the probability halves each time we delay the first head by one more toss.

Example

Suppose $P(H) = 0.3$. Then $P(T) = 0.7$, and

$$P(N = 4) = 0.7^3 \cdot 0.3.$$

This is the probability of the sequence

$$TTTH.$$

The first three tosses must fail, and the fourth toss must succeed.

Checking that the probabilities add to one

A probability assignment on a countable sample space should assign total probability one to the whole space. Here the elementary probabilities are

$$p, \quad (1-p)p, \quad (1-p)^2 p, \quad (1-p)^3 p, \quad \dots$$

Their sum is

$$p + (1-p)p + (1-p)^2 p + (1-p)^3 p + \cdots.$$

Factor out p:

$$p\left[1 + (1-p) + (1-p)^2 + (1-p)^3 + \cdots\right].$$

The expression in brackets is a geometric series with ratio $1-p$. Since $0 < p \le 1$, we have $0 \le 1-p < 1$, and

$$1 + (1-p) + (1-p)^2 + \cdots = \frac{1}{1-(1-p)} = \frac{1}{p}.$$

Therefore

$$p \cdot \frac{1}{p} = 1.$$

Thus the probabilities of all possible stopping times add to one.

Remark

This calculation also explains why we require $p > 0$. If $p = 0$, success never occurs, and the experiment "wait until the first success" does not produce a finite stopping time.

Events involving the stopping time

Once the stopping time model is built, events are subsets of

$$\{1, 2, 3, \dots\}.$$

For example, the event that the first success occurs no later than the third trial is

$$\{N \le 3\} = \{1, 2, 3\}.$$

Its probability is

$$P(N \le 3) = p + (1-p)p + (1-p)^2 p.$$

There is also a faster way to compute it. The complement of $\{N \le 3\}$ is

$$\{N > 3\},$$

which means that the first three trials are all failures. Hence

$$P(N > 3) = (1-p)^3.$$

Therefore

$$P(N \leq 3) = 1 - (1-p)^3.$$

More generally,

$$P(N \leq m) = 1 - (1-p)^m.$$

This is the probability that at least one success occurs in the first m trials.

Similarly,

$$P(N > m) = (1-p)^m,$$

because the first m trials must all be failures.

Waiting for the r-th success

A natural extension is to wait not for the first success, but for the r-th success. Let

$$N_r = \text{the trial on which the } r\text{-th success occurs.}$$

For $N_r = n$ to happen, the n-th trial must be a success, and among the first $n-1$ trials there must be exactly $r-1$ successes.

The first $n-1$ trials may contain those $r-1$ successes in

$$\binom{n-1}{r-1}$$

ways. Each full sequence with r successes and $n-r$ failures has probability

$$p^r(1-p)^{n-r}.$$

Therefore

$$P(N_r = n) = \binom{n-1}{r-1} p^r (1-p)^{n-r}, \qquad n = r, r+1, r+2, \ldots.$$

Example

Suppose we wait for the second head. The event $N_2 = 5$ means that the fifth toss is heads and exactly one of the first four tosses is heads. The position of that earlier head can be chosen in

$$\binom{4}{1} = 4$$

ways. Hence

$$P(N_2 = 5) = \binom{4}{1} p^2 (1-p)^3.$$

This extension is included to show that the first-success model is not an isolated trick. It is the first member of a family of waiting-time models. The logic is always the same: describe what must happen before the stopping trial, describe what must happen at the stopping trial, count the possible positions of earlier successes, and multiply the appropriate probabilities.

What waiting-time models teach

The waiting-time model teaches several new lessons.

First, a sample space may be infinite. The set $\{1, 2, 3, \ldots\}$ is a valid discrete sample space when it records a stopping time.

Second, an infinite sample space still requires probabilities that add to one. In the first-success model, this is guaranteed by the geometric series.

Third, the elementary outcome may be represented in different but equivalent ways. We may record the stopped sequence $T^{k-1}H$, or we may record the number k. The choice depends on which description is more useful.

Fourth, a formula such as

$$P(N = k) = (1 - p)^{k-1}p$$

should be read as a description of the required pattern: failures before the stopping trial and success at the stopping trial.

Remark

Waiting-time models are a bridge between elementary finite probability and later probability distributions. They show that probability spaces can be built from a process that stops at a random time, and that even an infinite model can be handled by careful construction from elementary outcomes.

4.7 A first continuous model: uniform distribution on an interval

All models constructed so far in this chapter were discrete. Some of them were finite, such as dice, cards, and urns. One of them was countably infinite: the waiting time until the first success. These examples are essential, but they do not yet represent the full scope of probability. Many random quantities are naturally modelled as points on a continuum.

The purpose of this section is to introduce one continuous model carefully: the uniform distribution on an interval. We shall not develop the full measure theory behind continuous probability. That would require a separate course. However, we must already see the basic change in viewpoint:

in a continuous model, individual points are not the right objects to count.

Instead of counting elementary outcomes, we measure sizes of sets. For an interval on the real line, the relevant size is length.

Choosing a point from an interval

Suppose an experiment produces a real number between 0 and 1. For example, imagine an idealized pointer that lands somewhere on a line segment of unit length. The sample space is

$$\Omega = [0, 1].$$

An elementary outcome is a single real number

$$x \in [0, 1].$$

This sample space is not finite and not countably infinite. It is uncountable. There is no list

$$x_1, x_2, x_3, \ldots$$

that contains all real numbers in $[0, 1]$. This already makes the model qualitatively different from coin tosses, dice rolls, card hands, urn selections, and waiting times.

Remark

The phrase "choose a point uniformly from $[0,1]$" cannot mean that every point receives the same positive probability. There are too many points. The probability is spread continuously over the interval.

Why points have probability zero

In a finite uniform model, each elementary outcome has probability

$$\frac{1}{|\Omega|}.$$

This idea cannot be copied directly to $[0,1]$, because $[0,1]$ has uncountably many points. If each point had the same positive probability, then the total probability would be far larger than 1. If each point had probability 0, then adding point probabilities one by one cannot recover the probability of an interval.

In the continuous uniform model, every individual point has probability zero:

$$P(\{x\}) = 0$$

for each $x \in [0,1]$. This does not mean that the experiment cannot produce a point. It means that probability is not assigned by counting individual points. The probability of landing in a region is determined by the length of that region.

For example, the event

$$A = [0.2, 0.5]$$

has length 0.3, so in the uniform model on $[0,1]$ we assign

$$P(A) = 0.3.$$

The event contains infinitely many points, but its probability is controlled by how much of the interval it occupies.

Intervals as events

The simplest events in a continuous interval model are intervals. In the uniform model on $[0,1]$, the probability of an interval is its length. Thus

$$P([c,d]) = d - c$$

whenever

$$0 \le c \le d \le 1.$$

The same value is assigned to

$$(c,d), \qquad [c,d), \qquad (c,d].$$

The endpoints do not change the probability, because individual points have probability zero.

Example

Let X be uniformly distributed on $[0,1]$. Then

$$P(0.25 \le X \le 0.75) = 0.75 - 0.25 = 0.5.$$

Also,

$$P(0.25 < X < 0.75) = 0.5.$$

The difference between the closed and open interval consists only of endpoints, and endpoints have probability zero in this continuous model.

This is a major difference from discrete probability. In a discrete model, removing one elementary outcome may change the probability. In a continuous uniform model, removing finitely many points from an interval does not change its probability.

Uniform distribution on $[a, b]$

The same idea works on any finite interval

$$[a, b], \qquad a < b.$$

The sample space is

$$\Omega = [a, b].$$

The total length of the sample space is

$$b - a.$$

If the model is uniform, then probability is proportional to length. For an interval $[c, d] \subseteq [a, b]$, we define

$$P([c, d]) = \frac{d - c}{b - a}.$$

The denominator is the length of the whole sample space. The numerator is the length of the event. Thus the formula is the continuous analogue of the uniform finite formula

$$P(A) = \frac{|A|}{|\Omega|}.$$

The symbol $|A|$ counted outcomes in a finite space. In the interval model, length replaces counting.

Definition

A random point X has the **uniform distribution on** $[a, b]$ if probabilities of intervals are proportional to their lengths:

$$P(c \leq X \leq d) = \frac{d - c}{b - a}$$

for every $[c, d] \subseteq [a, b]$.

Examples

Let X be uniformly distributed on $[2, 8]$. The whole interval has length

$$8 - 2 = 6.$$

The probability that X lies between 3 and 5 is

$$P(3 \leq X \leq 5) = \frac{5 - 3}{8 - 2} = \frac{2}{6} = \frac{1}{3}.$$

The probability that X is at least 6 is

$$P(X \geq 6) = P(6 \leq X \leq 8) = \frac{8 - 6}{8 - 2} = \frac{1}{3}.$$

The probability that X is exactly 4 is

$$P(X = 4) = 0.$$

This last statement is not a contradiction. The experiment produces some exact value, but no single prescribed value occupies positive length.

Unions of intervals

Events need not be single intervals. Suppose X is uniform on $[0, 1]$, and consider the event

$$A = [0, 0.2] \cup [0.7, 1].$$

The two intervals are disjoint. Their lengths are 0.2 and 0.3. Hence

$$P(A) = 0.2 + 0.3 = 0.5.$$

This is the same additivity principle used in discrete probability: disjoint parts have probabilities that add. The difference is that the probabilities are computed from lengths rather than from counts of elementary outcomes.

What are the events in a continuous model?

For a finite or countable sample space, it is often harmless at this level to think of every subset of Ω as an event. For continuous sample spaces, this becomes too optimistic. The interval $[0, 1]$ has extremely complicated subsets, and not every subset can be assigned a length in a way that is compatible with the usual rules of probability.

Therefore, a continuous probability model is not just

$$\Omega = [0, 1].$$

It also requires a specified collection of admissible events. For the interval model, the standard choice is the collection of Borel sets, denoted informally by

$$\mathcal{B}([0, 1]).$$

These are the sets obtained from intervals by the operations that probability needs: complements, countable unions, and countable intersections.

Definition

In a continuous model, the event space is usually not taken to be the set of all subsets of Ω. Instead, one specifies a suitable collection of events, usually a **sigma-algebra**. On the real line, the standard event collection is the Borel sigma-algebra generated by open intervals.

For this course, the technical construction is less important than the modelling message. Intervals, finite unions of intervals, countable unions of intervals, and sets obtained from them by complements are legitimate events. These are more than enough for the elementary continuous examples we need now.

Remark

The word "Borel" is a signal that continuous probability requires more care than discrete probability. We introduce it here so that the model is honest: continuous probability spaces need a sample space, an event collection, and a probability assignment.

The structure of the continuous uniform model

The continuous uniform model on $[a, b]$ has three pieces:

$$\Omega = [a, b],$$

$$\mathcal{F} = \mathcal{B}([a, b]),$$

and

$$P(A) = \frac{\text{length of } A}{b - a}$$

for events A whose length is defined in the standard sense.

At this stage, we mainly use this rule for intervals and simple unions of intervals. The notation $\mathcal{F}$ reminds us that events form a selected collection of subsets of Ω, not necessarily all subsets.

What this continuous model teaches

The uniform interval model teaches several lessons that are invisible in purely discrete examples.

First, a sample space may be uncountable. The interval $[0, 1]$ contains more points than any sequence can list.

Second, probability need not be built by assigning positive mass to individual elementary outcomes. In the uniform continuous model,

$$P(\{x\}) = 0$$

for every point x, while intervals can have positive probability.

Third, the probability of an event is measured by geometric size, not by the number of points. For the uniform distribution on $[a, b]$, interval probabilities are relative lengths.

Fourth, the event collection matters. In continuous probability, we must be more careful about which subsets of the sample space are allowed as events. The standard answer on the real line is to use Borel sets.

Theorem

The continuous uniform model replaces counting by length:

$$\text{finite uniform model: } P(A) = \frac{|A|}{|\Omega|},$$

$$\text{continuous uniform interval model: } P(A) = \frac{\text{length of } A}{\text{length of } \Omega}.$$

The idea is analogous, but the mathematics is different because the sample space is uncountable.

4.8 Checklist for building a probability model

Probability calculations should begin with a clearly specified model. The sample space may be finite, countably infinite, or uncountable, but the same modelling questions must be answered in every case.

Step 1: What is the experiment?

Describe the random procedure. State how many stages it has, whether it stops at a fixed or random time, and which assumptions govern the mechanism.

Step 2: What is recorded?

The recording rule determines the mathematical form of an outcome. Ask whether order, labels, complete sequences, numerical summaries, or stopping times are retained.

Remark

If the recording rule is not clear, then one elementary outcome is not clear and the sample space cannot be constructed correctly.

Step 3: What is one elementary outcome?

A complete elementary outcome may be a symbol, number, vector, sequence, set, count, stopping time, or point in an interval. It must contain exactly the information kept by the model.

Step 4: What is the sample space Ω?

The sample space is the set of all elementary outcomes. Typical structures are

$$A^n, \qquad \mathbb{N}, \qquad [a, b].$$

The first may be finite, the second is countably infinite, and the interval is uncountable. An uncountable sample space cannot be listed as a sequence of outcomes.

Step 5: What is the event space $\mathcal{F}$?

An event must belong to a specified collection $\mathcal{F}$ of admissible subsets of Ω. In elementary finite and countable models, all subsets are often used. In a continuous model, the standard event collection is a sigma-algebra, usually the Borel sigma-algebra generated by intervals.

Translate every verbal condition into an event $A \in \mathcal{F}$ before calculating.

Step 6: What probability assignment is justified?

For a finite uniform model,

$$P(A) = \frac{|A|}{|\Omega|}.$$

For a non-uniform discrete model,

$$P(A) = \sum_{\omega \in A} P(\{\omega\}).$$

For a uniform model on $[a, b]$, individual points have probability zero and interval probabilities are determined by relative length:

$$P(c \leq X \leq d) = \frac{d - c}{b - a}.$$

More general continuous models accumulate probability through a density or another probability measure.

Step 7: Use a method that matches the model

Use products for justified multistage structures, combinations for unordered selection, sums for discrete probability weights, and lengths or integrals for continuous regions. Independence must come from the probability model, not from notation alone.

Step 8: Interpret the answer

Explain which outcomes or regions form the event, which assumptions determine their probabilities, and what the final number means in the original situation.

Compact modelling checklist

Before calculating, ask:

1. What is the experiment?
2. What exactly is recorded?
3. What is one elementary outcome?
4. What is the sample space Ω?
5. Is Ω finite, countably infinite, or uncountable?
6. What event space $\mathcal{F}$ is being used?
7. What event $A \in \mathcal{F}$ is being studied?
8. Are elementary outcomes equally likely?
9. Is probability computed by counting, summing weights, measuring length, or integrating a density?
10. What does the result mean in the original context?

Theorem

The basic rule is

$$\text{construct } (\Omega, \mathcal{F}, P) \text{ first, calculate second.}$$

A probability calculation is meaningful only inside a clearly specified sample space, event collection, and probability assignment.

Chapter 5

Conditional Probability, Independence, and Bayes' Theorem

This chapter explores the algebra of events and the concept of conditional probability, which are fundamental to understanding how probabilities are updated with new information.

5.1 Information changes the space of reasoning

The previous chapter taught us how to construct probability models from experiments. We learned to identify elementary outcomes, build the sample space Ω, describe events as subsets of Ω, and assign probabilities only after the model has been made explicit.

In this chapter we add a new idea: sometimes probability is not computed from the original position of ignorance. Sometimes new information is given. When that happens, the question is no longer asked inside the whole sample space. The information restricts the set of possibilities that remain compatible with what we know.

The central message is:

new information changes the space in which we reason.

This idea is the conceptual beginning of conditional probability. We shall not begin by memorizing a formula. We shall first examine what information does to a probability model.

Information removes impossible cases

Suppose a die is rolled once. Before any information is given, the natural sample space is

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

If the die is fair, all six elementary outcomes are equally likely.

Now imagine that someone tells us:

the result is even.

This information changes what is still possible. The outcomes

$$1, 3, 5$$

are no longer compatible with the information. The only outcomes that remain possible are

$$2, 4, 6.$$

Thus we no longer reason inside the whole set Ω. We reason inside the smaller event

$$E = \{2, 4, 6\}.$$

If we now ask

what is the probability that the result is greater than 3?

we must not count favourable outcomes inside the original six outcomes. The information has already told us that the result is even. Therefore the relevant possibilities are only

$$2, 4, 6.$$

Among these, the outcomes greater than 3 are

$$4, 6.$$

So, after receiving the information that the result is even, the probability of "greater than 3" is obtained by comparing two favourable outcomes with the three outcomes still possible.

Remark

The old sample space has not disappeared from the model. It still explains what the experiment was. But for the new question, after the information is known, the active space of reasoning is the event compatible with that information.

A question may change its answer after information is given

Before any information is given, the event

$$A = \{4, 5, 6\}$$

that the die result is greater than 3 has three favourable outcomes out of six. In the fair die model, this gives probability $1/2$.

After the information

$$E = \{2, 4, 6\}$$

is given, the same verbal question is evaluated in a smaller space. The relevant outcomes are no longer

$$1, 2, 3, 4, 5, 6,$$

but only

$$2, 4, 6.$$

Inside that restricted space, the event "greater than 3" corresponds to

$$A \cap E = \{4, 6\}.$$

Thus the answer changes. The event A itself did not change as a subset of Ω. What changed is the set of outcomes that are still considered possible.

This is the basic phenomenon of conditioning: information changes the reference space.

Information as restriction to an event

In probability, information usually has the form

$$\text{we know that } B \text{ occurred.}$$

The event B then becomes the new space of reasoning. Outcomes outside B are no longer possible under the information.

If we are interested in another event A, we do not ask how much of Ω belongs to A. We ask how much of the still possible region B also belongs to A. The relevant part is therefore

$$A \cap B.$$

The geometry of the situation is:

$$\begin{aligned} \Omega &\quad \text{is the original universe,} \\ B &\quad \text{is the information now known,} \\ A \cap B &\quad \text{is the part of the information region where } A \text{ also holds.} \end{aligned}$$

Definition

To **condition on information** means to restrict attention to the outcomes compatible with that information and then measure the event of interest inside that restricted region.

This is not yet a computational formula. It is the idea that the formula must express.

Two dice: information about the sum

Consider two fair dice, with order recorded. The original sample space is

$$\Omega = \{(i, j) : i, j \in \{1, 2, 3, 4, 5, 6\}\}.$$

It contains 36 ordered pairs.

Let

$$B = \{\text{the sum is } 7\}.$$

Explicitly,

$$B = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}.$$

Suppose this information is given: the sum is 7. Then the six ordered pairs above are the only outcomes still compatible with what we know.

Now ask:

what is the probability that the first die shows 5?

Let

$$A = \{\text{the first die shows } 5\}.$$

Inside the whole sample space, A contains six outcomes:

$$(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6).$$

But most of these are irrelevant once we know that the sum is 7. Inside the restricted space B, the only outcome with first coordinate 5 is

$$(5, 2).$$

Thus, after the information is known, there is one favourable outcome among the six outcomes still possible.

This example shows why one must be careful with the phrase "the probability of A". Probability relative to what information? Before knowing the sum, the question is evaluated in Ω. After knowing the sum, the question is evaluated in B.

The event is not replaced by the information

A common misunderstanding is to think that once B is known, the event A is replaced by B. That is not correct. The event of interest remains A. The information is B. The favourable outcomes are those where both things are true:

$$A \cap B.$$

In the two-dice example:

- B says that the sum is 7;
- A says that the first die is 5;
- $A \cap B$ says that the first die is 5 and the sum is 7.

The information gives the new reference space. The intersection gives the part of that reference space where the target event occurs.

Remark

Conditioning always involves two roles: the event being asked about and the information being conditioned on. Confusing these roles is one of the most common sources of wrong answers.

Cards: the same card viewed after information

Draw one card from a standard deck. Suppose the exact card is recorded, so the sample space has 52 elementary outcomes.

Let

$$A = \{\text{the card is an ace}\}.$$

Before any additional information is given, there are 4 aces among 52 cards.

Now suppose we are told:

$$B = \{\text{the card is a heart}\}.$$

The information reduces the relevant possibilities to the 13 hearts. Inside this restricted set, only one card is an ace: the ace of hearts. Therefore, once we know that the card is a heart, the event "the card is an ace" is evaluated inside the set of hearts.

Now change the information. Suppose instead we are told:

$$C = \{\text{the card is a face card}\}.$$

The relevant possibilities are now the face cards. Inside that restricted set, there are no aces, because aces are not face cards in the usual convention. Thus the same event A, "the card is an ace", behaves differently under different information.

The event A did not change. The information changed the reference space.

Information may or may not change the probability

New information restricts the space of reasoning, but it does not always change the numerical probability of the event of interest.

For example, draw one card from a standard deck. Let

$$A = \{\text{the card is an ace}\},$$

and let

$$B = \{\text{the card is a heart}\}.$$

Knowing that the card is a heart leaves 13 possible cards, one of which is an ace. The probability of being an ace inside the hearts is therefore the same as the original proportion of aces in the whole deck:

one ace among thirteen hearts, just as four aces among fifty-two cards.

In this case the information "heart" does not change the probability of "ace".

But if the information is

$$C = \{\text{the card is a face card}\},$$

then the probability of being an ace becomes zero inside the restricted space. This information does change the probability.

This observation prepares the idea of independence. Two events are independent when learning that one occurred does not change the probability of the other. But before defining independence formally, we must first understand what it means to update the space of reasoning.

Renormalization: the whole information region becomes certain

When the information B is given, the event B itself becomes certain in the new situation. After being told that the die result is even, the statement "the die result is even" has probability 1 relative to the information. After being told that the card is a heart, the statement "the card is a heart" has probability 1 relative to the information.

This means that probabilities inside the restricted region must be rescaled so that the whole information region has total probability 1. In a uniform finite model, this rescaling is easy to see. If B contains several equally likely outcomes, then after we learn B, those outcomes form the new complete list of possibilities.

Example

A fair die is rolled and we learn that the result is even. The restricted space is

$$B = \{2, 4, 6\}.$$

Inside this new space, the three outcomes $2, 4, 6$ are treated as the complete possibilities. Therefore the event B itself has probability 1 relative to this information, and each of the three remaining outcomes has probability $1/3$ in the restricted uniform model.

In non-uniform models, the same idea holds, but the remaining probabilities must be rescaled proportionally. The original weights outside B are discarded because they correspond to outcomes no longer compatible with the information. The weights inside B are then adjusted so that their total becomes 1.

This rescaling principle is the reason the formal definition of conditional probability has the structure it has. The numerator must measure the part of the information region where the target event occurs. The denominator must measure the whole information region, because that whole region becomes the new unit of probability.

Why the information must have positive probability

Conditioning on information only makes sense if the information can occur in the model. If $P(B) = 0$, then the instruction "reason inside B" cannot be implemented by rescaling probabilities inside B, because there is no positive probability mass to rescale.

In finite elementary examples this is easy to understand. Suppose a standard die is rolled and someone says:

the result is 8.

This statement corresponds to the empty event in the die model $\Omega = \{1, 2, 3, 4, 5, 6\}$. There are no outcomes compatible with it. It cannot become a new space of reasoning.

More generally, conditional probability is defined only when the conditioning event has positive probability. This condition is not a technical decoration. It says that the information must leave a meaningful region in which probability can be redistributed.

The conceptual picture

The logic of conditioning can now be summarized as follows.

1. We begin with a probability model on Ω.
2. We receive information that an event B occurred.
3. Outcomes outside B are no longer compatible with the information.
4. The event B becomes the new reference space.

5. To study another event A, we look at the part of B where A is also true, namely $A \cap B$.
6. Probabilities inside B are rescaled so that B has total probability 1 in the new situation.

Theorem

The guiding principle of conditional probability is:

probability after information is probability inside the event described by that information.

The formal definition in the next section will turn this principle into a precise rule.

The important point is that the formula will not be arbitrary. It must express the geometry we have already seen: restrict to the information event, keep only the part where the target event also occurs, and renormalize the information region so that it becomes the new whole space.

5.2 Conditional probability

The previous section explained the idea before the formula: when information is received, the relevant space of reasoning is restricted. If we know that an event B occurred, then outcomes outside B are no longer compatible with what we know. If we then ask about another event A, the favourable outcomes are those in

$$A \cap B.$$

We now turn this idea into a precise definition. The definition should not be read as an arbitrary rule. It is the mathematical expression of two decisions: first restrict to the information event, then rescale so that the information event becomes the new whole space.

The finite uniform case first

Let Ω be a finite sample space in which all elementary outcomes are equally likely. Suppose we are told that an event B occurred, and assume B is not empty. After this information, the possible outcomes are the elements of B.

If we ask for the probability of A after knowing B, then only the outcomes in B matter. Among these, the favourable outcomes are precisely the elements of

$$A \cap B.$$

Thus, in the finite uniform case, the natural value is

$$\frac{|A \cap B|}{|B|}.$$

This expression is not guessed. It is the ordinary uniform probability formula, but applied in the restricted space B instead of the original space Ω.

Example

A fair die is rolled. Let

$$A = \{4, 5, 6\}$$

be the event that the result is greater than 3, and let

$$B = \{2, 4, 6\}$$

be the event that the result is even. If we know B, then the relevant space is $\{2, 4, 6\}$.

Inside this space, the event A occurs for 4 and 6. Therefore the probability of A after knowing B is

$$\frac{2}{3}.$$

This is the ratio $|A \cap B|/|B|$, because

$$A \cap B = \{4, 6\}.$$

From counting to probability mass

The expression

$$\frac{|A \cap B|}{|B|}$$

works only when the restricted elementary outcomes are equally likely. In a general probability model, outcomes may have different probabilities. Then counting outcomes is not enough. We must measure probability mass.

The information event B has probability $P(B)$. This is the total mass of all outcomes compatible with the information. The event $A \cap B$ has probability $P(A \cap B)$. This is the part of that mass for which the event A also occurs.

After learning B, the whole region B becomes the new reference space. It must therefore have total probability 1 in the updated situation. The part of B where A occurs should occupy the same relative share that it had inside B before rescaling. That relative share is

$$\frac{P(A \cap B)}{P(B)}.$$

This gives the definition.

Definition

Let A and B be events in a probability space, and assume

$$P(B) > 0.$$

The **conditional probability of A given B** is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

The condition $P(B) > 0$ is essential. If the information event has probability zero, then it cannot be made into a new probability-one reference region by ordinary rescaling.

Why the denominator is $P(B)$

The denominator is not a technical detail. It represents the whole amount of probability still compatible with the information. Once we know B, the event B must become certain in the new calculation:

$$P(B \mid B) = 1.$$

The definition gives exactly this:

$$P(B \mid B) = \frac{P(B \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1.$$

Similarly, an event impossible under the information should have conditional probability zero. If $A \cap B = \varnothing$, then

$$P(A \mid B) = \frac{P(\varnothing)}{P(B)} = 0.$$

This matches the idea that, once B is known, an event disjoint from B cannot occur.

Conditional probability as a new probability assignment

For a fixed event B with $P(B) > 0$, the rule

$$A \mapsto P(A \mid B)$$

defines a new probability assignment on the original events, but interpreted under the information B.

It has the usual properties. First,

$$P(A \mid B) \geq 0,$$

because probabilities are non-negative. Second,

$$P(\Omega \mid B) = \frac{P(\Omega \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1.$$

Third, if A_1 and A_2 are disjoint, then

$$P((A_1 \cup A_2) \mid B) = \frac{P((A_1 \cup A_2) \cap B)}{P(B)}.$$

Since

$$(A_1 \cup A_2) \cap B = (A_1 \cap B) \cup (A_2 \cap B),$$

and the two pieces on the right are disjoint, additivity gives

$$P((A_1 \cup A_2) \mid B) = \frac{P(A_1 \cap B) + P(A_2 \cap B)}{P(B)}.$$

Therefore

$$P((A_1 \cup A_2) \mid B) = P(A_1 \mid B) + P(A_2 \mid B).$$

This confirms that conditional probability is not a separate kind of arithmetic. It is ordinary probability after changing the reference space.

Multiplication rule

The definition can be rewritten as

$$P(A \cap B) = P(A \mid B)P(B).$$

This is called the multiplication rule. It says: to get the probability that both A and B occur, first measure the probability of the information region B, and then multiply by the probability that A occurs inside that region.

The same intersection can also be read in the opposite order:

$$P(A \cap B) = P(B \mid A)P(A),$$

provided $P(A) > 0$.

Thus

$$P(A \mid B)P(B) = P(B \mid A)P(A).$$

This identity will later become the basis of Bayes' theorem.

Remark

The multiplication rule is not an additional assumption. It is the definition of conditional probability rearranged. Its usefulness comes from the fact that many multi-stage situations are naturally described by first entering one event and then asking for another event inside it.

Example: drawing two cards without replacement

Draw two cards from a standard deck without replacement, and record the ordered sequence. Let

$$A = \{\text{the first card is an ace}\},$$

and

$$B = \{\text{the second card is an ace}\}.$$

We want to understand the probability that both cards are aces.

First,

$$P(A) = \frac{4}{52}.$$

If A has occurred, then one ace has already been removed. There are now 51 cards left, of which 3 are aces. Therefore

$$P(B \mid A) = \frac{3}{51}.$$

By the multiplication rule,

$$P(A \cap B) = P(A)P(B \mid A) = \frac{4}{52} \cdot \frac{3}{51}.$$

This calculation is not merely a formula. It follows the physical structure of the experiment: first card, then second card after the first card has changed the deck.

Example: medical-style testing without Bayes yet

Suppose a population is divided into two groups: people who have a certain condition and people who do not. Let

$$C = \{\text{the person has the condition}\},$$

and let

$$T = \{\text{the test result is positive}\}.$$

The conditional probability

$$P(T \mid C)$$

means: among people with the condition, what proportion test positive?

The conditional probability

$$P(C \mid T)$$

means: among people with a positive test result, what proportion have the condition?

These are different questions. They use different reference spaces. The first uses C as the information event. The second uses T as the information event. There is no reason for the two numbers to be equal.

This distinction is one of the main reasons conditional probability must be introduced carefully. The vertical bar in $P(A \mid B)$ is not punctuation. It separates the event being measured from the information that defines the new space of reasoning.

Conditional probabilities in tables

Conditional probability can often be read from a table. Suppose a group of students is classified by whether they passed an exam and whether they attended review sessions:

	passed	did not pass	total
attended	36	9	45
did not attend	24	31	55
total	60	40	100

Let

$$A = \{\text{student passed}\},$$

and

$$B = \{\text{student attended}\}.$$

Then $P(A \mid B)$ is not computed from all 100 students. The information B restricts attention to the 45 students who attended. Among them, 36 passed. Thus

$$P(A \mid B) = \frac{36}{45}.$$

By contrast,

$$P(B \mid A) = \frac{36}{60},$$

because now the information is that the student passed, and among the 60 students who passed, 36 attended. Again, the two conditional probabilities answer different questions.

What this definition teaches

The definition

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

should be remembered through its meaning:

- B is the information that becomes the new reference space;
- $A \cap B$ is the part of that reference space where the event of interest also occurs;
- division by $P(B)$ rescales the information region so that it has total probability 1.

Theorem

Conditional probability is ordinary probability after the space of reasoning has been restricted by information. The formula records this restriction and the necessary rescaling.

5.3 Independence and dependence

Conditional probability lets us ask how the probability of one event changes after another event is known. The next natural question is therefore:

when does information change the probability, and when does it not?

This is the conceptual origin of independence. Independence is not first of all a multiplication formula. It is a statement that learning one event occurred does not change the probability of another event.

Information that changes probability

Draw one card from a standard deck. Let

$$A = \{\text{the card is an ace}\}.$$

Before any information is given,

$$P(A) = \frac{4}{52} = \frac{1}{13}.$$

Now suppose we are told that the card is a face card. Let

$$B = \{\text{the card is a face card}\}.$$

Inside the face cards there are no aces, using the standard convention that face cards are jacks, queens, and kings. Therefore

$$P(A \mid B) = 0.$$

The information B changes the probability of A. It changes it from $1/13$ to 0. In this case, the events are dependent.

Information that does not change probability

Now let

$$C = \{\text{the card is a heart}\}.$$

If we are told that the card is a heart, the relevant space consists of the 13 hearts. Among them, exactly one is an ace. Hence

$$P(A \mid C) = \frac{1}{13}.$$

But this is the same as the original probability $P(A)$. Knowing that the card is a heart does not change the probability that it is an ace.

This is the basic meaning of independence:

> the information occurred, but it did not change the probability of the event of interest.

Definition

Events A and B, with $P(B) > 0$, are **independent in the conditional sense** if

$$P(A \mid B) = P(A).$$

They are **dependent** if knowing B changes the probability of A, that is, if

$$P(A \mid B) \neq P(A).$$

This definition expresses the intuitive meaning directly. It says that B brings no probabilistic information about A.

The multiplication form of independence

The conditional definition is conceptually clear, but it requires $P(B) > 0$. Using the definition of conditional probability, the equality

$$P(A \mid B) = P(A)$$

becomes

$$\frac{P(A \cap B)}{P(B)} = P(A).$$

Multiplying both sides by $P(B)$, we obtain

$$P(A \cap B) = P(A)P(B).$$

This gives the standard symmetric definition.

Definition

Events A and B are **independent** if

$$P(A \cap B) = P(A)P(B).$$

If this equality fails, the events are **dependent**.

This form is symmetric in A and B. It makes clear that if A gives no information about B, then B gives no information about A, provided the relevant conditional probabilities are defined.

Remark

The product formula should not be treated as the first meaning of independence. It is the algebraic form of the idea that conditioning on one event does not change the probability of the other.

Example: two coin tosses

Toss a fair coin twice and record the ordered sequence:

$$\Omega = \{HH, HT, TH, TT\}.$$

Let

$$A = \{\text{the first toss is heads}\} = \{HH, HT\},$$

and

$$B = \{\text{the second toss is heads}\} = \{HH, TH\}.$$

Then

$$P(A) = \frac{1}{2}, \qquad P(B) = \frac{1}{2}.$$

The intersection is

$$A \cap B = \{HH\},$$

so

$$P(A \cap B) = \frac{1}{4}.$$

Since

$$P(A)P(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4},$$

we have

$$P(A \cap B) = P(A)P(B).$$

Thus A and B are independent.

The conditional interpretation says the same thing. If we know the second toss is heads, the remaining possible sequences are

$$HH, TH.$$

Among these, exactly one has first toss heads. Hence

$$P(A \mid B) = \frac{1}{2} = P(A).$$

The information about the second toss does not change the probability of the first toss.

Example: cards and dependence

Draw one card from a deck. Let

$$A = \{\text{the card is an ace}\},$$

and

$$B = \{\text{the card is a face card}\}.$$

Then

$$A \cap B = \varnothing,$$

so

$$P(A \cap B) = 0.$$

But

$$P(A)P(B) = \frac{4}{52} \cdot \frac{12}{52} > 0.$$

Therefore

$$P(A \cap B) \neq P(A)P(B).$$

The events are dependent. Knowing that the card is a face card makes the event "ace" impossible.

Disjointness is not independence

Students often confuse disjoint events with independent events. These ideas are very different.

Events A and B are disjoint if

$$A \cap B = \varnothing.$$

This means they cannot occur together.

Events A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

This means that knowing one occurred does not change the probability of the other.

If A and B are disjoint and both have positive probability, then they cannot be independent. Indeed,

$$P(A \cap B) = 0,$$

but

$$P(A)P(B) > 0.$$

So the equality required for independence fails.

Remark

If two positive-probability events are disjoint, then learning that one occurred makes the other impossible. That is the strongest possible form of dependence, not independence.

Independence is not causation

Dependence means that probabilities change under information. It does not by itself identify a causal mechanism. If

$$P(A \mid B) \neq P(A),$$

then B is informative about A, but this does not automatically mean that B causes A.

For example, suppose A is the event that a randomly selected student passes an exam and B is the event that the student attended review sessions. If $P(A \mid B) > P(A)$, then attendance is associated with a higher pass probability in the model or data. But this observation alone does not prove that attendance caused the result. There may be other factors, such as preparation, motivation, or prior knowledge.

Probability detects changes of likelihood under information. Causation requires additional assumptions and is a separate question.

Independence of repeated trials as a modelling assumption

When we say that repeated coin tosses are independent, we are making a modelling assumption. For two tosses, this assumption says, for example, that knowing the first toss was heads does not change the probability that the second toss is heads.

If $P(H) = p$ for each toss and the tosses are independent, then

$$P(HH) = p^2,$$

$$P(HT) = p(1-p),$$
$$P(TH) = (1-p)p,$$
$$P(TT) = (1-p)^2.$$

Without the independence assumption, we cannot automatically multiply the single-toss probabilities. The second toss might depend on the first in some artificial mechanism. Independence is not guaranteed by notation such as $\{H,T\}^2$; it must be part of the probability assignment.

Example

Suppose a device produces two symbols, each either H or T, and the sample space is $\{HH, HT, TH, TT\}$. If the device always repeats the first symbol, then only HH and TT have positive probability. The sample space still uses two positions, but the two positions are not independent.

More than two events

For several events, independence becomes more subtle. It is not enough that each pair behaves independently. For three events A, B, C, full mutual independence requires the product rule for every relevant intersection:

$$P(A \cap B) = P(A)P(B),$$
$$P(A \cap C) = P(A)P(C),$$
$$P(B \cap C) = P(B)P(C),$$

and also

$$P(A \cap B \cap C) = P(A)P(B)P(C).$$

At this stage, the main point is conceptual: independence means that information from some events does not change probabilities of the others. For many events, this must hold consistently across combinations of events.

What independence teaches

Independence is a statement about information. It says that the occurrence of one event does not alter the probability assigned to another event. The product formula is important because it gives an algebraic test, but the meaning remains conditional:

knowing B does not change the probability of A.

Theorem

For events of positive probability, the following ideas are equivalent:

$$P(A \mid B) = P(A),$$

$$P(B \mid A) = P(B),$$

$$P(A \cap B) = P(A)P(B).$$

Each expresses the same idea: the events do not provide probabilistic information about each other.

5.4 Partitions and the law of total probability

Conditional probability becomes especially useful when a problem can be split into cases. Many probability questions have the following structure: we want the probability of an event A, but the mechanism that produces A depends on which one of several possible background cases occurred.

The mathematical language for such a situation is a partition.

Splitting the sample space into cases

Suppose the sample space Ω is split into events

$$B_1, B_2, \dots, B_n.$$

These events should represent mutually exclusive and exhaustive cases. Mutually exclusive means that no two cases can occur together:

$$B_i \cap B_j = \varnothing \qquad \text{when } i \neq j.$$

Exhaustive means that every possible outcome belongs to one of the cases:

$$B_1 \cup B_2 \cup \cdots \cup B_n = \Omega.$$

Definition

A finite collection of events $B_1, \dots, B_n$ is a **partition** of Ω if the events are pairwise disjoint and their union is Ω. In applications of conditional probability, we usually also require

$$P(B_i) > 0$$

for each case that appears in a conditional probability.

A partition is a disciplined way to say: exactly one of these cases occurs.

Example

If a person is selected from a population, the events

$$B_1 = \{\text{person is in group 1}\}, \quad B_2 = \{\text{person is in group 2}\}, \quad B_3 = \{\text{person is in group 3}\}$$

form a partition if every person belongs to exactly one of the three groups.

How an event is split by a partition

Let A be any event. If $B_1, \ldots, B_n$ is a partition of Ω, then A can be split according to the cases:

$$A = (A \cap B_1) \cup (A \cap B_2) \cup \cdots \cup (A \cap B_n).$$

These pieces are disjoint because the B_i's are disjoint. Thus the probability of A is the sum of the probabilities of the pieces:

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + \cdots + P(A \cap B_n).$$

This identity is just additivity applied after decomposing A into disjoint parts.

Replacing intersections by conditional probabilities

For each case B_i, the multiplication rule gives

$$P(A \cap B_i) = P(A \mid B_i)P(B_i).$$

Substituting this into the previous sum gives

$$P(A) = P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2) + \cdots + P(A \mid B_n)P(B_n).$$

Equivalently,

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i).$$

Theorem

Law of total probability. If $B_1, \ldots, B_n$ is a partition of Ω and $P(B_i) > 0$, then for every event A,

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i).$$

The formula says that the total probability of A is obtained by looking at all possible cases. In each case, we multiply the probability of the case by the probability of A inside that case, and then add the contributions.

Remark

The law of total probability is not a new mysterious rule. It is additivity over a partition, with each piece $P(A \cap B_i)$ rewritten as $P(A \mid B_i)P(B_i)$.

Weighted averages of conditional probabilities

The expression

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i)$$

can be read as a weighted average of the conditional probabilities

$$P(A \mid B_1), \ldots, P(A \mid B_n).$$

The weights are

$$P(B_1), \ldots, P(B_n).$$

They are non-negative and add to one because the B_i's form a partition:

$$P(B_1) + \cdots + P(B_n) = 1.$$

This interpretation is useful. If A is likely in a case that rarely occurs, its contribution to the total probability may still be small. If A is only moderately likely in a case that occurs often, its contribution may be large.

Example: choosing one of two coins

Suppose a box contains two coins. Coin 1 is fair:

$$P(H \mid B_1) = \frac{1}{2}.$$

Coin 2 is biased toward heads:

$$P(H \mid B_2) = \frac{3}{4}.$$

A coin is chosen at random, with

$$P(B_1) = P(B_2) = \frac{1}{2},$$

and then the chosen coin is tossed once. What is the probability of heads?

Let

$$A = \{\text{heads occurs}\}.$$

The events B_1 and B_2 form a partition of the background mechanism: we either chose coin 1 or coin 2. The law of total probability gives

$$P(A) = P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2).$$

Thus

$$P(A) = \frac{1}{2} \cdot \frac{1}{2} + \frac{3}{4} \cdot \frac{1}{2} = \frac{1}{4} + \frac{3}{8} = \frac{5}{8}.$$

The answer is between $1/2$ and $3/4$, because before the toss we do not know which coin was chosen. The total probability is the average of the head probabilities of the two coins, weighted by the probabilities of choosing them.

Example: urn chosen before drawing

Suppose one urn is selected first, and then one object is drawn from that urn. Urn 1 contains 3 red objects and 1 blue object. Urn 2 contains 1 red object and 4 blue objects. The probability of choosing urn 1 is $2/3$, and the probability of choosing urn 2 is $1/3$.

Let

$$A = \{\text{the drawn object is red}\},$$

$$B_1 = \{\text{urn 1 was chosen}\}, \qquad B_2 = \{\text{urn 2 was chosen}\}.$$

Then

$$P(A \mid B_1) = \frac{3}{4}, \qquad P(A \mid B_2) = \frac{1}{5}.$$

The cases B_1 and B_2 form a partition. Therefore

$$P(A) = P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2).$$

Substituting the values,

$$P(A) = \frac{3}{4} \cdot \frac{2}{3} + \frac{1}{5} \cdot \frac{1}{3}.$$

Hence

$$P(A) = \frac{1}{2} + \frac{1}{15} = \frac{17}{30}.$$

The result is not obtained by averaging the contents of the two urns equally, because the urns are not chosen equally. The weights $2/3$ and $1/3$ matter.

Example: data table

Suppose a population consists of two departments. Department 1 contains 40% of the population, and department 2 contains 60%. In department 1, 70% of people use a certain system. In department 2, 30% use it.

Let

$$A = \{\text{the selected person uses the system}\},$$
$$B_1 = \{\text{the selected person is in department 1}\},$$
$$B_2 = \{\text{the selected person is in department 2}\}.$$

Then

$$P(B_1) = 0.4, \qquad P(B_2) = 0.6,$$
$$P(A \mid B_1) = 0.7, \qquad P(A \mid B_2) = 0.3.$$

Therefore

$$P(A) = 0.7 \cdot 0.4 + 0.3 \cdot 0.6 = 0.28 + 0.18 = 0.46.$$

The overall usage rate is 46%. It is a weighted average of the department usage rates, weighted by department sizes.

Choosing the right partition

The law of total probability is useful only after the cases have been chosen well. A good partition should satisfy three conditions.

First, the cases must be mutually exclusive. If two cases can occur together, then adding their contributions may double-count outcomes.

Second, the cases must be exhaustive. If some outcomes are not covered by any case, then the sum will miss part of the probability.

Third, the conditional probabilities inside the cases should be meaningful and usually easier to understand than the original probability.

Example

For a test problem, a natural partition may be

$$B_1 = \{\text{person has the condition}\}, \qquad B_2 = \{\text{person does not have the condition}\}.$$

These two events are disjoint and exhaustive. They also match how test accuracy is usually described: sensitivity gives $P(\text{positive} \mid B_1)$, and false positive rate gives $P(\text{positive} \mid B_2)$.

Common mistake: forgetting the weights

A common mistake is to average conditional probabilities without using the case probabilities. For example, if

$$P(A \mid B_1) = 0.9, \qquad P(A \mid B_2) = 0.1,$$

one might be tempted to say that $P(A) = 0.5$. This is correct only if

$$P(B_1) = P(B_2) = 0.5.$$

If instead

$$P(B_1) = 0.1, \qquad P(B_2) = 0.9,$$

then

$$P(A) = 0.9 \cdot 0.1 + 0.1 \cdot 0.9 = 0.18.$$

The rare high-probability case contributes less than the common low-probability case.

What the law of total probability teaches

The law of total probability teaches how to assemble a probability from cases. The event A is decomposed into disjoint pieces $A \cap B_i$. Each piece is measured by first entering case B_i, then measuring A within that case.

Theorem

To use total probability, ask:

Which mutually exclusive and exhaustive cases can produce the event?

Then add the case contributions:

$$\text{probability of case} \times \text{probability of event inside case.}$$

This prepares the next idea. If total probability lets us compute the probability of an observation from possible causes or cases, Bayes' theorem lets us reverse the question: after the observation is known, how likely is each case?

5.5 Bayes' theorem

The law of total probability lets us compute the probability of an observation by splitting the world into possible cases. Bayes' theorem answers the reverse question. After an observation is known, how should we update the probabilities of the possible cases?

This is not a separate trick. It is conditional probability applied in the reverse direction.

The basic reversal problem

Suppose B is a possible background case and A is an observed event. We may know or be able to model

$$P(A \mid B),$$

the probability of observing A if case B is true. But after observing A, we may want

$$P(B \mid A),$$

the probability that case B is true given that A was observed.

These two probabilities are not the same. They use different reference spaces. The first reasons inside B. The second reasons inside A.

From the multiplication rule,

$$P(A \cap B) = P(A \mid B)P(B).$$

Also,

$$P(A \cap B) = P(B \mid A)P(A).$$

Since both expressions describe the same intersection, we get

$$P(B \mid A)P(A) = P(A \mid B)P(B).$$

If $P(A) > 0$, then

$$P(B \mid A) = \frac{P(A \mid B)P(B)}{P(A)}.$$

Theorem

Bayes' theorem, two-event form. If $P(A) > 0$ and $P(B) > 0$, then

$$P(B \mid A) = \frac{P(A \mid B)P(B)}{P(A)}.$$

This formula says that the updated probability of B after observing A is proportional to two quantities:

- how plausible B was before observing A, namely $P(B)$;
- how well B explains the observation A, namely $P(A \mid B)$.

Why the denominator is often expanded

In many applications, $P(A)$ is not known directly. Instead, the observation A can occur under several possible cases. If $B_1, \ldots, B_n$ form a partition of Ω, then the law of total probability gives

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i).$$

Substituting this into Bayes' theorem gives

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{\sum_{i=1}^{n} P(A \mid B_i)P(B_i)}.$$

Theorem

Bayes' theorem over a partition. If $B_1, \ldots, B_n$ is a partition of Ω, $P(B_i) > 0$, and $P(A) > 0$, then

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{\sum_{i=1}^{n} P(A \mid B_i)P(B_i)}.$$

The denominator is not an arbitrary normalizing term. It is the total probability of the observation A, obtained by adding all the ways in which A can occur through the cases B_i.

Prior, likelihood, posterior

Bayes' theorem is often described using three words.

The number

$$P(B_j)$$

is the **prior probability** of case B_j. It is the probability before observing A.

The number

$$P(A \mid B_j)$$

is the **likelihood** of the observation under case B_j. It measures how compatible the observation is with that case.

The number

$$P(B_j \mid A)$$

is the **posterior probability**. It is the updated probability of the case after observing A.

The logic is:

$$\text{prior} \times \text{likelihood} \quad \longrightarrow \quad \text{posterior after normalization.}$$

The normalization is necessary because the posterior probabilities over all cases must add to one.

Example: choosing an urn after observing a colour

Suppose one of two urns is chosen. Urn 1 is chosen with probability 2/3, and urn 2 is chosen with probability 1/3. Urn 1 contains 3 red objects and 1 blue object. Urn 2 contains 1 red object and 4 blue objects. One object is drawn from the chosen urn, and it is red. Which urn is now more likely to have been chosen?

Let

$$B_1 = \{\text{urn 1 was chosen}\}, \qquad B_2 = \{\text{urn 2 was chosen}\},$$

and

$$A = \{\text{the drawn object is red}\}.$$

We have

$$P(B_1) = \frac{2}{3}, \qquad P(B_2) = \frac{1}{3},$$
$$P(A \mid B_1) = \frac{3}{4}, \qquad P(A \mid B_2) = \frac{1}{5}.$$

First compute the total probability of drawing red:

$$P(A) = P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2).$$

Thus

$$P(A) = \frac{3}{4} \cdot \frac{2}{3} + \frac{1}{5} \cdot \frac{1}{3} = \frac{1}{2} + \frac{1}{15} = \frac{17}{30}.$$

Now Bayes' theorem gives

$$P(B_1 \mid A) = \frac{P(A \mid B_1)P(B_1)}{P(A)}.$$

Therefore

$$P(B_1 \mid A) = \frac{\frac{3}{4} \cdot \frac{2}{3}}{\frac{17}{30}} = \frac{\frac{1}{2}}{\frac{17}{30}} = \frac{15}{17}.$$

Similarly,

$$P(B_2 \mid A) = \frac{\frac{1}{5} \cdot \frac{1}{3}}{\frac{17}{30}} = \frac{\frac{1}{15}}{\frac{17}{30}} = \frac{2}{17}.$$

The posterior probabilities add to one:

$$\frac{15}{17} + \frac{2}{17} = 1.$$

The red observation strongly supports urn 1 because urn 1 was already more likely before the draw and also has a higher red proportion.

Example: diagnostic testing

Suppose a condition occurs in 1% of a population:

$$P(C) = 0.01, \qquad P(C^c) = 0.99.$$

A test has sensitivity 95%, meaning

$$P(T \mid C) = 0.95,$$

and false positive rate 5%, meaning

$$P(T \mid C^c) = 0.05.$$

Here T is the event that the test result is positive.

We want

$$P(C \mid T),$$

the probability that a person has the condition after receiving a positive test result.

By Bayes' theorem over the partition C, C^c,

$$P(C \mid T) = \frac{P(T \mid C)P(C)}{P(T \mid C)P(C) + P(T \mid C^c)P(C^c)}.$$

Substitute the values:

$$P(C \mid T) = \frac{0.95 \cdot 0.01}{0.95 \cdot 0.01 + 0.05 \cdot 0.99}.$$

The numerator is

$$0.0095.$$

The denominator is

$$0.0095 + 0.0495 = 0.059.$$

Therefore

$$P(C \mid T) = \frac{0.0095}{0.059} \approx 0.161.$$

So the probability is about 16.1%, not 95%.

This example is important because it shows the role of the base rate. A positive test result is evidence for the condition, but when the condition is rare, false positives may still be numerous among all positive results.

Natural frequency interpretation

The diagnostic example becomes clearer if we imagine 10000 people.

About 1% have the condition, so approximately

$$100$$

people have it, and

$$9900$$

do not.

Among the 100 people with the condition, the test is positive for about

$$0.95 \cdot 100 = 95$$

people.

Among the 9900 people without the condition, the test is positive for about

$$0.05 \cdot 9900 = 495$$

people.

So the total number of positive tests is about

$$95 + 495 = 590.$$

Among these, only 95 are true positives. Hence

$$P(C \mid T) \approx \frac{95}{590} \approx 0.161.$$

This is the same calculation as Bayes' theorem, but written as counts. It often makes the denominator easier to understand: the positive tests come from two sources, true positives and false positives.

Bayes' theorem as redistribution of probability over cases

Suppose $B_1, \ldots, B_n$ are possible cases. Before observing A, the probability assigned to case B_i is $P(B_i)$. After observing A, each case receives support proportional to

$$P(A \mid B_i)P(B_i).$$

The product has a clear meaning:

$$P(A \mid B_i)P(B_i) = P(A \cap B_i).$$

It is the probability that case B_i occurs and produces the observation A.

After observing A, all posterior probabilities must add to one. Therefore we divide each contribution by the total probability of A:

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i).$$

This turns the contributions into posterior probabilities.

Common reversal error

The most common mistake is to confuse

$$P(A \mid B)$$

with

$$P(B \mid A).$$

For example, test sensitivity $P(T \mid C)$ is not the same as the probability of having the condition after a positive test $P(C \mid T)$. The first uses the condition group as the reference space. The second uses the positive-test group as the reference space.

Bayes' theorem exists precisely because these two directions are different but related through the intersection $A \cap B$.

Remark

Whenever a conditional probability appears, read the phrase after the vertical bar first. It tells you the reference space. Then read the event before the bar. It tells you what is being measured inside that reference space.

What Bayes' theorem teaches

Bayes' theorem teaches how probability is updated by evidence. It combines three ingredients:

- prior probabilities of the possible cases;
- likelihoods of the observation under those cases;
- normalization by the total probability of the observation.

Theorem

Bayes' theorem is conditional probability used backward. It converts

$$P(\text{observation} \mid \text{case})$$

into

$$P(\text{case} \mid \text{observation}),$$

using the prior probabilities of the cases and the total probability of the observation.

5.6 Diagnostic examples and common traps

Conditional probability is conceptually simple once the reference space is clear, but it is easy to make mistakes when the roles of events are not stated explicitly. This section collects common traps and shows how to diagnose them.

The guiding question is always:

what is the information, and what is being measured inside that information?

Trap 1: confusing $P(A \mid B)$ with $P(B \mid A)$

The conditional probability

$$P(A \mid B)$$

means that B is the information and A is the event measured inside that information. The conditional probability

$$P(B \mid A)$$

reverses these roles.

These two numbers are usually different.

Example

Let

$$A = \{\text{a person is a mathematics student}\},$$

and

$$B = \{\text{a person takes a probability course}\}.$$

The probability $P(B \mid A)$ asks: among mathematics students, what proportion take a probability course? The probability $P(A \mid B)$ asks: among students in the probability course, what proportion are mathematics students?
These are different reference groups, so the answers need not be the same.

Diagnostic question:

Which event is after the vertical bar?

That event is the new reference space.

Trap 2: ignoring base rates

Bayesian problems often involve a rare event and an imperfect observation. The observation may be strong evidence, but the prior probability still matters. Ignoring the prior probability is called ignoring the base rate.

Suppose a condition occurs in 1% of a population. A test is positive for 99% of people with the condition and for 5% of people without the condition. A positive test does not mean the person has probability 99% of having the condition. The number 99% is

$$P(\text{positive} \mid \text{condition}).$$

The desired probability after a positive test is

$$P(\text{condition} \mid \text{positive}).$$

These are different.

Diagnostic question:

How common was the case before the evidence was observed?

Bayes' theorem combines the prior probability with the likelihood of the observation.

Trap 3: treating disjoint events as independent

If two events cannot occur together, they are disjoint:

$$A \cap B = \varnothing.$$

If both have positive probability, then learning that one occurred makes the other impossible. Therefore they are dependent, not independent.

For example, in one die roll, let

$$A = \{\text{the result is even}\} = \{2, 4, 6\},$$

and

$$B = \{\text{the result is odd}\} = \{1, 3, 5\}.$$

The events are disjoint. If we know B, then the probability of A becomes zero:

$$P(A \mid B) = 0.$$

But before knowing B,

$$P(A) = \frac{1}{2}.$$

The information changes the probability, so the events are dependent.

Diagnostic question:

If one event occurs, does the other become impossible?

If yes, then positive-probability events are not independent.

Trap 4: assuming independence without justification

Independence is not created by notation. Writing a sample space as a product, such as

$$\{H, T\}^2$$

or

$$\{1, 2, 3, 4, 5, 6\}^2,$$

describes possible ordered outcomes. It does not by itself prove that the stages are independent.

Independence is a property of the probability assignment. It says that knowing one event occurred does not change the probability of another.

Example

A mechanism produces two symbols, each either H or T. The possible outputs are

$$HH, HT, TH, TT.$$

If the mechanism chooses the first symbol fairly and then always repeats it, the only outputs with positive probability are HH and TT. The two positions are not independent, even though the notation has two coordinates.

Diagnostic question:

Where in the model is independence stated or justified?

If it is not stated or implied by the mechanism, do not use product rules as if it were automatic.

Trap 5: using total probability without a partition

The law of total probability requires cases that are mutually exclusive and exhaustive. If the cases overlap, adding contributions may double-count. If the cases do not cover the whole sample space, the sum may miss part of the event.

Before writing

$$P(A) = \sum_i P(A \mid B_i)P(B_i),$$

check that the events B_i form a partition.

Diagnostic questions:

- Can two of the cases occur at the same time?
- Is every possible outcome covered by one of the cases?
- Do all conditioning events have positive probability?

Trap 6: forgetting the denominator in Bayes' theorem

In Bayes' theorem, the denominator is the total probability of the observation. It includes all ways the observation can occur. Omitting some cases changes the reference space and gives the wrong posterior probability.

For a partition $B_1, \dots, B_n$, the denominator is

$$P(A) = \sum_i P(A \mid B_i)P(B_i).$$

It is not merely the likelihood under the case we are interested in.

Diagnostic question:

What are all possible sources of the observation?

The denominator must include them all.

Trap 7: treating a conditional probability as a causal statement

If

$$P(A \mid B) > P(A),$$

then B is informative about A. But this alone does not prove that B causes A. Conditional probability describes how probabilities change under information. Causal interpretation requires additional structure.

For example, if students who attend review sessions pass more often, then attendance is associated with passing. But from this probability statement alone we cannot conclude that attendance caused the higher pass rate. Students who attend may differ in other ways.

Diagnostic question:

Am I making a probability statement or a causal claim?

These are not the same kind of statement.

Trap 8: conditioning on an event of probability zero

The elementary formula

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

requires

$$P(B) > 0.$$

If $P(B) = 0$, the denominator vanishes and the formula is not defined.

In finite examples this often corresponds to impossible information. For a die roll, conditioning on the event "the result is 8" is not meaningful in the usual die model because that event is empty.

Diagnostic question:

Does the conditioning event have positive probability in this model?

A diagnostic workflow

When a conditional probability problem appears, use the following workflow.

1. Identify the original sample space or probability model.
2. Identify the information event: this is the event after the vertical bar.
3. Identify the target event: this is the event before the vertical bar.
4. Check that the information event has positive probability.
5. Replace the active space of reasoning by the information event.
6. Locate the target event inside that information event by taking the intersection.
7. Decide whether the problem needs direct conditioning, independence, total probability, or Bayes' theorem.

Theorem

Most mistakes with conditional probability are mistakes about reference spaces. Before calculating, say explicitly: "I am reasoning inside this event."

5.7 Final checklist

This chapter developed conditional probability as probability after information has changed the active space of reasoning. The central habit is to identify the reference space before calculating.

The checklist below summarizes the method.

Step 1: Identify the original model

Before conditioning, there must be a probability model. Identify:

- the sample space Ω;
- the elementary outcomes or relevant events;
- the probability assignment;
- whether the model is uniform or non-uniform.

Conditional probability does not replace model construction. It uses the model that has already been built.

Step 2: Identify the information event

In an expression

$$P(A \mid B),$$

the event after the vertical bar is the information event. It is the event that is assumed to have occurred.

Say explicitly:

$$\text{we are now reasoning inside } B.$$

Check that

$$P(B) > 0.$$

If $P(B) = 0$, the elementary conditional probability formula is not defined.

Step 3: Identify the target event

The event before the vertical bar is the event being measured. In

$$P(A \mid B),$$

the target event is A. The favourable part of the information region is

$$A \cap B.$$

Do not replace A by B. Do not reverse the order unless the problem asks for the reverse conditional probability.

Step 4: Use the definition when conditioning directly

If the problem asks directly for $P(A \mid B)$, use

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

In a finite uniform model this becomes

$$P(A \mid B) = \frac{|A \cap B|}{|B|}.$$

The denominator is the size or probability of the information region. The numerator is the part of that region where the target event also occurs.

Step 5: Use independence only when justified

Events A and B are independent when information about one does not change the probability of the other. Algebraically,

$$P(A \cap B) = P(A)P(B).$$

When probabilities are positive, this is equivalent to

$$P(A \mid B) = P(A)$$

and

$$P(B \mid A) = P(B).$$

Do not assume independence merely because there are two stages or two coordinates. Independence belongs to the probability assignment, not just to the notation.

Step 6: Use total probability when cases produce the event

If an event A may occur through several mutually exclusive and exhaustive cases $B_1, \ldots, B_n$, use the law of total probability:

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i).$$

Before using it, check:

- the cases are disjoint;
- the cases cover the whole sample space;
- the conditional probabilities inside the cases are meaningful.

Read the formula as:

$$\text{total probability} = \sum \text{case probability} \times \text{probability inside case}.$$

Step 7: Use Bayes' theorem when reversing information

Use Bayes' theorem when the problem gives or suggests probabilities of the form

$$P(\text{observation} \mid \text{case})$$

but asks for

$$P(\text{case} \mid \text{observation}).$$

For a partition $B_1, \ldots, B_n$,

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{\sum_{i=1}^{n} P(A \mid B_i)P(B_i)}.$$

Remember the roles:

- $P(B_j)$ is the prior probability of the case;
- $P(A \mid B_j)$ is the likelihood of the observation in that case;
- $P(B_j \mid A)$ is the posterior probability of the case after the observation.

Step 8: Interpret the result

After computing a conditional probability, translate it back into words.

For example,

$$P(A \mid B) = 0.7$$

means:

$$\text{among the outcomes compatible with } B, \text{ the event } A \text{ has probability } 0.7.$$

It does not mean that B was caused by A, nor that $P(B \mid A) = 0.7$.

Compact checklist

For any problem in this chapter, ask:

1. What is the original probability model?
2. What information has been given?
3. What event is after the vertical bar?
4. What event is before the vertical bar?
5. What is the intersection of the target event with the information event?
6. Is the conditioning event positive-probability?
7. Is independence stated or justified?
8. Is there a partition of cases?
9. Is the problem asking for a direct conditional probability or a reverse probability requiring Bayes' theorem?
10. What does the final number mean in words?

Theorem

The most important sentence in conditional probability is:

the event after the vertical bar is the space in which we now reason.

Once this is clear, the formulas become expressions of that change of reference space.

Chapter 6

Discrete Random Variables and Probability Laws

This chapter introduces random variables through the discrete case. A random variable is viewed as a function that extracts a numerical quantity from an elementary outcome and allows probabilistic phenomena to be studied through numerical characteristics.

The chapter develops discrete probability laws, probability mass functions, cumulative distribution functions, expectation, variance, moments, and the interpretation of these quantities. Classical discrete models are presented as mathematical descriptions of specific random mechanisms.

6.1 Random variables as functions on the sample space

Until now, probability models have been built around sample spaces and events. We described an experiment, decided what counts as an elementary outcome, constructed a sample space Ω, and assigned probabilities to events. A random variable adds a new layer: it extracts a numerical quantity from each outcome.

This is the central idea:

a random variable is a measurable function on the sample space.

Its randomness comes from the random elementary outcome to which the function is applied.

Definition

Let $(\Omega, \mathcal{F}, P)$ be a probability space. A **random variable** is a measurable function

$$X : \Omega \to \mathbb{R}.$$

For each elementary outcome $\omega \in \Omega$, the value $X(\omega)$ is a real number extracted from that outcome.

Measurability means that numerical conditions such as

$$\{\omega \in \Omega : X(\omega) \leq t\}$$

are events in $\mathcal{F}$, so their probabilities are defined. In the finite and countable examples of this chapter, this condition is automatic when every subset of Ω is an event. It becomes important in continuous models, where the event collection need not contain every subset of Ω.

Why introduce random variables?

Many experiments produce outcomes that contain more information than the numerical quantity we want to study. If a coin is tossed five times, one outcome may be

$$\omega = HHTTH.$$

The outcome records the complete order, while the random variable

$$X(\omega) = \text{the number of heads in } \omega$$

extracts only the count. For this outcome,

$$X(HHTTH) = 3.$$

The random variable does not replace the sample space. It is built on top of it.

Example: the sum of two dice

Roll two fair dice and record the ordered pair. The sample space is

$$\Omega = \{(i, j) : i, j \in \{1, 2, 3, 4, 5, 6\}\}.$$

Define

$$S : \Omega \to \mathbb{R}, \qquad S(i, j) = i + j.$$

Then

$$S(1, 6) = 7, \qquad S(3, 4) = 7, \qquad S(6, 6) = 12.$$

Different elementary outcomes may produce the same value of the random variable. The value 7 is not one elementary outcome in this model; it is the common value of S on several outcomes.

Remark

A random variable often aggregates information. The sample space distinguishes $(1, 6)$ from $(6, 1)$, but the sum assigns both outcomes the value 7.

Example: a numerical consequence

Suppose one observation has sample space

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

A numerical consequence Y may be defined by

$$Y(6) = 10, \qquad Y(2) = Y(4) = 2, \qquad Y(1) = Y(3) = Y(5) = -3.$$

The probability model describes which outcome occurs. The random variable describes the numerical value attached to that outcome.

Indicator random variables

Let $A \in \mathcal{F}$ be an event. Its indicator is

$$I_A(\omega) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$$

It records numerically whether A occurred.

Indicators are powerful because counts can be written as sums. For example, the number of heads in n tosses is a sum of n indicators, one for each toss.

Inverse images of numerical conditions

If B is a Borel subset of $\mathbb{R}$, then

$$\{X \in B\} = \{\omega \in \Omega : X(\omega) \in B\}$$

is the inverse image of B under X. Measurability guarantees that this set belongs to $\mathcal{F}$.

For the sum of two dice,

$$\{S = 7\} = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}.$$

Thus an expression such as $P(S = 7)$ is the probability of an event in the original sample space.

Remark

Every expression such as $P(X = x)$, $P(X \le t)$, or $P(X \in B)$ is a probability of an event in Ω. A random variable translates numerical conditions into events.

From outcomes to distributions

The workflow is

$$\text{experiment} \longrightarrow (\Omega, \mathcal{F}, P) \longrightarrow X(\omega) \longrightarrow \text{probabilities of numerical values or regions.}$$

The probability law transferred from outcomes to numerical values is called the distribution of X.

Theorem

A random variable does not create probability. It transfers the probability already defined on $(\Omega, \mathcal{F})$ to numerical events through a measurable function.

6.2 Discrete random variables and their support

A random variable is a function from the sample space to the real line. The next question is: what kinds of values can this function produce? In this chapter we focus on the discrete case, where the possible values form a finite or countable set.

Definition

A random variable X is **discrete** if the set of values it can take is finite or countably infinite. Equivalently, its possible values can be written as

$$x_1, x_2, x_3, \dots$$

with the list either ending after finitely many terms or continuing indefinitely.

The word discrete refers to the values of the random variable, not necessarily to how complicated the original experiment is. The sample space may contain detailed objects, but the random variable may record only a count, a sum, an indicator, or a payoff.

The range and the support

If $X : \Omega \to \mathbb{R}$, then the set of values that X actually takes is

$$X(\Omega) = \{X(\omega) : \omega \in \Omega\}.$$

This is the range of the function. In probability, we usually focus on values that have positive probability.

Definition

The **support** of a discrete random variable X is the set

$$\operatorname{supp}(X) = \{x \in \mathbb{R} : P(X = x) > 0\}.$$

It is the set of values that occur with positive probability.

In elementary finite models, the range and support are often the same. But the definition using positive probability is more robust, because it ignores values that may be formally possible in notation but receive probability zero.

Example: number of heads

Toss a coin n times and record the ordered sequence. The sample space is

$$\Omega = \{H, T\}^n.$$

Define

$$X(\omega) = \text{the number of heads in } \omega.$$

Then the possible values are

$$0, 1, 2, \ldots, n.$$

Thus

$$\operatorname{supp}(X) = \{0, 1, 2, \ldots, n\},$$

provided each sequence has positive probability.

This example shows a key point: Ω contains sequences, but the support of X contains numbers. The random variable turns a sequence into a count.

Example: sum of two dice

Roll two dice and record the ordered pair:

$$\Omega = \{1, 2, 3, 4, 5, 6\}^2.$$

Define the sum random variable

$$S(i, j) = i + j.$$

The smallest possible value is 2, and the largest possible value is 12. Every integer between them can occur, so

$$\operatorname{supp}(S) = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}.$$

Notice that the support has 11 values, while the original sample space has 36 elementary outcomes. Many different pairs lead to the same sum. For example,

$$S^{-1}(7) = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}.$$

Example: waiting time

In the experiment of tossing a coin until the first head appears, define

$$N = \text{the number of tosses needed to obtain the first head.}$$

Then

$$\operatorname{supp}(N) = \{1, 2, 3, \ldots\}.$$

This support is countably infinite. The random variable is still discrete, because its possible values can be listed.

Discrete does not mean finite

A discrete random variable may have infinitely many possible values. The waiting time N is the first important example. There is no largest possible waiting time: success may occur on the first trial, the second trial, the third trial, and so on. Still, the values are separated and countable:

$$1, 2, 3, \ldots$$

This is very different from a continuous variable such as a uniformly chosen point in $[0, 1]$, whose possible values cannot be listed in a sequence.

Remark

Discrete means finite or countably infinite. Continuous means that probability is spread over a continuum of values. The distinction concerns the structure of the possible values of the random variable.

Events generated by values of a random variable

Once X is defined, statements about X create events in the original sample space. For example,

$$\{X = 3\} = \{\omega \in \Omega : X(\omega) = 3\},$$
$$\{X \le 3\} = \{\omega \in \Omega : X(\omega) \le 3\}.$$

For the number of heads in five tosses, the event $\{X = 3\}$ contains all sequences with exactly three heads. For the sum of two dice, the event $\{S \le 4\}$ contains all ordered pairs whose sum is 2, 3, or 4.

What the support tells us

The support tells us where the probability law of the random variable lives. It is the list of values for which we must assign probabilities. Once the support is known, the next task is to determine

$$P(X = x)$$

for each x in the support.

Theorem

To study a discrete random variable, first identify its support. The support is the bridge between the original sample space and the numerical probability law of X.

6.3 Probability mass functions

Once a discrete random variable X has been defined, we want to know how probability is distributed among its possible values. This information is encoded by the probability mass function.

The basic quantity is

$$P(X = x).$$

This notation is convenient, but it must be read carefully. The expression $X = x$ describes an event in the original sample space:

$$\{X = x\} = \{\omega \in \Omega : X(\omega) = x\}.$$

Thus $P(X = x)$ means the probability of all elementary outcomes that are mapped to the value x.

Definition

Let X be a discrete random variable. Its **probability mass function** p_X is defined by

$$p_X(x) = P(X = x).$$

For values outside the support, $p_X(x) = 0$.

The word mass is used because, in a discrete distribution, probability is concentrated on separate values. Each value in the support carries a certain amount of probability mass.

The two basic properties

A probability mass function must satisfy two conditions:

$$p_X(x) \geq 0$$

for every x, and

$$\sum_{x \in \operatorname{supp}(X)} p_X(x) = 1.$$

These are not arbitrary requirements. They come from the axioms of probability. The events $\{X = x\}$ for different support values are disjoint, and their union is the whole sample space up to events of probability zero.

Remark

A proposed formula for a probability mass function is not valid until its values are non-negative and sum to 1 over the support.

Example: indicator of an event

Let A be an event and let I_A be its indicator:

$$I_A(\omega) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$$

The support of I_A is contained in $\{0, 1\}$. Its mass function is

$$p_{I_A}(1) = P(I_A = 1) = P(A),$$

and

$$p_{I_A}(0) = P(I_A = 0) = P(A^c) = 1 - P(A).$$

Thus an event can be represented as a zero-one random variable. This will be important when we study expectations and counts.

Example: sum of two dice

Let $S(i, j) = i + j$ be the sum of two fair dice. The sample space has 36 equally likely ordered pairs. To find $p_S(s)$, we count how many pairs have sum s.

The counts are

s	2	3	4	5	6	7	8	9	10	11	12
$\#\{(i,j) : i + j = s\}$	1	2	3	4	5	6	5	4	3	2	1

Therefore

s	2	3	4	5	6	7	8	9	10	11	12
$p_S(s)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

For example,

$$p_S(7) = P(S = 7) = \frac{6}{36} = \frac{1}{6}.$$

The value 7 has the largest probability because it can be produced in the largest number of ordered ways.

Example: number of heads in n tosses

Toss a coin n times independently, with

$$P(H) = p, \qquad P(T) = 1 - p.$$

Let

$$X = \text{the number of heads.}$$

The support is

$$\{0, 1, 2, \ldots, n\}.$$

For $X = k$, exactly k positions among the n positions must contain heads. There are

$$\binom{n}{k}$$

ways to choose those positions. Each sequence with k heads and $n - k$ tails has probability

$$p^k (1-p)^{n-k}.$$

Hence

$$p_X(k) = P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

This formula is not just a formula to remember. It contains the whole mechanism: choose positions of successes, multiply probabilities along each sequence, and sum over all sequences with the same count.

Example: urn model without replacement

Suppose an urn contains K objects of type success and $N - K$ objects of type failure. We select n objects without replacement and ignore order. Let

$$X = \text{the number of successes selected.}$$

If $X = k$, then we choose k successes from the K success objects and $n - k$ failures from the $N - K$ failure objects. The total number of possible samples is $\binom{N}{n}$. Therefore

$$p_X(k) = P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}},$$

for values of k for which the binomial coefficients are meaningful.

This distribution comes from the mechanism of sampling without replacement. It is different from the binomial model because the selections are not independent: a success removed from the population changes what remains.

Reading a mass function as a model

A mass function is a compact description of a probability law. But it should not be separated from the mechanism that produced it. When reading or constructing a mass function, ask:

- what is the original experiment?
- what is the random variable?
- what support values are possible?
- which elementary outcomes produce each value?
- how are their probabilities added?

Theorem

For a discrete random variable, the probability mass function is obtained by transferring probability from elementary outcomes to numerical values:

$$p_X(x) = P(\{\omega \in \Omega : X(\omega) = x\}).$$

6.4 Cumulative distribution functions

A probability mass function records probability at individual support values. A cumulative distribution function stores the same law by accumulating probability up to a threshold.

Definition

For a real-valued random variable X, the **cumulative distribution function**, or **CDF**, is

$$F_X(t) = P(X \le t), \qquad t \in \mathbb{R}.$$

If X is discrete with probability mass function p_X, then

$$F_X(t) = \sum_{x \le t} p_X(x).$$

Thus the CDF is the accumulated PMF.

Example: an indicator variable

Let I_A be the indicator of an event A, and write $p = P(A)$. Then

$$F_{I_A}(t) = \begin{cases} 0, & t < 0, \\ 1 - p, & 0 \le t < 1, \\ 1, & t \ge 1. \end{cases}$$

The jump at 0 has size $1 - p$, and the jump at 1 has size p.

Jumps and probability masses

For every real number x,

$$P(X = x) = F_X(x) - F_X(x^-),$$

where

$$F_X(x^-) = \lim_{t \uparrow x} F_X(t)$$

is the left-hand limit. The probability mass at a point is therefore the size of the jump of the CDF at that point.

Basic properties of every CDF

Every cumulative distribution function satisfies:

- $0 \le F_X(t) \le 1$ for every $t \in \mathbb{R}$;
- F_X is non-decreasing;
- F_X is right-continuous;
- $\lim_{t \to -\infty} F_X(t) = 0$;

- $\lim_{t\to\infty} F_X(t) = 1$.

Right-continuity means
$$\lim_{s\downarrow t} F_X(s) = F_X(t).$$

For a discrete random variable, the CDF is constant between support values and jumps upward at values with positive probability. The right-continuous convention includes the mass at the point where the jump occurs.

Computing probabilities from a CDF

Once F_X is known,
$$P(X > t) = 1 - F_X(t).$$

For $a < b$,
$$P(a < X \le b) = F_X(b) - F_X(a).$$

Endpoint conventions matter in the discrete case because an endpoint may carry positive probability.

Theorem

For a discrete random variable, the PMF and CDF describe the same probability law. The CDF accumulates the masses, and the masses are recovered from its jumps.

6.5 Expectation as a weighted average

A probability mass function tells us how probability is distributed over the possible values of a random variable. The next question is how to summarize that whole distribution by a single representative number. The most important such summary is expectation.

Expectation should not be introduced as a formula to memorize. It is a weighted average. Values that occur with larger probability receive larger weight.

Averages with unequal weights

Suppose a random variable takes values
$$x_1, x_2, \ldots, x_n$$

with probabilities
$$p_1, p_2, \ldots, p_n,$$

where
$$p_i \ge 0, \qquad p_1 + p_2 + \cdots + p_n = 1.$$

If all values were equally likely, their average would be
$$\frac{x_1 + x_2 + \cdots + x_n}{n}.$$

But if some values are more likely than others, the average should count those values more strongly. This leads to the weighted average
$$x_1p_1 + x_2p_2 + \cdots + x_np_n.$$

Definition

For a discrete random variable X with probability mass function p_X, the **expectation** of X is

$$\mathbb{E}[X] = \sum_{x \in \mathrm{supp}(X)} x\, p_X(x),$$

provided the sum is well-defined.

For finite support, the sum is always finite. For countably infinite support, one must check convergence; at this stage we mainly work with examples where the sum is meaningful.

Expectation from the sample space

If the original sample space is finite, expectation can also be computed by summing over elementary outcomes:

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega) P(\{\omega\}).$$

This formula averages the numerical value $X(\omega)$ over all elementary outcomes, weighted by their probabilities.

The formula using the mass function is obtained by grouping together all outcomes that produce the same value. If $X = x$, then all outcomes in the event $\{X = x\}$ contribute the same value x. The total probability of that event is $P(X = x) = p_X(x)$. Thus the grouped contribution is $xp_X(x)$.

Remark

The expectation can be computed either by averaging over elementary outcomes or by averaging over values of the random variable. The second method is usually more compact because it uses the distribution of X.

Example: one fair die

Let X be the result of one fair die roll. Then

$$P(X = k) = \frac{1}{6}, \qquad k = 1, 2, 3, 4, 5, 6.$$

Therefore

$$\mathbb{E}[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6}.$$

Hence

$$\mathbb{E}[X] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{21}{6} = 3.5.$$

The expectation is not necessarily a value that the random variable can take. A die never shows 3.5. The expectation is a center of balance of the probability distribution.

Example: payoff of a game

Suppose a game has payoff random variable X defined by

$$X = 10 \quad \text{with probability } \frac{1}{6},$$

$$X = 2 \quad \text{with probability } \frac{2}{6},$$

and

$$X = -3 \quad \text{with probability } \frac{3}{6}.$$

Then
$$\mathbb{E}[X] = 10 \cdot \frac{1}{6} + 2 \cdot \frac{2}{6} + (-3) \cdot \frac{3}{6}.$$
Thus
$$\mathbb{E}[X] = \frac{10}{6} + \frac{4}{6} - \frac{9}{6} = \frac{5}{6}.$$
The expected payoff is positive, even though some outcomes produce losses. Interpreted over many repetitions, $5/6$ is the average payoff per play that the model predicts in the long run.

Expectation of an indicator

Let I_A be the indicator of an event A. Then

$$I_A = 1 \quad \text{with probability } P(A),$$

and

$$I_A = 0 \quad \text{with probability } 1 - P(A).$$

Therefore
$$\mathbb{E}[I_A] = 1 \cdot P(A) + 0 \cdot (1 - P(A)) = P(A).$$

Theorem

For every event A,
$$\mathbb{E}[I_A] = P(A).$$

This identity is simple but fundamental. It says that probabilities can be read as expectations of indicator variables.

Linearity of expectation

Expectation has an important algebraic property: it is linear. If X and Y are random variables and a, b are constants, then
$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$
In finite models, this follows directly from the sample-space formula:

$$\mathbb{E}[aX + bY] = \sum_{\omega \in \Omega} (aX(\omega) + bY(\omega))P(\{\omega\}).$$

Distributing the sum gives

$$\mathbb{E}[aX + bY] = a \sum_{\omega \in \Omega} X(\omega)P(\{\omega\}) + b \sum_{\omega \in \Omega} Y(\omega)P(\{\omega\}).$$

Hence
$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

The remarkable point is that linearity does not require independence. It is a property of averaging.

Counting successes by indicators

Toss a coin n times. Let
$$X = \text{the number of heads.}$$

Define indicator variables
$$I_k = \begin{cases} 1, & \text{the } k\text{-th toss is heads,} \\ 0, & \text{the } k\text{-th toss is tails.} \end{cases}$$

Then

$$X = I_1 + I_2 + \cdots + I_n.$$

If each toss has probability p of heads, then

$$\mathbb{E}[I_k] = p$$

for every k. By linearity,

$$\mathbb{E}[X] = \mathbb{E}[I_1] + \cdots + \mathbb{E}[I_n] = np.$$

This derivation is often clearer than summing the binomial mass function. It shows that the expected number of successes is the sum of the expected contributions from each trial.

What expectation measures

Expectation is not the most likely value. It is not necessarily a value that can occur. It is a weighted average, or center of mass, of the distribution. In applications, it often represents a long-run average under repeated independent repetitions of the same model.

Summary

For a discrete random variable, expectation is computed by

$$\mathbb{E}[X] = \sum_x xP(X = x).$$

It summarizes the distribution by averaging values according to their probability weights. It is best understood as a consequence of the probability law of X, not as a separate rule detached from the model.

6.6 Variance, standard deviation, and moments

Expectation describes the center of a probability distribution, but it does not describe how spread out the values are. Two random variables may have the same expectation and very different variability. Variance is the basic measure of spread around the expectation.

Let

$$\mu = \mathbb{E}[X].$$

The quantity

$$X - \mu$$

measures the deviation of X from its mean. Positive values are above the mean; negative values are below the mean.

Why square the deviation?

One might first try to average the deviations:

$$\mathbb{E}[X - \mu].$$

But by linearity of expectation,

$$\mathbb{E}[X - \mu] = \mathbb{E}[X] - \mu = 0.$$

The positive and negative deviations cancel. This does not measure spread.

To avoid cancellation, we square the deviations. Large deviations then contribute more strongly, and all contributions are non-negative.

Definition

The **variance** of a random variable X is

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2],$$

provided this expectation exists.

For a discrete random variable, this becomes

$$\mathrm{Var}(X) = \sum_{x \in \mathrm{supp}(X)} (x - \mu)^2 p_X(x),$$

where $\mu = \mathbb{E}[X]$.

Standard deviation

Variance is measured in squared units. If X is measured in euros, then $\mathrm{Var}(X)$ is measured in squared euros. To return to the original units, we take the square root.

Definition

The **standard deviation** of X is

$$\sigma_X = \sqrt{\mathrm{Var}(X)}.$$

The standard deviation is a typical scale of deviation from the mean. It is not the same as the average absolute distance, but it is one of the most important measures of spread in probability and statistics.

A useful computational formula

The definition

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mu)^2]$$

is conceptually clear. For calculations, another form is often easier. Expand the square:

$$(X - \mu)^2 = X^2 - 2\mu X + \mu^2.$$

Taking expectation gives

$$\mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2].$$

Using linearity,

$$\mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2.$$

Since $\mu = \mathbb{E}[X]$, we get

$$\mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - 2\mu^2 + \mu^2.$$

Therefore

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Theorem

For every random variable for which the quantities are defined,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

This formula should not replace the meaning of variance. It is a computational consequence of the definition.

Example: one fair die

Let X be the result of one fair die roll. We already know

$$\mathbb{E}[X] = \frac{1+2+3+4+5+6}{6} = \frac{7}{2}.$$

Now compute

$$\mathbb{E}[X^2] = \frac{1^2+2^2+3^2+4^2+5^2+6^2}{6}.$$

The numerator is

$$1+4+9+16+25+36 = 91.$$

Thus

$$\mathbb{E}[X^2] = \frac{91}{6}.$$

Therefore

$$\mathrm{Var}(X) = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{91}{6} - \frac{49}{4}.$$

Using denominator 12,

$$\mathrm{Var}(X) = \frac{182}{12} - \frac{147}{12} = \frac{35}{12}.$$

The standard deviation is

$$\sigma_X = \sqrt{\frac{35}{12}}.$$

Example: an indicator

Let I_A be the indicator of an event A, and let

$$p = P(A).$$

Then

$$\mathbb{E}[I_A] = p.$$

Because I_A takes only the values 0 and 1, we have

$$I_A^2 = I_A.$$

Hence

$$\mathbb{E}[I_A^2] = \mathbb{E}[I_A] = p.$$

Therefore

$$\mathrm{Var}(I_A) = p - p^2 = p(1-p).$$

This result is important because many counts are sums of indicators. It also shows that the variance of a zero-one variable is largest when $p = 1/2$, and small when the event is almost impossible or almost certain.

Moments

Expectations of powers of X are called moments. The k-th raw moment is

$$\mathbb{E}[X^k].$$

For a discrete random variable,

$$\mathbb{E}[X^k] = \sum_x x^k p_X(x),$$

provided the sum is defined.

The first raw moment is the expectation:

$$\mathbb{E}[X].$$

The second raw moment $\mathbb{E}[X^2]$ is used in the computation of variance. Higher moments describe further aspects of the shape of the distribution, but in this course the most important quantities at this stage are expectation and variance.

Interpreting variance carefully

Variance is always non-negative because it is the expectation of a square:

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mu)^2] \geq 0.$$

It is zero exactly when X is constant with probability one. If there is no spread, every possible value is equal to the mean.

A larger variance indicates more spread, but it must be interpreted relative to the scale of the problem. A variance of 100 is large for exam scores out of 10, but small for yearly income measured in euros.

Summary

Expectation measures center. Variance measures average squared distance from the center. Standard deviation returns this spread to the original units:

$$\sigma_X = \sqrt{\mathrm{Var}(X)}.$$

Together, expectation and variance give the first basic numerical summary of a probability law.

6.7 Classical discrete probability laws as mechanisms

Classical discrete distributions should not be learned as a table of formulas. Each one is a compact description of a random mechanism. The formula for the probability mass function is important, but it should be read as the consequence of how the experiment is constructed.

In this section we collect several basic discrete laws and emphasize the mechanism behind each one.

Bernoulli law: one success-or-failure trial

The simplest discrete random variable records whether one event occurred. Let a single trial have two possible outcomes: success and failure. Define

$$X = \begin{cases} 1, & \text{success occurs,} \\ 0, & \text{failure occurs.} \end{cases}$$

If

$$P(\text{success}) = p,$$

then

$$P(X = 1) = p, \qquad P(X = 0) = 1 - p.$$

Definition

A random variable X has a **Bernoulli distribution** with parameter p, written $X \sim \mathrm{Bernoulli}(p)$, if

$$P(X = 1) = p, \qquad P(X = 0) = 1 - p.$$

The Bernoulli law is the probability law of an indicator variable. If $X = I_A$, then $X \sim \mathrm{Bernoulli}(P(A))$.

Its expectation and variance are

$$\mathbb{E}[X] = p, \qquad \mathrm{Var}(X) = p(1 - p).$$

Binomial law: number of successes in fixed independent trials

Now repeat a Bernoulli trial n times. Assume that each trial has success probability p, and that the trials are independent. Let

$$X = \text{the number of successes in } n \text{ trials.}$$

Then

$$\mathrm{supp}(X) = \{0, 1, 2, \ldots, n\}.$$

To have $X = k$, exactly k of the n positions must be successes. There are

$$\binom{n}{k}$$

ways to choose those positions. Each sequence with k successes and $n - k$ failures has probability

$$p^k(1 - p)^{n-k}.$$

Therefore

$$P(X = k) = \binom{n}{k} p^k(1 - p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

Definition

A random variable X has a **binomial distribution** with parameters n and p, written $X \sim \mathrm{Binomial}(n, p)$, if

$$P(X = k) = \binom{n}{k} p^k(1 - p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

The mechanism matters. Binomial means fixed number of trials, two outcomes per trial, constant success probability, independence, and counting successes.

Using indicators, if

$$X = I_1 + I_2 + \cdots + I_n,$$

where each $I_i \sim \mathrm{Bernoulli}(p)$, then

$$\mathbb{E}[X] = np.$$

With independence one also obtains

$$\mathrm{Var}(X) = np(1 - p).$$

Geometric law: waiting for the first success

In a binomial model, the number of trials is fixed and the number of successes is random. In the geometric model, the number of successes is fixed at one first success, and the waiting time is random.

Repeat independent Bernoulli trials with success probability p. Let

$$N = \text{the trial on which the first success occurs.}$$

Then

$$\operatorname{supp}(N) = \{1, 2, 3, \dots\}.$$

For $N = k$, the first $k-1$ trials must be failures and the k-th trial must be a success. Hence

$$P(N = k) = (1-p)^{k-1}p, \qquad k = 1, 2, 3, \dots.$$

Definition

A random variable N has a **geometric distribution** with parameter p, in the waiting-time convention, if

$$P(N = k) = (1-p)^{k-1}p, \qquad k = 1, 2, 3, \dots.$$

The geometric distribution is a model for waiting until the first success. Its support is infinite because success may be delayed for any number of trials.

Hypergeometric law: sampling without replacement

Suppose a population contains N objects. Among them, K are classified as successes and $N-K$ as failures. We sample n objects without replacement and ignore order. Let

$$X = \text{the number of successes in the sample.}$$

To get $X = k$, choose k successes from the K success objects and choose $n-k$ failures from the $N-K$ failure objects. The total number of samples is $\binom{N}{n}$. Therefore

$$P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}.$$

Definition

A random variable X has a **hypergeometric distribution** with parameters N, K, n if

$$P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$$

for the admissible values of k.

This model differs from the binomial model because sampling is done without replacement. The draws are dependent: after one object is selected, the composition of the remaining population changes.

Poisson law: counts of rare events

The Poisson distribution is used to model counts of events occurring in a fixed region of time, space, area, or volume, when events occur with a stable average rate and without strong clustering beyond what the model allows.

Let

$$X = \text{the number of events in one observation window.}$$

The support is

$$\{0, 1, 2, \ldots\}.$$

A Poisson model with parameter $\lambda > 0$ has mass function

$$P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots.$$

Definition

A random variable X has a **Poisson distribution** with parameter $\lambda > 0$, written $X \sim \text{Poisson}(\lambda)$, if

$$P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots.$$

The parameter λ is both the expectation and the variance:

$$\mathbb{E}[X] = \lambda, \qquad \text{Var}(X) = \lambda.$$

In this chapter, the main point is interpretation: the Poisson law models a count, not a waiting time and not a fixed number of trials.

Comparing the mechanisms

The following table summarizes the roles of the classical laws.

Law	Random variable	Mechanism
Bernoulli	0 or 1	one success-or-failure trial
Binomial	number of successes	fixed independent trials
Geometric	waiting time	repeat until first success
Hypergeometric	number of successes	sampling without replacement
Poisson	count of events	events in a fixed window

The same numerical value may appear in several models, but the interpretation is different. A value k in a binomial model means k successes among a fixed number of trials. A value k in a geometric model means the first success occurred on trial k. A value k in a Poisson model means k events were counted in a window.

Remark

Choosing a distribution is not choosing a formula. It is choosing a mechanism. Before using a classical law, identify what is being counted, what is fixed, what is random, and what assumptions connect the experiment to the model.

6.8 Final checklist

This chapter introduced discrete random variables as functions on sample spaces. The central workflow was

$$\Omega \longrightarrow X(\omega) \longrightarrow P(X = x).$$

The sample space describes possible outcomes of the experiment. The random variable extracts a numerical quantity. The distribution describes how probability is transferred from outcomes to values.

Checklist for constructing a discrete random variable

When a problem involves a discrete random variable, proceed as follows.

1. Identify the original experiment.
2. Construct or recall the sample space Ω.
3. Define the random variable as a function
$$X : \Omega \to \mathbb{R}.$$
4. State clearly what $X(\omega)$ means for an elementary outcome ω.
5. Find the support:
$$\mathrm{supp}(X) = \{x : P(X = x) > 0\}.$$
6. For each support value x, identify the event
$$\{X = x\} = \{\omega \in \Omega : X(\omega) = x\}.$$
7. Compute the probability mass function
$$p_X(x) = P(X = x).$$
8. Check that
$$\sum_x p_X(x) = 1.$$

Checklist for interpreting the distribution

After the probability law has been found, ask:

1. What does each value of X mean in the original experiment?
2. Which values are most likely?
3. Is the support finite or countably infinite?
4. Does the mass function come from counting, weighting, or both?
5. What is the cumulative distribution function
$$F_X(t) = P(X \le t)?$$
6. How do jumps of the CDF correspond to masses of the PMF?

Checklist for numerical summaries

For expectation and variance, use the distribution of X:

$$\mathbb{E}[X] = \sum_x xP(X = x),$$

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

For computation, one may also use

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Interpret these quantities carefully:

- expectation is a weighted average or center of mass;
- expectation need not be a possible value of X;
- variance measures average squared deviation from the expectation;
- standard deviation is the square root of variance and uses the original units of the variable.

Checklist for classical laws

Before applying a named distribution, identify the mechanism.

- Bernoulli: one success-or-failure trial.
- Binomial: number of successes in a fixed number of independent trials.
- Geometric: waiting time until the first success.
- Hypergeometric: number of successes in sampling without replacement.
- Poisson: count of events in a fixed observation window.

Do not select a distribution because a formula looks familiar. Select it because the assumptions of the mechanism match the experiment.

Summary

A discrete random variable is a numerical view of an experiment. The probability law of that variable is obtained by collecting all elementary outcomes that produce the same value and adding their probabilities. Classical distributions are reusable models of common mechanisms, not isolated formulas.

Chapter 7

Continuous Random Variables and Probability Laws

This chapter develops the continuous counterpart of discrete probability laws. The central conceptual change is that probability is no longer concentrated at individual points; instead, probabilities of intervals become the fundamental objects.

The chapter introduces probability density functions, cumulative distribution functions, integrals, expectation, variance, moments, quantiles, and the interpretation of probability as area under a density. Continuous models such as the uniform, exponential, normal, gamma, and beta distributions are treated as tools for modelling continuous random phenomena.

7.1 From point probabilities to interval probabilities

The passage from discrete random variables to continuous random variables is one of the most important conceptual changes in elementary probability. In the discrete case, probability can be assigned directly to separate values:

$$P(X = x).$$

For example, if X is the result of one fair die roll, then

$$P(X = 4) = \frac{1}{6}.$$

The value 4 carries a positive amount of probability mass.

For many real measurements this picture is no longer appropriate. If X is a waiting time, a height, a temperature, a measurement error, or a lifetime, then there are not just a few separated possible values. Between any two possible values there are many other possible values. The natural questions are then not usually point questions such as

$$P(X = 2.0000000000),$$

but interval questions such as

$$P(1.9 \leq X \leq 2.1),$$

or

$$P(X > 5).$$

The purpose of this chapter is to build the correct language for such questions. The main idea is simple to state but easy to misunderstand:

Summary

For a continuous random variable, probability is not concentrated at individual points. Probabilities of intervals are the basic objects.

A first example: choosing a point from an interval

Imagine that a point is chosen uniformly from the interval $[0, 1]$. Let

$$X = \text{the chosen point.}$$

The model says that all subintervals of the same length should have the same probability. Therefore the interval $[0.2, 0.5]$, whose length is 0.3, should have probability 0.3:

$$P(0.2 \leq X \leq 0.5) = 0.3.$$

Likewise,

$$P(0.7 \leq X \leq 0.9) = 0.2.$$

Now ask for the probability of one exact point, for instance $X = 0.4$. If that point had some positive probability, say $c > 0$, then every other point should also have the same probability by uniformity. But the interval $[0, 1]$ contains infinitely many points. Assigning the same positive mass to every point would make the total probability exceed 1. The only reasonable value is

$$P(X = 0.4) = 0.$$

This does not mean that the value 0.4 is impossible in the ordinary verbal sense. The model allows 0.4 as a possible value. It means that a single point has no length and therefore receives no probability mass in this continuous model.

Remark

Probability zero does not always mean logical impossibility. In continuous models, individual points may have probability zero even though they belong to the set of possible values.

Why intervals replace points

In the uniform model on $[0, 1]$, probability is proportional to length:

$$P(a \leq X \leq b) = b - a,$$

whenever $0 \leq a \leq b \leq 1$. Thus

$$P(0.1 \leq X \leq 0.4) = 0.3,$$

while

$$P(0.100 \leq X \leq 0.400) = 0.300.$$

The number of decimal places used to write the endpoints is irrelevant. What matters is the interval itself.

A single point is an interval whose length has collapsed to zero:

$$\{a\} = [a, a].$$

Therefore, in this model,

$$P(X = a) = P(a \leq X \leq a) = a - a = 0.$$

This is the first fundamental difference between the discrete and continuous cases.

Comparison with a discrete model

Suppose Y is the result of one fair die roll. Then the possible values are

$$1, 2, 3, 4, 5, 6.$$

The probability law is determined by the point probabilities

$$P(Y = k) = \frac{1}{6}, \qquad k = 1, 2, 3, 4, 5, 6.$$

An event such as $Y \in \{2, 3, 4\}$ is handled by adding point masses:

$$P(Y \in \{2, 3, 4\}) = P(Y = 2) + P(Y = 3) + P(Y = 4) = \frac{3}{6}.$$

For the uniform random point X on $[0, 1]$, the corresponding calculation is not a sum over points. The event $0.2 \le X \le 0.5$ contains infinitely many points, but each individual point has probability zero. The event has positive probability because the whole interval has positive length:

$$P(0.2 \le X \le 0.5) = 0.5 - 0.2 = 0.3.$$

Situation	Basic object	Probability is computed by
Discrete model	Points	Adding probability masses
Continuous model	Intervals	Measuring accumulated density

The phrase "accumulated density" will be made precise in the next sections. For now, the key idea is that a continuous probability law spreads probability across a continuum of values.

Endpoint conventions

In discrete probability, the difference between

$$P(Y < 4)$$

and

$$P(Y \le 4)$$

can be important, because the value 4 may have positive probability.

In continuous models with no point masses, endpoints do not change interval probabilities. For example, if X is uniformly distributed on $[0, 1]$, then

$$P(0.2 < X < 0.5) = P(0.2 \le X \le 0.5) = 0.3.$$

The only difference between the two events is the possible inclusion of the two endpoints 0.2 and 0.5, and each endpoint has probability zero.

More generally, for the continuous random variables studied in this chapter,

$$P(a < X < b) = P(a \le X \le b) = P(a < X \le b) = P(a \le X < b).$$

This is convenient, but it should not be used blindly in every probability model. It relies on the absence of point masses.

What must be learned again

The discrete chapter introduced random variables, support, probability mass functions, cumulative distribution functions, expectation, variance, moments, and classical models. The continuous chapter revisits these ideas, but several concepts must be reinterpreted:

- point probabilities are replaced by interval probabilities;
- probability mass functions are replaced by density functions;
- sums are replaced by integrals;
- probabilities become areas under curves;
- expectation becomes a continuous weighted average;
- quantiles become especially natural because intervals and cumulative probabilities are central.

The conceptual difficulty is not the notation itself. The difficulty is that the same symbols can suggest the wrong discrete intuition. For instance, the symbol $f_X(x)$ will describe a density, but $f_X(x)$ is not usually a probability. A height of a curve and an area under a curve are different objects.

Theorem

The continuous case should not be read as the discrete case with more values. It is a different way of distributing probability: probability is assigned to regions by accumulation, not to individual points by mass.

7.2 Probability density functions

The continuous analogue of a probability mass function is a probability density function. This sentence is useful, but it can also be misleading if it is read too quickly. A mass function gives probabilities of individual values:

$$p_X(x) = P(X = x).$$

A density function does not usually do that. A density describes how probability is spread along the real line so that probabilities of intervals can be obtained by integration.

Definition

A random variable X is called **continuous with density** f_X if there is a non-negative function $f_X : \mathbb{R} \to [0, \infty)$ such that for every interval $[a, b]$,

$$P(a \leq X \leq b) = \int_a^b f_X(x)\, dx.$$

The function f_X is called a **probability density function** of X.

There are two requirements hidden in this definition. First,

$$f_X(x) \geq 0$$

for every x, because probabilities cannot be negative. Second, the total area under the density must be 1:

$$\int_{-\infty}^{\infty} f_X(x)\, dx = 1.$$

This expresses the fact that X must take some real value with total probability 1.

Remark

A proposed density is not valid until it is non-negative and has total integral 1. These are the continuous analogues of the two basic properties of a probability mass function.

Density is not probability

The value $f_X(x)$ is a height. It is not, by itself, the probability that $X = x$. The probability of an interval is an area:

$$P(a \leq X \leq b) = \int_a^b f_X(x)\, dx.$$

A height becomes a probability only after it is accumulated over a region.

This distinction is essential. If $f_X(2) = 0.4$, this does not mean

$$P(X = 2) = 0.4.$$

It means that near $x = 2$, probability is being distributed with local density approximately 0.4 per unit of length. To turn this into a probability, one must choose an interval, for instance

$$P(2 \leq X \leq 2.1) = \int_2^{2.1} f_X(x)\, dx.$$

If the density is nearly constant on that small interval, this probability is approximately

$$0.4 \cdot 0.1 = 0.04.$$

Example: uniform density on $[0, 1]$

For a point chosen uniformly from $[0, 1]$, the density is

$$f_X(x) = \begin{cases} 1, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

The total area is

$$\int_{-\infty}^{\infty} f_X(x)\, dx = \int_0^1 1\, dx = 1.$$

For an interval $[a, b] \subseteq [0, 1]$,

$$P(a \leq X \leq b) = \int_a^b 1\, dx = b - a.$$

Thus

$$P(0.2 \leq X \leq 0.5) = \int_{0.2}^{0.5} 1\, dx = 0.3.$$

This reproduces the length interpretation from the previous section.

Example: uniform density on $[2, 6]$

Suppose X is uniformly distributed on $[2, 6]$. The interval has length 4, so the density must have constant height $1/4$ on that interval:

$$f_X(x) = \begin{cases} \frac{1}{4}, & 2 \leq x \leq 6, \\ 0, & \text{otherwise.} \end{cases}$$

The total area is

$$\int_2^6 \frac{1}{4}\,dx = \frac{1}{4}(6-2) = 1.$$

For example,

$$P(3 \le X \le 5) = \int_3^5 \frac{1}{4}\,dx = \frac{1}{2}.$$

The height $1/4$ is not a probability. The probability $1/2$ comes from the area of a rectangle with width 2 and height $1/4$.

Can a density be larger than one?

Yes. A density value can be larger than 1. This often surprises students because probabilities cannot be larger than 1. But density is not probability.

For example, consider the uniform distribution on $[0, 0.2]$. The interval has length 0.2, so the constant density must be

$$f_X(x) = 5,$$

for $0 \le x \le 0.2$. The total area is still

$$\int_0^{0.2} 5\,dx = 5 \cdot 0.2 = 1.$$

The height is 5, but no probability has exceeded 1. The probability of a smaller interval, say $[0, 0.05]$, is

$$P(0 \le X \le 0.05) = \int_0^{0.05} 5\,dx = 0.25.$$

Remark

The units of a density are "probability per unit of x". If x is measured in hours, the density is measured per hour. This is another reason why a density height is not itself a probability.

Example: a triangular density

Let

$$f_X(x) = \begin{cases} 2x, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

This is a valid density because it is non-negative and

$$\int_0^1 2x\,dx = \left[x^2\right]_0^1 = 1.$$

The density is small near 0 and larger near 1, so values close to 1 are more likely to occur than values close to 0, in the sense that intervals near 1 have larger probability than intervals of the same length near 0.

For example,

$$P(0 \le X \le 0.5) = \int_0^{0.5} 2x\,dx = 0.25,$$

whereas

$$P(0.5 \le X \le 1) = \int_{0.5}^1 2x\,dx = 0.75.$$

Both intervals have the same length, but the second has larger probability because the density is higher there.

What the graph of a density tells us

A density graph should be read by areas, not by point heights alone. High parts of the graph indicate regions where probability accumulates quickly. Low parts indicate regions where probability accumulates slowly. Regions outside the support, where the density is zero, receive no probability.

When looking at a density, ask:

- where is the density non-zero?
- is the density constant, increasing, decreasing, or concentrated in a particular region?
- does the total area under the curve equal 1?
- which intervals should have large probability?
- which intervals should have small probability?

Theorem

For a continuous random variable with density f_X, probabilities are computed by area:

$$P(X \in A) = \int_A f_X(x)\,dx$$

for intervals and, more generally, for sets for which the integral is defined. The value $f_X(x)$ is a density height, not a point probability.

7.3 Probability as area under a density

A density becomes useful only when it is integrated. The integral collects the small contributions of density over an interval. Geometrically, this integral is an area under the density curve.

If X has density f_X, then

$$P(a \le X \le b) = \int_a^b f_X(x)\,dx.$$

This formula is the continuous counterpart of adding probability masses in the discrete case.

Sums and integrals

For a discrete random variable, probabilities are obtained by summing masses:

$$P(X \in A) = \sum_{x \in A} p_X(x).$$

For a continuous random variable with density, probabilities are obtained by integrating density:

$$P(X \in A) = \int_A f_X(x)\,dx.$$

The two formulas have the same purpose: both accumulate probability over a set of values. The difference is the way probability is distributed.

Case	Local description	Accumulation rule
Discrete	mass $p_X(x)$ at points	sum masses
Continuous	density $f_X(x)$ over intervals	integrate density

Small intervals and local approximation

Suppose f_X is continuous near a point x. For a small positive number h, the probability of the interval $[x, x+h]$ is

$$P(x \le X \le x+h) = \int_x^{x+h} f_X(t)\, dt.$$

If the interval is very short, the density does not change much across it, so

$$\int_x^{x+h} f_X(t)\, dt \approx f_X(x)h.$$

This approximation explains why density can be read as local concentration of probability. A larger density means that probability accumulates faster per unit length near that point.

The approximation must not be confused with an equality in general. The exact probability is the integral. The product $f_X(x)h$ is a useful local approximation when h is small and the density varies slowly over the interval.

Example: constant density

Let X be uniformly distributed on $[0, 10]$. Then

$$f_X(x) = \begin{cases} \frac{1}{10}, & 0 \le x \le 10, \\ 0, & \text{otherwise.} \end{cases}$$

For the interval $[2, 7]$,

$$P(2 \le X \le 7) = \int_2^7 \frac{1}{10}\, dx = \frac{5}{10} = 0.5.$$

The probability is the area of a rectangle whose width is 5 and whose height is $1/10$.

For the shorter interval $[2, 3]$,

$$P(2 \le X \le 3) = \int_2^3 \frac{1}{10}\, dx = 0.1.$$

In a constant-density model, probability is proportional to interval length.

Example: increasing density

Let

$$f_X(x) = \begin{cases} 2x, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

The probability of the first half of the interval is

$$P(0 \le X \le 0.5) = \int_0^{0.5} 2x\, dx = 0.25.$$

The probability of the second half is

$$P(0.5 \le X \le 1) = \int_{0.5}^1 2x\, dx = 0.75.$$

The two intervals have the same length, but the second interval receives three times as much probability. The reason is not that it contains more points; both intervals contain infinitely many points. The reason is that the density is larger on the second interval.

Events formed from intervals

If an event is a union of disjoint intervals, its probability is the sum of the corresponding areas. For example, if X is uniformly distributed on $[0, 1]$, then

$$P(X \in [0.1, 0.2] \cup [0.7, 0.9]) = P(0.1 \leq X \leq 0.2) + P(0.7 \leq X \leq 0.9).$$

The intervals are disjoint, so

$$P(X \in [0.1, 0.2] \cup [0.7, 0.9]) = 0.1 + 0.2 = 0.3.$$

This is the same additivity principle used in earlier chapters. What changes is how the probability of each interval is computed.

Why single points have probability zero

For a continuous random variable with density f_X, the probability of a single point a is

$$P(X = a) = P(a \leq X \leq a) = \int_a^a f_X(x)\,dx = 0.$$

The interval has zero width, so the area is zero.

This argument is precise for random variables described by densities. It is not a statement about every possible random variable. Some random variables may have both continuous parts and point masses. In this chapter, unless stated otherwise, continuous random variables are treated through densities.

Remark

For a density model, changing or removing finitely many individual points does not change the probability of an interval. Area is not affected by individual points.

Checking probabilities by geometry

A useful way to check continuous probability calculations is to compare the area with the total area. Since the total area under a density is 1, an interval cannot have probability larger than 1. If a computed interval probability is negative or larger than 1, something is wrong.

Also, if one interval lies inside another, its probability cannot be larger:

$$[a, b] \subseteq [c, d] \quad \Longrightarrow \quad P(a \leq X \leq b) \leq P(c \leq X \leq d).$$

This monotonicity comes from the non-negativity of the density.

Summary

A density describes how probability is spread. An integral accumulates that spread over an interval. The probability is the area, not the height.

7.4 Cumulative distribution functions

The cumulative distribution function is the most universal way to describe a real-valued random variable. It works for discrete, continuous, and mixed probability laws.

Definition

For a real-valued random variable X, the **cumulative distribution function**, or **CDF**, is

$$F_X(x) = P(X \le x), \qquad x \in \mathbb{R}.$$

The value $F_X(x)$ is the probability accumulated at or below x.

CDF from a density

If X has density f_X, then

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt.$$

The variable t is a dummy integration variable. The CDF is therefore an accumulated area function.

Recovering interval probabilities

For $a < b$,

$$\{X \le b\} = \{X \le a\} \cup \{a < X \le b\},$$

and the two events on the right are disjoint. Hence

$$P(a < X \le b) = F_X(b) - F_X(a).$$

If X has a density, then point probabilities are zero, so endpoint choices do not change the result:

$$P(a \le X \le b) = F_X(b) - F_X(a).$$

Theorem

For a random variable with a density,

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt, \qquad P(a \le X \le b) = F_X(b) - F_X(a).$$

Example: uniform distribution on $[0, 1]$

For $X \sim \mathrm{Unif}(0, 1)$,

$$f_X(x) = \begin{cases} 1, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore

$$F_X(x) = \begin{cases} 0, & x < 0, \\ x, & 0 \le x \le 1, \\ 1, & x > 1. \end{cases}$$

For example,

$$P(0.2 \le X \le 0.7) = F_X(0.7) - F_X(0.2) = 0.5.$$

Example: a triangular density

Let

$$f_X(x) = \begin{cases} 2x, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$F_X(x) = \begin{cases} 0, & x < 0, \\ x^2, & 0 \leq x \leq 1, \\ 1, & x > 1. \end{cases}$$

Consequently,

$$P(0.5 \leq X \leq 1) = 1 - 0.5^2 = 0.75.$$

Basic properties of every CDF

Every cumulative distribution function satisfies:

- $0 \leq F_X(x) \leq 1$ for every x;
- F_X is non-decreasing;
- F_X is right-continuous;
- $\lim_{x\to-\infty} F_X(x) = 0$;
- $\lim_{x\to\infty} F_X(x) = 1$.

Right-continuity is a property of every CDF, not only of density models. When X has a density, its CDF is absolutely continuous and therefore continuous. At every point where f_X is continuous, the fundamental theorem of calculus gives

$$F_X'(x) = f_X(x).$$

Remark

The density is the local rate at which the CDF accumulates probability. A high density makes the CDF rise quickly; a low density makes it rise slowly.

CDF versus density

The density and the CDF describe the same distribution from different points of view. The density describes local concentration per unit length. The CDF records accumulated probability. A density need not exist for every probability law, but a CDF always exists.

Summary

The CDF is the universal description

$$F_X(x) = P(X \leq x).$$

For a density model it is obtained by integration, while interval probabilities are recovered by differences of CDF values.

7.5 Expectation as a continuous weighted average

Expectation was introduced in the discrete case as a weighted average:

$$\mathbb{E}[X] = \sum_x xP(X = x).$$

Each possible value contributes its value multiplied by its probability mass. For a continuous random variable, individual values usually have probability zero, so this formula cannot be copied literally. The idea of a weighted average remains, but the weights are supplied by density and accumulated by integration.

Definition

Let X be a continuous random variable with density f_X. If the integral is well-defined, the **expectation** of X is

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx.$$

This formula should be read as a continuous weighted average. The value x is weighted by the density near x, and the integral adds all those infinitesimal contributions.

Why the formula uses $x f_X(x)$

In a small interval near x of width dx, the probability is approximately

$$f_X(x)\, dx.$$

Values in that very small interval are approximately equal to x. Therefore the contribution of that small interval to the average is approximately

$$x f_X(x)\, dx.$$

Adding such contributions across all possible values gives

$$\int_{-\infty}^{\infty} x f_X(x)\, dx.$$

This is not a proof at the level of measure theory, but it gives the correct interpretation: expectation is still an average, and density provides the continuous weights.

Example: uniform distribution on $[0, 1]$

Let X be uniformly distributed on $[0, 1]$. Then

$$f_X(x) = 1$$

for $0 \le x \le 1$. Hence

$$\mathbb{E}[X] = \int_0^1 x\, dx = \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2}.$$

This agrees with the geometric intuition: the interval $[0, 1]$ is symmetric, and its center is $1/2$.

Example: uniform distribution on $[a, b]$

If X is uniformly distributed on $[a, b]$, then

$$f_X(x) = \frac{1}{b-a}, \qquad a \le x \le b.$$

Therefore

$$\mathbb{E}[X] = \int_a^b x \frac{1}{b-a}\, dx = \frac{1}{b-a}\left[\frac{x^2}{2}\right]_a^b.$$

Thus

$$\mathbb{E}[X] = \frac{b^2 - a^2}{2(b-a)} = \frac{(b-a)(a+b)}{2(b-a)} = \frac{a+b}{2}.$$

The expected value is the midpoint of the interval. This is not because the midpoint is the most likely value; in a continuous uniform distribution no single point has positive probability. It is the center of balance of the whole interval.

Example: triangular density

Let

$$f_X(x) = \begin{cases} 2x, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$\mathbb{E}[X] = \int_0^1 x \cdot 2x \, dx = \int_0^1 2x^2 \, dx.$$

Hence

$$\mathbb{E}[X] = \left[\frac{2x^3}{3} \right]_0^1 = \frac{2}{3}.$$

This expectation is larger than $1/2$, which matches the shape of the density. The density is higher near 1, so the distribution places more probability on larger values.

Expectation of a function of a random variable

Often we need the expectation of a transformed quantity, such as X^2, $\sqrt{X}$, or e^X. If g is a suitable function and X has density f_X, then

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) \, dx,$$

provided the integral is well-defined.

For example, if X is uniformly distributed on $[0, 1]$, then

$$\mathbb{E}[X^2] = \int_0^1 x^2 \, dx = \frac{1}{3}.$$

This is different from $(\mathbb{E}[X])^2 = (1/2)^2 = 1/4$. In general,

$$\mathbb{E}[X^2] \ne (\mathbb{E}[X])^2.$$

The first quantity averages squared values. The second squares the average. These are different operations.

Linearity of expectation

Linearity remains true in the continuous case. If the expectations exist, then

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

This property does not require independence. It is a property of averaging, not a property of separate random mechanisms.

For a single continuous random variable and constants a, b, one can see this from the integral formula:

$$\mathbb{E}[aX + b] = \int_{-\infty}^{\infty} (ax + b) f_X(x) \, dx.$$

Distributing the integral gives

$$\mathbb{E}[aX + b] = a \int_{-\infty}^{\infty} x f_X(x) \, dx + b \int_{-\infty}^{\infty} f_X(x) \, dx.$$

Since the total integral of the density is 1,

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b.$$

Expectation may not exist

The integral defining expectation must be meaningful. Some distributions have heavy tails, and the integral of $|x|f_X(x)$ may be infinite. In such cases, the expectation is not finite. This is not a technical detail to ignore: writing down the formula is not enough; one must know whether the integral converges.

In this introductory course, most examples will have finite expectation. Still, the condition is part of the mathematical definition.

Summary

For a continuous random variable, expectation is a continuous weighted average:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx.$$

The density supplies the weights, and the integral accumulates them. The interpretation is the same as in the discrete case, but sums are replaced by integrals and point masses are replaced by density.

7.6 Variance, standard deviation, and moments

Expectation describes a center of balance of a distribution. It does not describe how spread out the distribution is. Two continuous random variables can have the same expectation and very different variability. To measure spread, we use variance and standard deviation.

Deviation from the mean

Let

$$\mu = \mathbb{E}[X].$$

The quantity

$$X - \mu$$

measures how far the random variable is from its mean. But the average deviation $\mathbb{E}[X - \mu]$ is always zero whenever the expectation exists, because

$$\mathbb{E}[X - \mu] = \mathbb{E}[X] - \mu = 0.$$

Positive and negative deviations cancel each other. To measure size of deviation without cancellation, we square it.

Definition

Let X be a continuous random variable with density f_X and finite mean $\mu = \mathbb{E}[X]$. The **variance** of X is

$$\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f_X(x)\, dx,$$

provided the integral is finite. The **standard deviation** is

$$\sigma_X = \sqrt{\mathrm{Var}(X)}.$$

Variance is measured in squared units of X. Standard deviation returns to the original units, which often makes it easier to interpret.

Computational formula

Variance can also be computed using

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2,$$

provided the relevant expectations exist.

Indeed,

$$\mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2].$$

By linearity of expectation,

$$\mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2.$$

Since $\mathbb{E}[X] = \mu$, this becomes

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - 2\mu^2 + \mu^2 = \mathbb{E}[X^2] - \mu^2.$$

Remark

The formula $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ is useful for computation, but the definition $\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2]$ explains the meaning: variance is the average squared distance from the mean.

Example: uniform distribution on $[0, 1]$

Let X be uniformly distributed on $[0, 1]$. We already know that

$$\mathbb{E}[X] = \frac{1}{2}.$$

Also,

$$\mathbb{E}[X^2] = \int_0^1 x^2\,dx = \frac{1}{3}.$$

Therefore

$$\mathrm{Var}(X) = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}.$$

The standard deviation is

$$\sigma_X = \sqrt{\frac{1}{12}} = \frac{1}{2\sqrt{3}}.$$

Example: uniform distribution on $[a, b]$

For $X \sim \mathrm{Unif}(a, b)$,

$$\mathbb{E}[X] = \frac{a+b}{2}.$$

The second moment is

$$\mathbb{E}[X^2] = \int_a^b x^2 \frac{1}{b-a}\,dx = \frac{1}{b-a}\left[\frac{x^3}{3}\right]_a^b = \frac{b^3-a^3}{3(b-a)}.$$

Using

$$b^3 - a^3 = (b-a)(a^2 + ab + b^2),$$

we obtain

$$\mathbb{E}[X^2] = \frac{a^2+ab+b^2}{3}.$$

Hence

$$\operatorname{Var}(X) = \frac{a^2 + ab + b^2}{3} - \left(\frac{a+b}{2}\right)^2.$$

After simplifying,

$$\operatorname{Var}(X) = \frac{(b-a)^2}{12}.$$

This formula shows that spread depends only on the length of the interval, not on its location.

Example: triangular density

Let

$$f_X(x) = \begin{cases} 2x, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

We previously found

$$\mathbb{E}[X] = \frac{2}{3}.$$

Now

$$\mathbb{E}[X^2] = \int_0^1 x^2 \cdot 2x \, dx = \int_0^1 2x^3 \, dx = \frac{1}{2}.$$

Thus

$$\operatorname{Var}(X) = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}.$$

The density is concentrated toward the right side of the interval, so the mean is larger than 1/2, while the variance reflects the remaining spread around that shifted center.

Moments

Moments summarize different aspects of a distribution. The k-th raw moment, when it exists, is

$$\mathbb{E}[X^k] = \int_{-\infty}^{\infty} x^k f_X(x) \, dx.$$

The first raw moment is the expectation. The second raw moment helps compute variance. Higher moments describe additional features of the shape of the distribution.

Central moments are moments around the mean:

$$\mathbb{E}[(X-\mu)^k].$$

The second central moment is variance. Higher central moments are used to study asymmetry and tail behaviour, although this course mainly uses the first two moments.

What variance does and does not say

Variance measures average squared distance from the mean. A small variance means that values tend to stay close to the mean. A large variance means that values are more spread out. However, variance alone does not determine the entire shape of a distribution. Different distributions may have the same mean and variance but very different densities.

Summary

For a continuous random variable with density f_X,

$$\operatorname{Var}(X) = \int_{-\infty}^{\infty} (x-\mu)^2 f_X(x) \, dx, \qquad \mu = \mathbb{E}[X].$$

Equivalently,
$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$
Variance measures average squared spread around the mean; standard deviation is its square root and is expressed in the original units.

7.7 Quantiles and medians

Expectation and variance summarize a distribution by numerical averages. Another important way to summarize a continuous distribution is to ask where a given amount of probability has accumulated. This leads to quantiles.

The cumulative distribution function is the natural tool here. Since
$$F_X(x) = P(X \le x),$$
the value $F_X(x)$ tells us how much probability lies to the left of x. A quantile reverses this question.

Definition

Let X be a random variable with CDF F_X. A number q_p is called a p**-quantile** of X, where $0 < p < 1$, if
$$P(X \le q_p) \ge p$$
and
$$P(X \ge q_p) \ge 1 - p.$$
When the CDF is continuous and strictly increasing, this is equivalent to
$$F_X(q_p) = p.$$

The second formulation, $F_X(q_p) = p$, is often the one used in calculations. The more careful definition is included because not every distribution has a strictly increasing CDF. In this chapter, most examples are simple continuous models where the equation $F_X(q_p) = p$ gives the intended quantile.

Median

The most important quantile is the median. A median is a point that splits the distribution into two parts of probability at least one half on each side. For a continuous distribution with strictly increasing CDF, the median m satisfies
$$F_X(m) = \frac{1}{2}.$$
Equivalently,
$$P(X \le m) = \frac{1}{2}.$$

The median is not the same concept as the expectation. The expectation is a center of balance. The median is a point of probability balance: half of the probability lies to the left and half to the right.

Example: uniform distribution on $[0, 1]$

For $X \sim \mathrm{Unif}(0, 1)$, the CDF is
$$F_X(x) = x,$$

for $0 \leq x \leq 1$. The p-quantile satisfies

$$q_p = p.$$

For example, the median is

$$q_{0.5} = 0.5,$$

and the first and third quartiles are

$$q_{0.25} = 0.25, \qquad q_{0.75} = 0.75.$$

This agrees with the interpretation of probability as length.

Example: triangular density

Let

$$f_X(x) = \begin{cases} 2x, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

The CDF on $[0, 1]$ is

$$F_X(x) = x^2.$$

The p-quantile solves

$$x^2 = p.$$

Since $x \geq 0$,

$$q_p = \sqrt{p}.$$

The median is therefore

$$q_{0.5} = \sqrt{0.5} = \frac{1}{\sqrt{2}} \approx 0.707.$$

This median is greater than 0.5. That makes sense because the density is larger near 1, so probability accumulates more slowly at first and more quickly later.

Expectation versus median

For symmetric distributions, expectation and median often coincide. For skewed distributions, they may differ.

In the triangular example with density $f_X(x) = 2x$ on $[0, 1]$, we found

$$\mathbb{E}[X] = \frac{2}{3} \approx 0.667,$$

while the median is

$$\frac{1}{\sqrt{2}} \approx 0.707.$$

These numbers are close but not equal. They summarize the distribution in different ways. The expectation balances the density as a physical mass would be balanced. The median divides probability into two equal halves.

Quartiles and interquartile range

The quartiles are quantiles corresponding to probabilities 0.25, 0.5, and 0.75. The first quartile Q_1 satisfies approximately

$$F_X(Q_1) = 0.25,$$

the median Q_2 satisfies

$$F_X(Q_2) = 0.5,$$

and the third quartile Q_3 satisfies

$$F_X(Q_3) = 0.75.$$

The interquartile range is

$$IQR = Q_3 - Q_1.$$

It measures the length of the central interval containing about half of the probability.

These ideas will return when empirical distributions and descriptive statistics are studied. There, quantiles are computed from data. Here, quantiles are computed from a theoretical distribution.

Reading quantiles as probability statements

A quantile should always be interpreted probabilistically. If $q_{0.9} = 12$, then the statement means that approximately 90% of the probability lies at or below 12, and approximately 10% lies above 12, assuming a continuous model with no ambiguity at the quantile.

It does not mean that 12 is likely as an exact value. In a continuous model, $P(X = 12) = 0$ when X has a density. The quantile is important because of the area accumulated up to it.

Summary

Quantiles invert the cumulative distribution function. For continuous models with strictly increasing CDF,

$$F_X(q_p) = p.$$

The median corresponds to $p = 0.5$. Quantiles describe probability positions, not point probabilities.

7.8 Classical continuous laws as modelling tools

Classical continuous distributions should not be learned as isolated formulas. Each distribution is a mathematical model for a certain kind of random phenomenon. The formula for the density is important, but the modelling idea is more important: what mechanism or structure does the distribution represent?

This section introduces several important continuous laws at an interpretive level. The aim is not to exhaust all their properties, but to understand why such models appear and what questions they help answer.

Uniform distribution

A random variable has a uniform distribution on $[a, b]$ if probability is spread evenly over the interval. Its density is

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise.} \end{cases}$$

The total area is

$$\int_a^b \frac{1}{b-a}\,dx = 1.$$

For any subinterval $[c, d] \subseteq [a, b]$,

$$P(c \le X \le d) = \frac{d-c}{b-a}.$$

Thus probability is proportional to length.

The uniform distribution is appropriate when the model assumes that no region of the interval is favoured over another. This assumption should be justified by the context; it should not be used simply because the interval is known.

Exponential distribution

The exponential distribution is commonly used to model waiting times. Its density with rate parameter $\lambda > 0$ is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

The density is largest at 0 and decreases as x increases. This means that short waiting times are more likely than long waiting times, although long waits remain possible.

The CDF is

$$F_X(x) = 1 - e^{-\lambda x}, \qquad x \geq 0.$$

Therefore

$$P(X > t) = 1 - F_X(t) = e^{-\lambda t}.$$

The parameter λ controls the scale of the waiting time. Larger λ means that the density decreases more quickly and waiting times tend to be shorter.

The exponential distribution is closely connected with the idea of a constant instantaneous rate. In later applications, it appears as a model for time until a random event occurs under a stable rate assumption.

Normal distribution

The normal distribution is one of the central continuous models. Its density has the form

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad x \in \mathbb{R},$$

where $\mu \in \mathbb{R}$ and $\sigma > 0$.

The parameter μ is the center of the distribution, and σ controls its spread. The density is symmetric around μ. Values close to μ have higher density than values far away from μ.

The normal distribution is used when many small sources of variation combine or when a distribution is approximately symmetric and bell-shaped. Its deeper importance comes from limit theorems, especially the central limit theorem. This connection will be developed later; for now, the normal distribution should be read as a smooth, symmetric model for continuous variation.

Remark

The normal density has no elementary antiderivative. Probabilities for normal random variables are usually computed using tables, software, or standardized normal distribution functions.

Gamma distribution

The gamma distribution is a flexible model on $[0, \infty)$. It is often used for positive quantities such as waiting times, lifetimes, or accumulated amounts. One common parametrization has density

$$f_X(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \qquad x > 0,$$

where $\alpha > 0$ is a shape parameter and $\lambda > 0$ is a rate parameter. Here $\Gamma(\alpha)$ is the gamma function, which generalizes factorials.

When $\alpha = 1$, the gamma distribution becomes the exponential distribution. For larger shape parameters, the density can rise from zero, reach a peak, and then decay. This makes the gamma family more flexible than the exponential family.

A useful interpretation is that certain gamma distributions model the waiting time until several events have occurred, not just the first one. Thus the gamma law extends the waiting-time idea behind the exponential distribution.

Beta distribution

The beta distribution is a flexible family of distributions on $[0,1]$. Its density has the form

$$f_X(x) = \frac{1}{B(\alpha,\beta)} x^{\alpha-1}(1-x)^{\beta-1}, \qquad 0 < x < 1,$$

where $\alpha > 0$ and $\beta > 0$. The constant $B(\alpha,\beta)$ normalizes the density so that the total area is 1.

The beta distribution is useful for modelling proportions, probabilities, and other quantities constrained between 0 and 1. Depending on the parameters, the density can be concentrated near 0, near 1, near the middle, or even near both endpoints.

For example, the uniform distribution on $[0,1]$ is a special beta distribution with $\alpha = 1$ and $\beta = 1$. In that case,

$$f_X(x) = 1, \qquad 0 < x < 1.$$

Choosing a model

A distribution should be chosen because its assumptions match the phenomenon being modelled. The following questions are useful:

- What values are possible? Are they all real numbers, only positive numbers, or values between 0 and 1?
- Is the distribution expected to be symmetric or skewed?
- Are extreme values plausible or very unlikely?
- Is the variable a measurement, a waiting time, a proportion, or an accumulated quantity?
- Which assumptions justify the chosen family?

Distribution	Support	Typical modelling idea
Uniform	bounded interval	no region favoured
Exponential	$[0,\infty)$	waiting time to first event
Normal	$\mathbb{R}$	symmetric continuous variation
Gamma	$[0,\infty)$	positive waiting or accumulated quantity
Beta	$[0,1]$	proportion or probability-like quantity

Summary

Classical continuous laws are modelling tools. Their densities are formulas, but the formulas should be connected to support, shape, parameters, and the mechanism or interpretation that makes the model appropriate.

7.9 Final checklist

This chapter introduced continuous random variables as probability models in which probability is spread over intervals rather than concentrated at individual points. The following checklist summarizes the ideas that should be understood before moving on.

Conceptual understanding

You should be able to explain the following points in words:

- why a continuous random variable can satisfy $P(X = x) = 0$ for every individual value x, while still taking values in a whole interval;
- why probabilities of intervals, not probabilities of individual points, are the fundamental objects in continuous models;
- why a density value $f_X(x)$ is not usually a probability;
- why probability is represented by area under a density curve;
- why changing endpoints usually does not change probabilities in continuous density models;
- how the continuous case parallels the discrete case, and where the analogy must be treated carefully.

Definitions and formulas

You should know the meaning of the following formulas:

$$P(a \leq X \leq b) = \int_a^b f_X(x)\,dx,$$

$$\int_{-\infty}^{\infty} f_X(x)\,dx = 1,$$

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(t)\,dt,$$

$$P(a \leq X \leq b) = F_X(b) - F_X(a),$$

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\,dx,$$

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

You should also remember that these formulas require the relevant integrals to exist. In particular, expectation and variance are not guaranteed to be finite for every possible distribution.

Typical tasks

After studying this chapter, you should be able to:

- check whether a proposed function is a valid density;
- compute interval probabilities by integrating a density;
- compute interval probabilities by using a CDF;
- derive a CDF from a density in simple examples;
- recover a density from a differentiable CDF;
- compute expectation and variance for simple continuous distributions;
- find quantiles by solving $F_X(q_p) = p$ when the CDF is strictly increasing;
- interpret the support, shape, and parameters of a continuous model.

Common mistakes

The most common mistakes in this topic are not algebraic; they are conceptual. Avoid the following errors:

- treating $f_X(x)$ as if it were $P(X = x)$;
- thinking that a density cannot be larger than 1;
- forgetting to check that the total area under a density is 1;
- using sums where integrals are needed;
- assuming that probability zero always means impossibility;
- confusing the expectation with the most likely value or with the median;
- choosing a classical distribution by name without checking whether its support and modelling assumptions match the situation.

Questions for self-explanation

A good test of understanding is whether you can answer the following questions without immediately searching for a formula:

1. If $P(X = x) = 0$ for every x, how can X still have total probability 1?
2. What is the difference between the height of a density and the area under a density?
3. Why does $P(a < X < b) = P(a \le X \le b)$ for the density models studied here?
4. How is a CDF built from a density?
5. How is an interval probability read from a CDF?
6. Why is expectation an average even though no point has positive probability?
7. What does variance measure geometrically or intuitively?
8. What modelling situation might suggest a uniform, exponential, normal, gamma, or beta distribution?

Summary

The essential message of the chapter is this: in continuous probability, point probabilities disappear, but probability itself does not disappear. It is accumulated over intervals as area under a density, and the familiar ideas of distribution, expectation, variance, and quantiles are rebuilt using integrals.

Chapter 8

Empirical Distributions and Descriptive Statistics

This chapter studies empirical distributions and descriptive statistics obtained from data. The concepts developed earlier for theoretical probability laws are now examined through observed datasets.

Frequency tables, histograms, empirical distribution functions, means, medians, variances, standard deviations, quantiles, and related descriptive measures are introduced as tools for describing and understanding data.

8.1 From probability laws to observed data

The previous chapters studied probability laws as mathematical models. A random variable X had a probability law described by a probability mass function, a density function, or a cumulative distribution function. We defined quantities such as

$$\mathbb{E}[X], \qquad \mathrm{Var}(X), \qquad \sigma_X,$$

and we interpreted them as properties of the probability model.

This chapter changes the point of view. We now begin with data: a finite list of observed values. The central question is not only what model could have produced those values, but also how to describe the values that have actually been observed.

Summary

A probability law is an ideal mathematical object. A dataset is a finite list of observations. Descriptive statistics are numerical and graphical summaries computed from that finite list.

This distinction is essential. If a model says that

$$X \sim N(0, 1),$$

then the model has theoretical expectation 0 and theoretical variance 1. But an observed sample generated from that model might be

$$-0.31, \quad 0.84, \quad 1.12, \quad -1.45, \quad 0.07.$$

The average of these five observed values will not usually be exactly 0, and their empirical variance will not usually be exactly 1. This is not a contradiction. The model describes the random mechanism; the sample is one finite outcome of that mechanism.

The same words, but a different status

Many words from the previous chapters appear again:

- distribution,
- mean,

- variance,
- standard deviation,
- quantile,
- cumulative distribution function.

However, their status changes. Earlier, they were properties of a probability law. Now they are summaries computed from data.

For example, the theoretical expectation

$$\mathbb{E}[X]$$

is a property of the random variable X. The sample mean

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

is a number computed from observed values

$$x_1, x_2, \ldots, x_n.$$

These two objects are related, but they are not the same object.

Remark

A common mistake is to write as if the sample mean $\bar{x}$ were the same as $\mathbb{E}[X]$. The sample mean is computed from data. The expectation belongs to a probability model.

Why empirical summaries matter

Empirical summaries serve several purposes. They help us see the structure of a dataset, detect unusual observations, compare groups, and form questions about possible probability models. They also prepare the ground for statistical inference, where one uses data to reason about an unknown population or an unknown model parameter.

At this stage, however, we are not yet proving inferential statements. We are learning the language of data description. The goal is to understand what a summary measures before using it for estimation or decision making.

A small example

Suppose the observed waiting times, in minutes, are

$$2.1, \quad 3.4, \quad 1.8, \quad 4.0, \quad 2.7.$$

From these five numbers, we can compute a sample mean, a sample variance, a median, and other summaries. We can also make a histogram or an empirical cumulative distribution function.

But these five numbers are not themselves an exponential distribution, a normal distribution, or any other theoretical distribution. They are data. A theoretical model may later be proposed to explain or approximate how such data arise. The empirical summaries describe what has been observed.

The guiding comparison

Throughout this chapter, we repeatedly compare theoretical and empirical objects:

Theoretical probability	Empirical data	Interpretation
Probability law of X	empirical distribution	how mass is assigned
CDF F_X	empirical CDF F_n	accumulated probability or proportion
$\mathbb{E}[X]$	sample mean $\bar{x}$	center
$\mathrm{Var}(X)$	sample variance	spread
Theoretical quantile	empirical quantile	probability position
Density	histogram estimate	shape of distribution

The table is only a guide. It does not say that the empirical object is equal to the theoretical object. It says that empirical summaries are data-based counterparts of the theoretical concepts developed earlier.

Theorem

The main conceptual task of this chapter is to keep two levels separate while understanding their relationship: the level of the probability model and the level of the observed sample.

8.2 Samples and observations

A dataset is usually written as a finite list

$$x_1, x_2, \ldots, x_n.$$

The number n is called the sample size. Each x_i is one observed value. For example, if five waiting times are recorded, then a dataset might be

$$2.1, \quad 3.4, \quad 1.8, \quad 4.0, \quad 2.7.$$

Here $n = 5$, and each number is one observation.

Definition

A **sample** is a finite collection of observed values

$$x_1, x_2, \ldots, x_n.$$

The values x_i are called **observations**, and n is the **sample size**.

This definition is deliberately simple. At the descriptive level, the data are already observed. We can sort them, count them, average them, plot them, and summarize them.

Random variables before observation, numbers after observation

In probability modelling, before data are observed, we often write

$$X_1, X_2, \ldots, X_n.$$

These symbols represent random variables. After observation, their realized values are written as

$$x_1, x_2, \ldots, x_n.$$

This distinction is small in notation but large in meaning.

Notation	Meaning	Level
X_i	random variable before observation	model
x_i	observed numerical value	data

For instance, before measuring tomorrow's temperature at noon, the temperature can be modelled as a random variable X. After the measurement is made, the recorded value, say 18.6, is a number. The random variable describes uncertainty before observation; the observed value is what actually appeared.

Remark

The same letter in uppercase and lowercase often marks this difference: uppercase for the random variable, lowercase for the observed value.

Samples from a population

In applications, data are often interpreted as observations from a larger population. The population may be a real finite population, such as all students in a university, or an idealized population represented by a probability model.

For example:

- a sample of heights may be taken from a population of people;
- a sample of delivery times may be taken from a logistics process;
- a sample of simulated values may be generated from a normal distribution;
- a sample of measurement errors may be collected from repeated measurements of the same quantity.

In each case, the observed data are finite. The population or probability model is usually larger than the sample and may not be fully known.

Independent and identically distributed observations

A common modelling assumption is that observations arise from random variables

$$X_1, X_2, \ldots, X_n$$

that are independent and have the same probability law. This is often abbreviated as **i.i.d.**, meaning **independent and identically distributed**.

This assumption means two things:

- the observations are produced by the same probabilistic mechanism;
- knowing one observation does not change the probability law of the others.

The i.i.d. assumption is powerful, but it is an assumption. It should not be silently imposed on every dataset. Measurements over time may be dependent. People in the same household may be similar. Repeated measurements may be affected by the same instrument error. The model must be checked against the context.

Real data and computer-generated data

This chapter uses both real-data language and computer-generated samples. A computer-generated sample is not real-world evidence by itself, but it is useful for learning. It allows us to see how definitions behave when the underlying model is known.

For example, a computer may generate values from a uniform distribution on $[0, 1]$. Then we know the theoretical CDF is

$$F(x) = x, \qquad 0 \le x \le 1.$$

After generating a finite sample, we can compute the empirical CDF and compare it with the theoretical one. This comparison helps us understand the relationship between a model and a finite sample from that model.

Summary

At the model level, we work with random variables $X_1, \ldots, X_n$. At the data level, we work with observed values $x_1, \ldots, x_n$. Descriptive statistics are computed from the observed values, while probability laws describe the random mechanism that may have produced them.

8.3 Frequency tables and empirical probabilities

For discrete or grouped data, the first natural summary is a frequency table. A frequency table records how often each value, or each class of values, occurs in the dataset.

Suppose the observed values are

$$2, \quad 3, \quad 2, \quad 5, \quad 3, \quad 3, \quad 4, \quad 2.$$

There are $n = 8$ observations. The value 2 appears three times, the value 3 appears three times, the value 4 appears once, and the value 5 appears once.

Value a	Count n_a	Relative frequency n_a/n
2	3	3/8
3	3	3/8
4	1	1/8
5	1	1/8

Definition

For a dataset $x_1, \ldots, x_n$, the **frequency** of a value a is

$$n_a = \#\{i : x_i = a\}.$$

The **relative frequency** of a is

$$\frac{n_a}{n}.$$

The relative frequencies add to 1, because every observation contributes to exactly one observed value:

$$\sum_a \frac{n_a}{n} = 1.$$

This makes relative frequencies look like probabilities. Indeed, they define an empirical probability law on the observed values.

Empirical probability

The empirical probability of observing the value a in the dataset is the relative frequency

$$P_n(\{a\}) = \frac{n_a}{n}.$$

The notation P_n emphasizes that this probability is built from a sample of size n. It is not necessarily the true probability from the population or from the theoretical model.

For the example above,

$$P_n(\{2\}) = \frac{3}{8}, \qquad P_n(\{4\}) = \frac{1}{8}.$$

These are exact statements about the observed data. They do not prove that the true probability of observing 2 is $3/8$. They say that $3/8$ of this particular sample consists of the value 2.

Remark

Relative frequency is data-based. Probability in a model is theory-based. The relationship between them is one of the central themes of statistics, but the two objects should not be identified without explanation.

Grouped frequencies

For continuous data, exact repeated values may be rare. If waiting times are recorded with many decimal places, it may happen that no two observations are exactly equal. A table of exact values is then not very informative.

Instead, values are often grouped into intervals. For example, suppose waiting times are grouped into bins

$$[0, 2), \quad [2, 4), \quad [4, 6), \quad [6, 8).$$

For each bin, we count how many observations fall inside it:

$$n_j = \#\{i : a_j \le x_i < a_{j+1}\}.$$

The relative frequency of the bin is

$$\frac{n_j}{n}.$$

This is the empirical proportion of observations in that interval.

Connection with probability laws

If a theoretical model says that a random variable X has probability law P, then the model probability of a set A is

$$P(X \in A).$$

For a dataset, the empirical counterpart is

$$P_n(A) = \frac{\#\{i : x_i \in A\}}{n}.$$

Thus, for an interval $[a, b]$,

$$P_n([a, b]) = \frac{\#\{i : a \le x_i \le b\}}{n}.$$

This number tells us what proportion of the observed data lies in the interval. It is the data-based analogue of the theoretical probability

$$P(a \le X \le b).$$

A warning about small samples

In a small sample, relative frequencies can be unstable. If a coin is tossed ten times and heads appears seven times, the relative frequency of heads is 0.7. This does not by itself prove that the probability of heads is 0.7. It says that in this sample, heads appeared in 70% of the trials.

As sample size increases, relative frequencies often stabilize under appropriate probabilistic assumptions. The mathematical explanation of that stabilization belongs to later chapters on sampling and limit theorems. Here we use relative frequencies descriptively.

Summary

A frequency counts observations. A relative frequency divides that count by the sample size. Relative frequencies form an empirical probability law on the data, but they should not be confused with the unknown probabilities of the population or model.

8.4 The empirical distribution

A dataset can be turned into a probability distribution of its own. This does not mean that the dataset becomes the true population law. It means that we define a probability measure that gives equal weight to each observed data point.

Suppose the data are

$$x_1, x_2, \ldots, x_n.$$

The empirical distribution assigns mass $1/n$ to each observation. If the same value occurs several times, the masses at that value add together.

Definition

For a dataset $x_1, \ldots, x_n$, the **empirical distribution** P_n is defined by

$$P_n(A) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{\{x_i \in A\}}$$

for every set A of values under consideration.

The indicator $\mathbf{1}_{\{x_i \in A\}}$ equals 1 if the observation x_i lies in A, and equals 0 otherwise. Therefore the sum counts how many observations fall in A. Dividing by n gives the observed proportion.

Example: repeated values

Consider the dataset

$$2, \quad 3, \quad 2, \quad 5, \quad 3, \quad 3, \quad 4, \quad 2.$$

The empirical distribution assigns mass $1/8$ to each observation. Since the value 2 appears three times, its total empirical mass is

$$P_n(\{2\}) = \frac{3}{8}.$$

Similarly,

$$P_n(\{3\}) = \frac{3}{8}, \qquad P_n(\{4\}) = \frac{1}{8}, \qquad P_n(\{5\}) = \frac{1}{8}.$$

The empirical distribution is therefore a discrete probability law concentrated on the observed values.

A continuous sample still gives a discrete empirical distribution

Suppose the data are measurements from a continuous process:

$$1.82, \quad 2.17, \quad 1.96, \quad 2.04, \quad 2.31.$$

Even if the theoretical model is continuous, the empirical distribution of this finite dataset is discrete. It puts mass $1/5$ at each of the five observed numbers.

This is an important point. A continuous model may generate data, but once we observe a finite sample, the empirical distribution is a finite discrete object. The continuity belongs to the model, not to the finite list of observed values.

Remark

Every finite empirical distribution is discrete. This remains true even when the data are measurements from a continuous phenomenon.

Relation to the theoretical distribution

If data are generated from a random variable X with theoretical probability law P, then P describes the mechanism before observation. The empirical distribution P_n describes the observed sample after observation.

For a set A, compare

$$P(X \in A)$$

with

$$P_n(A) = \frac{\#\{i : x_i \in A\}}{n}.$$

The first is a model probability. The second is an observed proportion. If the model is correct and the sample is large, the observed proportion may be close to the model probability. But closeness is not automatic for every finite sample.

Empirical distribution as a model of the data

The empirical distribution is a perfect probability model for the observed data in the following limited sense: if one chooses one of the recorded observations uniformly at random, then the probability of choosing a value in A is exactly $P_n(A)$.

For example, if the dataset has 100 observations and 23 of them lie in the interval $[0, 1]$, then choosing one observation uniformly at random from the recorded dataset gives probability

$$\frac{23}{100}$$

of selecting an observation in $[0, 1]$.

This interpretation is useful, but it is not the same as saying that the true population probability of $[0, 1]$ is $23/100$. The empirical distribution is about the sample.

Summary

The empirical distribution assigns mass $1/n$ to each observation. It is the probability law of a randomly selected observed data point. It summarizes the sample exactly, while a theoretical distribution describes an idealized random mechanism that may have generated the sample.

8.5 The empirical distribution function

The cumulative distribution function of a random variable is

$$F_X(x) = P(X \leq x).$$

It tells us how much model probability lies to the left of x. For a dataset, the analogous object is the empirical distribution function. It tells us what proportion of the observed data lies to the left of x.

Definition

For a dataset $x_1, \ldots, x_n$, the **empirical distribution function**, or **empirical CDF**, is

$$F_n(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{\{x_i \leq x\}}.$$

The value $F_n(x)$ is the proportion of observations not larger than x. It is the CDF of the empirical distribution.

A hand calculation

Consider the dataset

$$2, \quad 4, \quad 4, \quad 7, \quad 9.$$

Here $n = 5$. Let us compute several values of F_n.

For $x = 3$, only the observation 2 is at most 3, so

$$F_n(3) = \frac{1}{5}.$$

For $x = 4$, the observations $2, 4, 4$ are at most 4, so

$$F_n(4) = \frac{3}{5}.$$

For $x = 8$, the observations $2, 4, 4, 7$ are at most 8, so

$$F_n(8) = \frac{4}{5}.$$

For $x = 10$, all observations are at most 10, so

$$F_n(10) = 1.$$

Step function interpretation

The empirical CDF is a step function. It is constant between observed values and jumps at observed values. In the example above, it jumps at 2, 4, 7, and 9. At the value 4, it jumps by $2/5$, because two observations are equal to 4.

This is the graphical form of the empirical distribution. Each observation contributes mass $1/n$, and the empirical CDF accumulates that mass from left to right.

Remark

The empirical CDF is always a step function for a finite dataset. The theoretical CDF of a continuous model may be smooth, but the empirical CDF of a finite sample from that model is still a step function.

Comparing F_n with F_X

If the data are generated from a theoretical distribution with CDF F_X, then we can compare the empirical CDF F_n with F_X. The two functions answer related but different questions:

$$F_X(x) = P(X \le x)$$

is a model probability, while

$$F_n(x) = \frac{\#\{i : x_i \le x\}}{n}$$

is an observed proportion.

For example, if the theoretical model is uniform on $[0, 1]$, then

$$F_X(x) = x, \qquad 0 \le x \le 1.$$

A sample of size 20 from this model gives an empirical CDF with jumps of size 1/20. It will not usually equal the line $F_X(x) = x$, but it may be close in many places. A larger sample often gives a closer visual approximation.

This observation is important, but at this stage it remains descriptive. Later, limit theorems will explain why empirical summaries stabilize under suitable assumptions.

Interval proportions from the empirical CDF

Just as a theoretical CDF gives interval probabilities by differences, the empirical CDF gives interval proportions. For $a < b$,

$$F_n(b) - F_n(a)$$

is the proportion of observations satisfying

$$a < x_i \le b.$$

Thus

$$F_n(b) - F_n(a) = \frac{\#\{i : a < x_i \le b\}}{n}.$$

Endpoint conventions matter for empirical data because observations can be exactly equal to endpoints. This is different from continuous density models, where individual endpoints usually have probability zero.

Summary

The empirical CDF is

$$F_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{\{x_i \le x\}}.$$

It is the observed proportion of data not exceeding x. It is the data-based counterpart of the theoretical CDF $F_X(x) = P(X \le x)$.

8.6 Histograms and binning

A histogram is a graphical summary of data. It is especially useful for continuous or nearly continuous observations, where exact repeated values may be rare and a table of exact values may not show the shape of the data clearly.

The basic idea is to divide the range of the data into intervals, called bins, and count how many observations fall into each bin.

Bins and counts

Suppose the bins are

$$[a_0, a_1), \quad [a_1, a_2), \quad \ldots, \quad [a_{m-1}, a_m).$$

The count in the j-th bin is

$$n_j = \#\{i : a_{j-1} \leq x_i < a_j\}.$$

The relative frequency in that bin is

$$\frac{n_j}{n}.$$

This is the proportion of observations that fall in the bin.

For example, if $n = 100$ and 18 observations fall in the interval $[2, 4)$, then the relative frequency of that bin is

$$\frac{18}{100} = 0.18.$$

This means that 18% of the observed data lie in that interval.

Frequency histograms and relative-frequency histograms

A frequency histogram uses bar heights equal to bin counts n_j. A relative-frequency histogram uses bar heights equal to n_j/n. Both are useful, but they answer slightly different visual questions.

Frequency histograms show how many observations are in each bin. Relative- frequency histograms show what proportion of the sample is in each bin. When comparing samples of different sizes, relative frequencies are usually easier to compare than raw counts.

Density-scaled histograms

When a histogram is compared with a probability density, the bar heights should usually be scaled so that bar areas represent relative frequencies. If the j-th bin has width

$$\Delta_j = a_j - a_{j-1},$$

then the density-scaled height is

$$h_j = \frac{n_j}{n\Delta_j}.$$

The area of the bar is then

$$h_j \Delta_j = \frac{n_j}{n}.$$

Thus the area of the bar equals the empirical probability of that bin.

Remark

A density-scaled histogram should be read by areas, just as a probability density is read by areas. The height of a bar depends on the bin width; the area of a bar represents the proportion of observations in that bin.

Histogram versus density

A histogram is computed from data. A density is part of a theoretical probability model. They may be compared, but they are not the same object.

For example, if a sample is generated from a normal distribution, its histogram may look bell-shaped. The normal density is a smooth theoretical curve. The histogram is a finite-sample visual summary. If another sample is generated from the same model, the histogram will usually look slightly different.

This variability is not an error. It is the natural effect of finite sampling.

Object	Built from	Meaning
Histogram	observed data	empirical visual summary
Density	probability model	theoretical distribution shape

The choice of bins matters

A histogram depends on the choice of bins. Wide bins produce a smoother but less detailed picture. Narrow bins show more detail but may look irregular, especially for small samples.

Therefore a histogram is not a purely mechanical truth about a dataset. It is a summary that depends on a modelling choice: how the data are grouped. Good bin choices make important structure visible without creating misleading patterns.

A computer experiment

A useful way to understand histograms is to generate samples from a known model. For example, one may generate 1000 values from a standard normal distribution and draw a density-scaled histogram. The histogram will usually resemble the normal density, but it will not be identical to it.

A typical Python-style workflow is:

```
import numpy as np

sample = np.random.normal(loc=0, scale=1, size=1000)
sample_mean = np.mean(sample)
sample_std = np.std(sample)
```

The computer experiment does not prove a theorem. It creates a controlled example where the theoretical model is known, so that the relationship between a finite sample and the underlying distribution can be observed.

Summary

A histogram groups data into bins. A density-scaled histogram is designed so that bar areas equal relative frequencies. Histograms are empirical and depend on bin choice; densities are theoretical objects from probability models.

8.7 The sample mean as an empirical expectation

The theoretical expectation $\mathbb{E}[X]$ is a property of a probability law. It was introduced as a weighted average of possible values. For observed data, the corresponding quantity is the sample mean.

Definition

For a dataset $x_1, x_2, \ldots, x_n$, the **sample mean** is

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

The sample mean adds all observed values and divides by the number of observations. It is a center of the observed data.

A small example

For the data

$$2, \quad 4, \quad 4, \quad 7, \quad 9,$$

the sample mean is

$$\bar{x} = \frac{2+4+4+7+9}{5} = \frac{26}{5} = 5.2.$$

The number 5.2 is not one of the observed values. This is not a problem. The mean is a balance point of the data, not necessarily a value that appears in the dataset.

Connection with the empirical distribution

The sample mean can be interpreted as the expectation with respect to the empirical distribution. If the empirical distribution assigns mass $1/n$ to each observation, then its expectation is

$$\sum_{i=1}^{n} x_i \frac{1}{n} = \frac{1}{n} \sum_{i=1}^{n} x_i = \bar{x}.$$

Thus the sample mean is not a new mysterious formula. It is the same idea of expectation applied to the empirical distribution of the data.

If the data have repeated values, the same idea can be written using frequencies. Suppose the distinct observed values are $a_1, \dots, a_k$, with counts $n_{a_1}, \dots, n_{a_k}$. Then

$$\bar{x} = \sum_{j=1}^{k} a_j \frac{n_{a_j}}{n}.$$

This is a weighted average of the distinct observed values, where the weights are relative frequencies.

Sample mean versus theoretical expectation

The sample mean $\bar{x}$ and the theoretical expectation $\mathbb{E}[X]$ are related but different.

The expression

$$\mathbb{E}[X]$$

belongs to a probability model. It is computed from a probability mass function, a density, or another description of the law of X. The expression

$$\bar{x}$$

belongs to observed data. It is computed from a finite list of numbers.

If the sample is generated from a model with expectation $\mu = \mathbb{E}[X]$, then $\bar{x}$ may be close to μ, especially for large samples under suitable assumptions. But for any particular finite sample, $\bar{x}$ need not equal μ.

Remark

The statement $\bar{x} = \mu$ is usually false for a finite random sample. The sample mean is an observed summary; the theoretical mean is a model quantity. Later chapters explain why sample means often become close to theoretical means as sample size grows.

Sensitivity to extreme values

The sample mean uses all observed values, and each value enters linearly. This makes it sensitive to extreme observations.

Compare the two datasets

$$1, \quad 2, \quad 2, \quad 3, \quad 4$$

and

$$1, \quad 2, \quad 2, \quad 3, \quad 100.$$

The first mean is

$$\frac{1+2+2+3+4}{5} = \frac{12}{5} = 2.4.$$

The second mean is

$$\frac{1+2+2+3+100}{5} = \frac{108}{5} = 21.6.$$

Changing one observation from 4 to 100 dramatically changes the mean. This is not a defect of the arithmetic; it is a property of what the mean measures. The mean is a balance point, and a large value pulls the balance point toward itself.

Summary

The sample mean

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

is the expectation of the empirical distribution. It is the data-based counterpart of $\mathbb{E}[X]$, but it is not identical to the theoretical expectation of a probability model.

8.8 Sample variance and standard deviation

For observed data $x_1, \ldots, x_n$, define

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

Definition

The **empirical variance** is

$$v_n = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

Its square root is the empirical standard deviation.

This is the variance of the empirical distribution, because every observation has empirical mass $1/n$.

A computational identity

Expanding the squares gives

$$v_n = \frac{1}{n}\sum_{i=1}^{n} x_i^2 - \bar{x}^2.$$

The definition explains the meaning, while this form is often convenient for calculation.

The corrected sample variance

For inference, the observations are treated before measurement as random variables $X_1, \dots, X_n$. Define

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \overline{X})^2.$$

After measurement, the observed value is

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2.$$

Assume that $X_1, \dots, X_n$ are independent and identically distributed with mean μ and variance $\sigma^2 < \infty$. The identity

$$\sum_{i=1}^{n} (X_i - \overline{X})^2 = \sum_{i=1}^{n} (X_i - \mu)^2 - n(\overline{X} - \mu)^2$$

implies

$$\mathbb{E}\left[\sum_{i=1}^{n} (X_i - \overline{X})^2\right] = n\sigma^2 - n\operatorname{Var}(\overline{X}).$$

Since

$$\operatorname{Var}(\overline{X}) = \frac{\sigma^2}{n},$$

we obtain

$$\mathbb{E}\left[\sum_{i=1}^{n} (X_i - \overline{X})^2\right] = (n-1)\sigma^2.$$

Therefore

$$\mathbb{E}[S^2] = \sigma^2.$$

This is why the denominator $n - 1$ is used when the aim is to estimate the population variance without systematic downward bias.

The random empirical variance

$$V_n = \frac{1}{n} \sum_{i=1}^{n} (X_i - \overline{X})^2$$

instead satisfies

$$\mathbb{E}[V_n] = \frac{n-1}{n}\sigma^2.$$

It remains the correct variance of the empirical distribution itself.

Remark

The deviations from the sample mean satisfy

$$\sum_{i=1}^{n} (X_i - \overline{X}) = 0.$$

Thus the final deviation is determined by the preceding deviations. The factor $n - 1$ reflects this constraint.

Two related quantities

Quantity	Formula	Main role
Empirical variance	$\frac{1}{n}\sum(x_i - \bar{x})^2$	describes the observed data
Corrected variance	$\frac{1}{n-1}\sum(x_i - \bar{x})^2$	estimates population variance

The theoretical variance σ^2 belongs to a probability model. The quantities v_n and s^2 are calculated from data and need not equal σ^2 in a finite sample.

Summary

The denominator n gives the variance of the empirical distribution. The denominator $n-1$ produces an unbiased estimator of the population variance under independent identical sampling with finite variance.

8.9 Empirical quantiles, median, and interquartile range

Quantiles were introduced for probability distributions through the cumulative distribution function. For data, quantiles describe positions in the ordered sample. They answer questions such as: where is the middle of the data? Where is the lower quarter? Where is the upper quarter?

To discuss empirical quantiles, we first sort the data:

$$x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}.$$

The parentheses indicate order statistics. The value $x_{(1)}$ is the smallest observation, and $x_{(n)}$ is the largest observation.

Median

The median is a value that divides the ordered data into two parts. Roughly half of the observations lie at or below the median, and roughly half lie at or above it.

For an odd sample size $n = 2m + 1$, the median is usually taken as

$$x_{(m+1)}.$$

For an even sample size $n = 2m$, a common convention is to take the average of the two middle observations:

$$\frac{x_{(m)} + x_{(m+1)}}{2}.$$

For example, for the ordered data

$$1, \quad 2, \quad 2, \quad 3, \quad 100,$$

the median is 2. For the ordered data

$$1, \quad 2, \quad 2, \quad 3,$$

a common median is

$$\frac{2+2}{2} = 2.$$

Median versus mean

The median and the mean both describe center, but they measure different kinds of center. The mean is a balance point and is sensitive to extreme values. The median is a positional center and is more resistant to extreme values.

Consider

$$1, \quad 2, \quad 2, \quad 3, \quad 100.$$

The mean is

$$\bar{x} = \frac{108}{5} = 21.6,$$

while the median is 2. The value 100 pulls the mean strongly to the right, but it does not move the median beyond the middle observation.

This does not mean that the median is always better. It means that the mean and median answer different descriptive questions.

Empirical quantiles

A p-quantile of a theoretical distribution is connected with the equation

$$F_X(q_p) = p.$$

For data, the empirical analogue is based on the empirical CDF. An empirical p-quantile is a value below which approximately a proportion p of the data lies.

There are several conventions for empirical quantiles, especially when interpolation is used. Different software packages may give slightly different answers for small datasets. The conceptual idea is stable: an empirical quantile marks a probability position in the ordered sample.

Remark

Empirical quantiles are not unique by one universal computational convention. When exact numerical agreement matters, the convention used by the software or textbook should be stated.

Quartiles and interquartile range

The first quartile Q_1 is an empirical 0.25-quantile. The median Q_2 is an empirical 0.5-quantile. The third quartile Q_3 is an empirical 0.75-quantile.

The interquartile range is

$$IQR = Q_3 - Q_1.$$

It measures the length of the central part of the data, ignoring the lowest and highest quarters.

The IQR is often used as a robust measure of spread. It is less sensitive to extreme observations than the standard deviation.

Relation to theoretical quantiles

If a sample is generated from a distribution with theoretical p-quantile q_p, then the empirical p-quantile may be close to q_p for a large sample. But the empirical quantile is computed from data, while the theoretical quantile is defined by the probability law.

For example, in a standard normal model, the theoretical median is 0. A finite sample from that model will usually have a sample median close to 0, but not exactly equal to 0.

Summary

Empirical quantiles are positions in the ordered sample. They are data-based counterparts of theoretical quantiles. The median describes positional center, and the interquartile range describes the spread of the central part of the data.

8.10 Outliers and robust summaries

An outlier is an observation that is far from the main body of the data. Outliers matter because they can strongly affect some summaries, especially the sample mean and standard deviation.

There is no single mathematical definition of an outlier that is correct in all contexts. Whether an observation is unusual depends on the scale of the data, the measurement process, and the question being asked.

A simple example

Compare the datasets

$$1, \quad 2, \quad 2, \quad 3, \quad 4$$

and

$$1, \quad 2, \quad 2, \quad 3, \quad 100.$$

In the second dataset, the value 100 is far from the other observations. It has a large effect on the mean:

$$\frac{1+2+2+3+4}{5} = 2.4,$$

whereas

$$\frac{1+2+2+3+100}{5} = 21.6.$$

The median, however, is 2 in both datasets.

This example shows that the mean is sensitive to extreme values, while the median is more robust.

Robustness

A summary is called robust if it is not strongly affected by a small number of extreme observations. The median is more robust than the mean. The interquartile range is more robust than the standard deviation.

This does not make robust summaries automatically superior. If extreme values are genuine and important, then a sensitive summary may reveal something that a robust summary hides. If extreme values are measurement errors, then a robust summary may better represent the main part of the data.

Feature	Sensitive summary	More robust summary
Center	mean	median
Spread	standard deviation	interquartile range

The 1.5 IQR rule

A common descriptive convention marks observations as potential outliers if they fall below

$$Q_1 - 1.5IQR$$

or above

$$Q_3 + 1.5IQR.$$

This rule is useful for exploratory analysis, especially in boxplots. It is not a law of probability. It is a convention for flagging observations that deserve attention.

Remark

Calling an observation an outlier does not mean it should automatically be removed. It means the observation should be investigated, interpreted, and checked in context.

Outliers and modelling

Outliers may have different explanations:

- a data entry error;
- a measurement error;
- a rare but genuine event;
- a mixture of different populations;
- an inappropriate theoretical model.

For example, if most delivery times are between 20 and 40 minutes but one recorded delivery time is 400 minutes, then the value might be a recording error, a severe disruption, or evidence that the process has occasional extreme delays. The correct interpretation cannot be decided from the number alone.

Relation to probability models

Some probability models naturally produce extreme observations. Heavy-tailed models produce large deviations more often than light-tailed models. Therefore an observation that looks like an outlier under one model may be unsurprising under another model.

This is another reason to distinguish data description from modelling. A boxplot or IQR rule can flag a value as unusual in the dataset. A probability model can then help decide whether such a value is plausible under a proposed mechanism.

Summary

Outliers are observations far from the main body of the data. They can strongly affect means and standard deviations. Robust summaries such as the median and IQR help describe the central part of the data, but unusual observations should be interpreted in context rather than removed mechanically.

8.11 Computer-generated samples

Computer-generated samples are useful because they let us study data when the underlying probability model is known. In real data analysis, the true model is usually unknown. In simulation, we choose the model ourselves and then observe what finite samples from that model look like.

This makes simulation an important learning tool. It helps us see the difference between a theoretical distribution and the empirical summaries computed from one finite sample.

What simulation does

Suppose we ask a computer to generate

$$X_1, X_2, \ldots, X_n$$

from a standard normal distribution. The model has

$$\mathbb{E}[X] = 0, \qquad \mathrm{Var}(X) = 1.$$

After generation, we obtain observed values

$$x_1, x_2, \ldots, x_n.$$

From these values we compute

$$\bar{x}, \qquad v_n, \qquad F_n,$$

and draw a histogram.

The sample mean will usually be close to 0 for large n, but it will not usually be exactly 0. The histogram may resemble the normal density, but it will not be identical. The empirical CDF may resemble the normal CDF, but it will be a step function.

A basic Python-style workflow

A typical computational experiment has the following structure:

```
import numpy as np

sample = np.random.normal(loc=0, scale=1, size=1000)
sample_mean = np.mean(sample)
sample_std = np.std(sample)
q1, median, q3 = np.quantile(sample, [0.25, 0.5, 0.75])
```

The commands are not the main point. The mathematical point is that the computer creates one finite sample from a specified model, and then the empirical summaries describe that sample.

Changing the sample size

One of the most useful experiments is to compare samples of different sizes. For example, generate samples from the same distribution with

$$n = 20, \qquad n = 100, \qquad n = 1000.$$

For each sample, compute the sample mean, draw a histogram, and plot the empirical CDF.

Typically, small samples look irregular. Larger samples often display the shape of the underlying distribution more clearly. The sample mean often gets closer to the theoretical expectation, and the empirical CDF often gets closer to the theoretical CDF.

This is an observed pattern, not yet a proof. Later chapters will explain it using the law of large numbers and related limit results.

Comparing several models

Simulation also helps compare different probability models. For example:

- samples from a uniform distribution tend to fill an interval evenly;
- samples from a normal distribution tend to cluster near the center and produce a bell-shaped histogram;
- samples from an exponential distribution are non-negative and often right-skewed;
- samples from a beta distribution lie between 0 and 1 and can take many shapes depending on the parameters.

Seeing these samples makes the earlier theoretical distributions less abstract. The density describes the ideal shape. The simulated sample shows one finite realization from that ideal model.

Simulation is not proof

A computer experiment can suggest a pattern, illustrate a definition, or reveal a mistake in intuition. It does not replace mathematical proof. For instance, if we generate many samples and see their means close to 0, this supports the idea that sample means tend to stabilize, but it does not prove the law of large numbers.

Simulation is best used together with definitions and theory:

- definitions tell us what to compute;
- simulation shows how the computations behave in examples;
- theory explains why the behaviour occurs under assumptions.

Summary

Computer-generated samples help connect probability models with empirical data. They show how histograms, empirical CDFs, sample means, variances, and quantiles behave for finite samples from known distributions. Simulation illustrates mathematics; it does not replace mathematical justification.

8.12 Comparing empirical and theoretical distributions

A central purpose of this chapter is to make the relationship between data and models visible. Empirical summaries are computed from a finite sample. Theoretical quantities belong to a probability law. The two levels can be compared, but they should not be confused.

Theoretical CDF and empirical CDF

Let F_X be the CDF of a theoretical model:

$$F_X(x) = P(X \le x).$$

Let F_n be the empirical CDF of observed data:

$$F_n(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{\{x_i \le x\}}.$$

The function F_X gives probabilities according to the model. The function F_n gives observed proportions according to the sample.

If the sample is generated from the model, then $F_n(x)$ may be close to $F_X(x)$ for many values of x, especially when n is large. But F_n is still a step function built from data, while F_X is the theoretical CDF.

Theoretical mean and sample mean

The theoretical mean is

$$\mu = \mathbb{E}[X].$$

The sample mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

If the data are generated from a model with mean μ, then $\bar{x}$ is an empirical approximation to μ. For a small sample, the difference may be large. For larger samples, the difference often becomes smaller under suitable assumptions.

At this stage, the correct language is:

> The sample mean is a data-based summary that may approximate the theoretical mean.

It is not correct to say that the sample mean is the theoretical mean.

Theoretical variance and sample variance

The same distinction applies to variance. The model variance is

$$\mathrm{Var}(X).$$

The empirical variance is computed from data:

$$v_n = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

The corrected sample variance

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

will become important when the aim is to estimate a population variance. But both v_n and s^2 are computed from the observed sample. They are not equal by definition to the theoretical variance.

Histogram and density

A histogram is built from data and depends on bin choices. A density is a function from a probability model. A density-scaled histogram can be compared with a density curve, but the comparison must be interpreted carefully.

If the histogram follows the shape of the density, then the sample is visually consistent with the model. If the histogram is very different from the density, then this may suggest that the model is inappropriate, the sample is small, the binning is misleading, or the data contain structure not captured by the model.

A visual comparison is a starting point, not a final conclusion.

A guiding table

Question	Model object	Data object
How much lies below x?	$F_X(x)$	$F_n(x)$
Where is the center?	$\mathbb{E}[X]$	$\bar{x}$, median
How spread out is it?	$\mathrm{Var}(X), \sigma_X$	empirical variance, sample standard deviation, IQR
What is the shape?	density or PMF	histogram, frequency table
Where are probability positions?	theoretical quantiles	empirical quantiles

This table summarizes the conceptual bridge built in this chapter. The next chapters will study this bridge more deeply by treating sample summaries as random variables.

Preparing for sampling distributions

Once a sample is viewed as the realization of random variables

$$X_1, \ldots, X_n,$$

statistics such as the sample mean also become random variables before the data are observed:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

After observation, $\bar{X}$ takes the value $\bar{x}$.

This is the next conceptual step. Descriptive statistics summarize the observed sample. Sampling distributions describe how those summaries vary from sample to sample before observation.

Summary

Empirical summaries connect data to probability models. They are not the same as theoretical quantities, but they are designed to approximate, visualize, or summarize the corresponding model concepts when data are generated from a model. The next step is to study the randomness of the summaries themselves.

8.13 Final checklist

This chapter introduced empirical distributions and descriptive statistics as ways of summarizing observed data. The most important idea is the distinction between theoretical objects from probability models and empirical objects computed from finite samples.

Conceptual understanding

You should be able to explain the following distinctions:

- a probability law versus a finite dataset;
- a random variable X_i before observation versus an observed value x_i after observation;
- theoretical probability $P(X \in A)$ versus empirical probability $P_n(A)$;
- theoretical CDF F_X versus empirical CDF F_n;
- density versus histogram;
- theoretical expectation $\mathbb{E}[X]$ versus sample mean $\bar{x}$;
- theoretical variance $\mathrm{Var}(X)$ versus empirical or corrected sample variance;
- theoretical quantiles versus empirical quantiles.

Definitions and formulas

You should know the meaning of the following formulas:

$$P_n(A) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{\{x_i \in A\}},$$

$$F_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{\{x_i \le x\}},$$

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i,$$

$$v_n = \frac{1}{n}\sum_{i=1}^{n} (x_i - \bar{x})^2,$$

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n} (x_i - \bar{x})^2,$$

$$IQR = Q_3 - Q_1.$$

You should also understand why the denominator n describes the empirical distribution, while the denominator $n-1$ is connected with later inferential use.

Typical tasks

After studying this chapter, you should be able to:

- build a frequency table from discrete data;
- compute relative frequencies and interpret them as empirical probabilities;
- compute values of the empirical CDF by hand for a small dataset;
- explain why the empirical CDF is a step function;
- draw or interpret a histogram;
- explain why bin choice affects a histogram;
- compute the sample mean, empirical variance, standard deviation, median, quartiles, and IQR;
- identify how outliers affect means and standard deviations;
- explain why computer-generated samples help illustrate the relationship between a model and data;
- compare empirical summaries with theoretical quantities without confusing them.

Common mistakes

Avoid the following mistakes:

- saying that a sample is itself a normal distribution, exponential distribution, or other theoretical law;
- treating $\bar{x}$ as if it were automatically equal to $\mathbb{E}[X]$;
- treating an empirical histogram as if it were exactly a density;
- forgetting that histograms depend on bin choices;
- ignoring the difference between n and $n-1$ in variance formulas;
- removing outliers mechanically without understanding their context;
- treating simulation as proof rather than illustration.

Questions for self-explanation

A good test of understanding is whether you can answer these questions clearly:

1. What is the empirical distribution of a dataset?
2. Why is every finite empirical distribution discrete?
3. What does $F_n(x)$ measure?
4. How is a histogram related to a density, and how is it different?

5. Why can two samples from the same model have different sample means?
6. What is the difference between describing a sample and estimating a population quantity?
7. Why do computer-generated samples help us understand probability models?
8. What will change in the next chapter when statistics such as $\bar{X}$ are treated as random variables?

Summary

Empirical distributions and descriptive statistics translate the language of probability into the language of observed data. They show how the concepts of distribution, center, spread, and quantiles begin to work when only a finite sample is available. The next chapter studies the randomness of these summaries under repeated sampling.

Chapter 9

Sampling and Distributions of Statistics

This chapter introduces a new level of probabilistic structure. Before observation, an entire sample is a random object with its own probability law. Statistics such as the sample mean, sample proportion, variance, and median are functions of that random sample and therefore have probability distributions of their own.

The purpose of the chapter is to make the idea of a sampling distribution concrete. The emphasis is on understanding how possible samples receive probabilities, how statistics group samples into numerical summaries, and why repeated sampling produces variation in those summaries. This prepares the transition from descriptive statistics to limit theorems and statistical inference.

9.1 From data summaries to random quantities

The previous chapter studied an observed dataset

$$x_1, \ldots, x_n.$$

After observation, its mean, variance, median, and empirical CDF are fixed summaries. Before observation, the entries are random variables

$$X_1, \ldots, X_n,$$

so a summary calculated from them is also random.

Definition

A **statistic** is a function of a random sample:

$$T = g(X_1, \ldots, X_n).$$

For observed values $x_1, \ldots, x_n$, the same rule gives the realized value

$$t = g(x_1, \ldots, x_n).$$

For the sample mean,

$$\overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

is a random variable, while

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

is one observed number.

Sampling variation

Another sample from the same probability model generally produces another value of the statistic. Statistical reasoning therefore requires a description of this sample-to-sample variation.

Definition

The **sampling distribution** of T is the probability law of

$$T = g(X_1, \dots, X_n)$$

under the specified sampling model.

Two distributions

The empirical distribution describes the values inside one observed dataset. A sampling distribution describes possible values of a statistic across possible samples.

Object	Level	Main question
Empirical distribution	observed data	How are the observations distributed?
Sampling distribution	random sampling	How does the statistic vary?

The next section studies the complete random sample. A later section then obtains the sampling distribution by applying a statistic to every possible sample.

Theorem

The central transition is

$$\text{observed data} \longrightarrow \text{fixed summaries}$$

and then

$$\text{random samples} \longrightarrow \text{random summaries.}$$

9.2 The sample itself is a random object

In descriptive statistics, we begin with observed values

$$x_1, x_2, \dots, x_n.$$

These values are already known. We can arrange them, draw a histogram, compute their mean, or describe their empirical distribution.

Before the observations are collected, however, there is no fixed list of numbers. There is a random procedure that will produce such a list. We represent its possible results by

$$X_1, X_2, \dots, X_n.$$

The whole sample is therefore the random vector

$$(X_1, X_2, \dots, X_n).$$

It is one random object with many possible values.

Definition

A **random sample** is a collection of random observations

$$X_1, X_2, \dots, X_n$$

generated according to a specified probabilistic model. Before observation, the sample has many possible values. After observation, one of these possibilities becomes the observed sample

$$(x_1, x_2, \ldots, x_n).$$

The word "sample" is therefore used at two related levels. Before the data are known, it means a random object. After the data are known, it means one realized list of values. The notation helps us keep these levels separate:

$$(X_1, \ldots, X_n) \quad \longrightarrow \quad (x_1, \ldots, x_n).$$

The arrow represents observation. It does not mean that randomness disappears from the model. It means that one possible random sample has become the sample we actually saw.

A small sample has a distribution

Consider a mechanism that produces either failure, recorded as 0, or success, recorded as 1. Suppose that the probability of success is

$$P(X_i = 1) = p,$$

and the probability of failure is

$$P(X_i = 0) = 1 - p.$$

Now take two observations and record their order. The random sample is

$$(X_1, X_2).$$

Its possible values are

$$(0,0), \qquad (0,1), \qquad (1,0), \qquad (1,1).$$

If the two observations are independent repetitions of the same mechanism, then these possible samples have probabilities

$$\begin{aligned} P((X_1, X_2) = (0,0)) &= (1-p)^2, \\ P((X_1, X_2) = (0,1)) &= (1-p)p, \\ P((X_1, X_2) = (1,0)) &= p(1-p), \\ P((X_1, X_2) = (1,1)) &= p^2. \end{aligned}$$

They add to one:

$$(1-p)^2 + (1-p)p + p(1-p) + p^2 = 1.$$

This is a probability distribution on possible samples. The elementary outcomes are no longer single zeros or ones. They are complete samples of size two.

Example

Let $p = 0.6$. Then

$$P((0,0)) = 0.16, \qquad P((0,1)) = 0.24,$$
$$P((1,0)) = 0.24, \qquad P((1,1)) = 0.36.$$

Before sampling, all four samples are possible. After the experiment, we may observe, for example,

$$(x_1, x_2) = (1, 0).$$

The observed pair is one realization of the random pair (X_1, X_2).

The samples $(0,1)$ and $(1,0)$ contain the same number of successes, but they are different ordered samples. They become indistinguishable only if we decide to record a summary such as the total number of successes.

The sampling model determines the law of the sample

The probabilities above did not come from the list of possible samples alone. They came from assumptions about how the observations were generated. We assumed that each observation had the same success probability p and that the two observations were independent.

In introductory models, we often assume that

$$X_1, X_2, \ldots, X_n$$

are independent and have the same probability law. This is commonly abbreviated by saying that the observations are **iid**: independent and identically distributed.

These assumptions are useful, but they should not be treated as automatic. If observations influence one another, or if objects are sampled without replacement from a small population, then the possible samples or their probabilities may change. The law of the sample always belongs to a specified sampling mechanism.

Remark

A dataset does not carry its sampling model inside itself. The model must state how the observations were generated. The same observed values may arise from different random mechanisms.

Three different distributions

At this point it is important to separate three objects.

First, the **population law** describes one random observation, for example

$$P(X_i = 1) = p.$$

Second, the **law of the sample** describes the random vector

$$(X_1, \ldots, X_n).$$

It assigns probabilities to possible complete samples.

Third, after one sample has been observed, its **empirical distribution** describes the values that occurred in that particular dataset.

For example, if the observed sample is

$$(1, 0, 1, 1, 0),$$

then its empirical distribution assigns relative frequency 3/5 to 1 and 2/5 to 0. This empirical distribution describes one realized sample. It is not the probability law of all possible samples.

Summary

The sampling process creates a new probability space:

$$\text{probability model for one observation} \longrightarrow \text{probability law for complete samples} \longrightarrow \text{one observed dataset.}$$

A statistic will be obtained by applying a rule to each possible sample. Because the sample is random, the statistic will also be random.

9.3 From possible samples to a sampling distribution

The previous section showed that the complete sample

$$(X_1, X_2, \ldots, X_n)$$

is a random object. It has many possible values, and the sampling model assigns probabilities to those values.

A statistic is obtained by applying the same rule to every possible sample. For example, the sample mean is the rule

$$(x_1, \ldots, x_n) \longmapsto \frac{x_1 + \cdots + x_n}{n}.$$

Before the sample is observed, this rule produces the random variable

$$\overline{X} = \frac{X_1 + \cdots + X_n}{n}.$$

Because different samples may give different values of $\overline{X}$, the sample mean has its own probability distribution.

Definition

The **sampling distribution** of a statistic is the probability law of that statistic under repeated sampling from a specified model.

The word "sampling" is important. The distribution does not describe the values inside one observed dataset. It describes the values that the statistic could take over all possible samples generated by the model.

A complete example with samples of size two

Return to two independent Bernoulli observations with

$$P(X_i = 1) = p, \qquad P(X_i = 0) = 1 - p.$$

The possible ordered samples are

$$(0, 0), \qquad (0, 1), \qquad (1, 0), \qquad (1, 1).$$

Now define the sample mean

$$\overline{X} = \frac{X_1 + X_2}{2}.$$

For zero–one data, this mean is also the proportion of successes in the sample. Applying the rule to every possible sample gives

possible sample	probability	$\overline{X}$
$(0, 0)$	$(1-p)^2$	0
$(0, 1)$	$(1-p)p$	$\frac{1}{2}$
$(1, 0)$	$p(1-p)$	$\frac{1}{2}$
$(1, 1)$	p^2	1

The statistic has only three possible values:

$$0, \qquad \frac{1}{2}, \qquad 1.$$

The middle value is produced by two different samples. Therefore

$$P(\overline{X} = 0) = (1-p)^2,$$

$$P\left(\overline{X} = \frac{1}{2}\right) = (1-p)p + p(1-p) = 2p(1-p),$$

and

$$P(\overline{X} = 1) = p^2.$$

These probabilities add to one:

$$(1-p)^2 + 2p(1-p) + p^2 = 1.$$

This is the sampling distribution of $\overline{X}$.

Example

For $p = 0.6$, the sampling distribution is

$$P(\overline{X} = 0) = 0.16,$$

$$P\left(\overline{X} = \frac{1}{2}\right) = 0.48,$$

$$P(\overline{X} = 1) = 0.36.$$

The value $1/2$ is most likely because it can be produced by either $(0,1)$ or $(1,0)$.

Several samples may produce the same statistic

A statistic usually records less information than the complete sample. The ordered samples

$$(0,1) \qquad \text{and} \qquad (1,0)$$

are different, but both have sample mean $1/2$. The statistic groups together all samples that produce the same summary value.

This is the same modelling idea that appeared earlier in the course. A detailed outcome is transformed into a coarser description:

$$\text{complete sample} \longrightarrow \text{statistic value}.$$

The probability of a statistic value is obtained by adding the probabilities of all samples that lead to that value.

Remark

A sampling distribution is not created by listing possible numerical values and assigning equal probabilities to them. Its probabilities are inherited from the probabilities of the samples that produce those values.

The same sample can produce many statistics

From one random sample we may compute many different summaries. For example, from

$$(X_1, X_2, X_3)$$

we may form

$$\overline{X} = \frac{X_1 + X_2 + X_3}{3},$$

$$M = \operatorname{median}(X_1, X_2, X_3),$$

or

$$R = \max(X_1, X_2, X_3) - \min(X_1, X_2, X_3).$$

Each rule produces a different random variable and therefore a different sampling distribution.

The purpose of this chapter is not to derive every such distribution. The important idea is more general:

$$\text{random sample} \longrightarrow \text{statistic} \longrightarrow \text{sampling distribution.}$$

Summary

A statistic has a probability law because it is a function of a random sample. To construct its sampling distribution in a small model:

1. list the possible samples;
2. assign probabilities to those samples;
3. compute the statistic for each sample;
4. add the probabilities of samples giving the same statistic value.

9.4 The sample mean and the sample proportion

The previous section constructed a sampling distribution by listing all possible samples. This works well for very small models, but the number of possible samples grows quickly. For larger samples, we often study a statistic through a few important characteristics of its distribution.

The two most important questions are:

Where is the sampling distribution centred?

and

How much does the statistic vary from sample to sample?

Expectation describes the centre, while variance and standard deviation describe the spread.

The sample mean

Let

$$X_1, X_2, \dots, X_n$$

be independent observations from the same distribution. Suppose

$$\mathbb{E}[X_i] = \mu \qquad \text{and} \qquad \mathrm{Var}(X_i) = \sigma^2$$

for every i. The sample mean is

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

It is a random variable because it depends on the random sample.

Using linearity of expectation,

$$\mathbb{E}[\overline{X}] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i].$$

Since every observation has expectation μ,

$$\mathbb{E}[\overline{X}] = \frac{1}{n}(n\mu) = \mu.$$

Theorem

If the observations all have expectation μ, then

$$\mathbb{E}[\overline{X}] = \mu.$$

Thus the sampling distribution of the sample mean is centred at the population mean.

This does not mean that every observed sample mean equals μ. Different samples usually produce different values. The statement concerns the centre of the entire sampling distribution.

Now consider its variance. Because the observations are independent,

$$\mathrm{Var}(X_1 + \cdots + X_n) = \mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_n).$$

Therefore

$$\mathrm{Var}(\overline{X}) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i).$$

Since every observation has variance σ^2,

$$\mathrm{Var}(\overline{X}) = \frac{1}{n^2}(n\sigma^2) = \frac{\sigma^2}{n}.$$

Hence

$$\mathrm{SD}(\overline{X}) = \frac{\sigma}{\sqrt{n}}.$$

Theorem

For independent observations with common variance σ^2,

$$\mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n}.$$

As the sample size increases, the sampling distribution of the mean becomes less spread out.

The individual observations do not become less variable when n increases. The population law has not changed. What becomes less variable is their average. A single unusually large or small observation has less influence when it is combined with many other observations.

Example

Suppose individual observations have

$$\mu = 50, \qquad \sigma = 12.$$

For samples of size $n = 4$,

$$\mathbb{E}[\overline{X}] = 50, \qquad \mathrm{SD}(\overline{X}) = \frac{12}{\sqrt{4}} = 6.$$

For samples of size $n = 36$,

$$\mathbb{E}[\overline{X}] = 50, \qquad \mathrm{SD}(\overline{X}) = \frac{12}{\sqrt{36}} = 2.$$

Both sampling distributions are centred at 50, but means from samples of size 36 vary less from sample to sample.

At this stage we are describing the centre and spread of the sampling distribution. We are not yet claiming that the distribution is normal. The shape of the distribution and normal approximations belong to the next chapter.

The sample proportion

Suppose each observation records success or failure:

$$X_i = \begin{cases} 1, & \text{if the } i\text{-th observation is a success,} \\ 0, & \text{if the } i\text{-th observation is a failure.} \end{cases}$$

If

$$P(X_i = 1) = p,$$

then

$$\mathbb{E}[X_i] = p \qquad \text{and} \qquad \mathrm{Var}(X_i) = p(1-p).$$

The sample proportion of successes is

$$\widehat{p} = \frac{X_1 + \cdots + X_n}{n}.$$

This is exactly the sample mean of zero–one observations. Therefore the general results for the sample mean apply immediately:

$$\mathbb{E}[\widehat{p}] = p$$

and

$$\mathrm{Var}(\widehat{p}) = \frac{p(1-p)}{n}.$$

Consequently,

$$\mathrm{SD}(\widehat{p}) = \sqrt{\frac{p(1-p)}{n}}.$$

Summary

The sample mean and the sample proportion illustrate the same principle:

a statistic varies because the sample varies.

For the sample mean,

$$\mathbb{E}[\overline{X}] = \mu, \qquad \mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n}.$$

For the sample proportion,

$$\mathbb{E}[\widehat{p}] = p, \qquad \mathrm{Var}(\widehat{p}) = \frac{p(1-p)}{n}.$$

In both cases, increasing the sample size reduces sampling variability without changing the centre of the distribution.

9.5 Repeated sampling and simulation

For a very small model, a sampling distribution can be constructed exactly by listing all possible samples. For larger models, this list may be too long to write down. The idea of a sampling distribution remains the same, but it can be explored by repeating the sampling procedure many times.

Suppose the model and the sample size are fixed. One repetition produces one sample and therefore one value of a statistic:

$$(X_1, \dots, X_n) \longrightarrow T = g(X_1, \dots, X_n).$$

Repeating the entire procedure produces many values:

$$T_1, T_2, T_3, \dots.$$

A histogram or frequency table of these values gives an empirical picture of the sampling distribution of T.

One dataset and many repeated samples

Consider a Bernoulli model with success probability p. From one sample of size n, we compute the sample proportion

$$\widehat{p} = \frac{X_1 + \cdots + X_n}{n}.$$

If the observed sample is

$$(1, 0, 1, 1, 0),$$

then the observed proportion is

$$\widehat{p} = \frac{3}{5}.$$

This is one value obtained from one sample.

If the experiment is repeated with new samples of size 5, other values may appear:

$$0, \quad \frac{1}{5}, \quad \frac{2}{5}, \quad \frac{3}{5}, \quad \frac{4}{5}, \quad 1.$$

The sampling distribution tells us how likely these values are before any particular sample is observed.

In this example the distribution can also be found exactly. The event

$$\widehat{p} = \frac{k}{n}$$

means that exactly k successes occurred. Therefore

$$P\left(\widehat{p} = \frac{k}{n}\right) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \dots, n.$$

The binomial formula is not a new rule for sampling distributions. It appears because all samples with exactly k successes are grouped into the same value k/n.

A simulation procedure

A computer can imitate repeated sampling. To explore the sampling distribution of a statistic T, we may use the following procedure:

1. Generate a sample of size n from the chosen probability model.
2. Compute the statistic T from that sample.
3. Store the value of T.
4. Repeat the first three steps many times.
5. Summarize the stored values by a table, histogram, mean, and standard deviation.

The resulting histogram is not a histogram of all observations from one large sample. Each plotted value is one statistic computed from one complete sample. For example, if $T = \overline{X}$, then every entry in the histogram is a sample mean.

Remark

Repeating one experiment many times and taking one large sample are related but not identical descriptions. A sampling distribution is built by repeatedly forming samples of a fixed size and recomputing the statistic for each sample.

Three distributions that should not be confused

Suppose a model generates observations X, and we repeatedly take samples of size n. Three different distributions may appear.

Distribution	What varies?	What it describes
Population distribution	one observation X	the model for individual values
Empirical distribution	values in one dataset	what was observed in one sample
Sampling distribution	a statistic T	variation across repeated samples

These distributions answer different questions. A population distribution asks what individual observations may look like. An empirical distribution describes the values actually seen. A sampling distribution asks how a summary would change if the sampling process were repeated.

For example, individual observations may be strongly skewed while averages of samples have a different shape. Conversely, a nearly symmetric observed histogram from one sample does not prove that a statistic has a symmetric sampling distribution. The levels must be kept separate.

Other statistics also have sampling distributions

The same repeated-sampling idea applies to any statistic. From each sample we may compute:

$$\overline{X}, \qquad \widehat{p}, \qquad S^2, \qquad M,$$

where these symbols may represent the sample mean, sample proportion, sample variance, and sample median.

Their sampling distributions need not have the same shape or variability. Some statistics react strongly to unusual observations; others are more resistant. Some distributions can be derived exactly; others are easier to study by simulation or approximation.

This chapter does not attempt to derive every possible distribution. Its purpose is to establish the level of reasoning:

$$\text{model} \longrightarrow \text{random samples} \longrightarrow \text{statistic values} \longrightarrow \text{sampling distribution.}$$

Summary

Simulation makes the idea of a sampling distribution visible. It does not change the definition and it does not replace mathematical reasoning. The theoretical sampling distribution belongs to the probability model. A simulation produces an empirical approximation by generating many possible samples and recomputing the statistic each time.

9.6 Final checklist

This chapter introduced a new level of randomness. Earlier chapters studied random observations and their probability laws. We now treat the entire sample as a random object and study quantities computed from that sample.

The central chain of ideas is

$$\text{probability model} \longrightarrow \text{random sample} \longrightarrow \text{statistic} \longrightarrow \text{sampling distribution.}$$

Conceptual understanding

You should be able to explain the following distinctions:

- a random observation X versus an observed value x;
- a random sample $(X_1, \dots, X_n)$ versus an observed sample $(x_1, \dots, x_n)$;
- a statistic $T = g(X_1, \dots, X_n)$ versus its observed value $t = g(x_1, \dots, x_n)$;
- the population distribution versus the law of the complete sample;
- the empirical distribution of one dataset versus the sampling distribution of a statistic;
- variability among individual observations versus variability among sample means or sample proportions.

Main definitions

A random sample is a collection

$$X_1, X_2, \dots, X_n$$

generated according to a specified sampling model.

A statistic is a function of the sample:

$$T = g(X_1, \dots, X_n).$$

The sampling distribution of T is the probability law of that statistic over all possible samples generated by the model.

Main examples

For the sample mean,

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

If the observations have common expectation μ, then

$$\mathbb{E}[\overline{X}] = \mu.$$

If they are independent and have common variance σ^2, then

$$\mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n}.$$

For zero–one observations, the sample proportion is

$$\widehat{p} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

If $P(X_i = 1) = p$, then

$$\mathbb{E}[\widehat{p}] = p, \qquad \mathrm{Var}(\widehat{p}) = \frac{p(1-p)}{n}.$$

These formulas describe the centre and spread of sampling distributions. They do not say that every observed statistic equals the corresponding population quantity.

Typical tasks

After studying this chapter, you should be able to:

- identify the complete random sample in a simple model;
- list all possible samples when the sample space is small;
- assign probabilities to complete samples;
- compute a statistic for each possible sample;
- construct a sampling distribution by grouping samples that give the same statistic value;
- explain why a statistic is a random variable;
- distinguish a histogram of observations from a histogram of statistics obtained from repeated samples;
- explain why larger samples produce less variable sample means and sample proportions;
- describe how simulation approximates a sampling distribution.

Common mistakes

Avoid the following mistakes:

- treating the observed dataset as the only possible sample;
- confusing the empirical distribution with the sampling distribution;
- assuming that possible statistic values are equally likely;
- forgetting that probabilities of statistic values come from the probabilities of the samples producing them;
- interpreting $\mathbb{E}[\overline{X}] = \mu$ as saying that every sample mean equals μ;
- claiming that the sample mean is normally distributed without an additional argument;
- treating a simulation as an exact proof of a probability law;
- forgetting that independence and a common distribution are modelling assumptions.

Questions for self-explanation

A good test of understanding is whether you can answer these questions clearly:

1. In what sense is a sample random before it is observed?
2. What is the difference between $(X_1, \ldots, X_n)$ and $(x_1, \ldots, x_n)$?
3. Why does a statistic have a probability distribution?
4. How is the probability of a statistic value obtained from the probabilities of complete samples?
5. What is the difference between a population distribution, an empirical distribution, and a sampling distribution?
6. Why do different samples from the same model produce different means?

7. Why does the variance of $\overline{X}$ decrease when n increases?
8. Why is a sample proportion a special case of a sample mean?
9. What does repeated sampling reveal that one observed dataset cannot?

Summary

A sample is not only a collection of observed numbers. Before observation it is a random object governed by a probability model. Every statistic computed from that sample is therefore also random and has its own probability law. This new level of randomness is the bridge from descriptive statistics to limit theorems and statistical inference.

Chapter 10

Limit Theorems and Statistical Inference

This chapter explains how statistical inference emerges from probability theory. Sampling distributions describe how statistics behave over repeated samples. When sample sizes become large, these distributions exhibit regular patterns that allow observed data to be compared with probabilistic models.

The law of large numbers explains why sample averages become stable, while the central limit theorem describes the scale and approximate shape of their random fluctuations. These ideas lead to standard errors, confidence intervals, and statistical tests. The emphasis is not on memorizing procedures, but on understanding how probability makes conclusions from data possible.

10.1 From sampling distributions to statistical inference

The previous chapter introduced a second level of randomness. A probability model describes individual observations, but it also determines the probability law of the entire sample and of every statistic computed from that sample.

If

$$X_1, X_2, \ldots, X_n$$

is a random sample and

$$T = g(X_1, X_2, \ldots, X_n)$$

is a statistic, then T has a sampling distribution. Before the data are observed, the statistic is random. After the sample has been collected, it takes one concrete value

$$t = g(x_1, x_2, \ldots, x_n).$$

This chapter asks how the observed value t can be used to learn about the unknown mechanism that generated the data.

Probability and inference move in opposite directions

In a probability problem, the model is known. We begin with assumptions such as

$$X_i \sim F$$

or

$$P(X_i = 1) = p,$$

and ask what kinds of samples and statistic values the model may produce.

The direction of reasoning is

$$\text{model} \longrightarrow \text{possible samples} \longrightarrow \text{possible statistic values}.$$

In statistical inference, the situation is reversed. We observe one sample and one statistic value, while some part of the model is unknown. We then ask what this observed result suggests about the mechanism that produced it.

The direction now appears to be

$$\text{observed sample} \longrightarrow \text{information about the model}.$$

This reversal is not automatic. One observed sample does not determine the model uniquely. Different models may produce similar data, and even a correct model may occasionally produce unusual samples. Statistical inference must therefore use probability to measure uncertainty rather than pretend that uncertainty has disappeared.

Summary

Probability asks what data a model may generate. Statistical inference asks what an observed dataset allows us to say about the model that generated it.

Parameters and statistics

A probability model often contains an unknown numerical characteristic called a **parameter**. We denote it by a symbol such as

$$\theta, \qquad \mu, \qquad \sigma^2, \qquad \text{or} \qquad p.$$

The parameter belongs to the model. It is not computed from the observed sample.

A statistic is computed from the sample and may be used to estimate the parameter. We often write

$$\widehat{\theta} = g(X_1, \dots, X_n).$$

Before observation, $\widehat{\theta}$ is random. After observation, it becomes a number

$$\widehat{\theta}_{\text{obs}} = g(x_1, \dots, x_n).$$

For example, in a Bernoulli model the unknown parameter is the success probability p. The sample proportion

$$\widehat{p} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

is a statistic used to estimate it. Similarly, the sample mean

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

may be used to estimate the population mean μ.

Remark

A parameter and a statistic may play similar roles in a formula, but they are mathematically different. The parameter is a fixed but unknown feature of the model. The statistic is a random quantity before the sample is observed.

Why large samples matter

The sampling distribution of a statistic may be complicated for small samples. As the sample size increases, however, important regularities begin to appear. Sample averages become more stable, their random fluctuations occur on a predictable scale, and in many situations their standardized distributions take an approximately universal form.

These regularities are the bridge from probability to inference. They allow us to answer questions such as:

- How close is a sample mean likely to be to the population mean?
- How much can a sample proportion vary from sample to sample?

- Which parameter values are compatible with the observed statistic?
- Would the observed result be plausible under a proposed model?

The first major regularity is the law of large numbers. It explains why averages and proportions become stable. The second is the central limit theorem. It describes the scale and approximate shape of their remaining fluctuations.

Inference is based on hypothetical repetition

Usually we observe only one dataset. Nevertheless, statistical reasoning asks what would happen if the same sampling procedure were repeated many times.

This does not mean that the actual experiment must literally be repeated. The repetition is part of the probability model. It allows us to describe how the statistic would vary across all samples that could have been observed.

Suppose a model with parameter θ implies that a statistic T is usually close to a certain value. After observing $T = t_{\text{obs}}$, we may ask whether this result is typical or unusual under that model. If it is typical, the data are compatible with the model. If it is extremely unusual, the data provide evidence against it.

The same sampling behaviour can also be used in the opposite way. If a statistic is usually close to the parameter it estimates, then an observed statistic can be surrounded by a range of plausible parameter values.

These two ideas lead to the main forms of inference developed later in the chapter:

$$\text{estimation and confidence intervals,}$$

and

$$\text{statistical hypotheses and tests.}$$

Theorem

Statistical inference is possible because statistics have predictable sampling behaviour. Large-sample regularities connect the unknown model parameter with the random statistic computed from observed data.

The logical structure of the chapter

The reasoning developed in this chapter follows the chain

$$\text{probability model} \longrightarrow \text{sampling distribution} \longrightarrow \text{large-sample regularity} \longrightarrow \text{observed statistic} \longrightarrow \text{statistical c}$$

The conclusion is never obtained from the observed number alone. It depends on the sampling model, the assumptions, and the mathematical description of how the statistic behaves across possible samples.

Summary

Statistical inference does not replace probability theory. It uses probability theory in reverse. We first understand how a model generates samples and statistics. Only then can one observed statistic be used to estimate an unknown parameter or assess the compatibility of a proposed model with the data.

10.2 The law of large numbers and the stability of averages

Let

$$X_1, X_2, \ldots, X_n$$

be independent observations from the same distribution, with

$$\mathbb{E}[X_i] = \mu \qquad \text{and} \qquad \mathrm{Var}(X_i) = \sigma^2 < \infty.$$

The sample mean is

$$\overline{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

The subscript emphasizes that the statistic changes when the sample size changes.

From the previous chapter,

$$\mathbb{E}[\overline{X}_n] = \mu$$

and

$$\mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n}.$$

The centre of the sampling distribution stays at μ, while its variance decreases as n increases.

This already suggests a fundamental regularity: averages from large samples should usually lie close to the population mean.

Randomness remains, but averages become stable

Increasing the sample size does not make the individual observations less random. Each X_i still follows the same probability law. What changes is the behaviour of their average.

For a small sample, one unusually large or small observation may strongly affect $\overline{X}_n$. In a large sample, the same observation is combined with many others and has less influence on the final average.

The reduction in variance gives the scale of this stabilization:

$$\mathrm{SD}(\overline{X}_n) = \frac{\sigma}{\sqrt{n}}.$$

Thus doubling the sample size does not halve the standard deviation of the sample mean. To halve it, the sample size must be multiplied by four.

Example

Suppose individual observations have standard deviation 20. Then

$$\mathrm{SD}(\overline{X}_{25}) = \frac{20}{5} = 4,$$

while

$$\mathrm{SD}(\overline{X}_{100}) = \frac{20}{10} = 2.$$

The observations themselves remain equally variable, but averages based on one hundred observations fluctuate less than averages based on twenty-five.

The law of large numbers

The law of large numbers makes the stabilization precise. For every fixed $\varepsilon > 0$,

$$P\left(|\overline{X}_n - \mu| > \varepsilon\right) \longrightarrow 0 \qquad \text{as } n \to \infty.$$

Equivalently,

$$P\left(|\overline{X}_n - \mu| \le \varepsilon\right) \longrightarrow 1.$$

Theorem

For independent identically distributed observations with finite mean μ and finite variance σ^2, the sample mean converges in probability to the population mean:

$$\overline{X}_n \xrightarrow{P} \mu.$$

This statement is the finite-variance form of the **law of large numbers**. More general versions require only a finite expectation, but their proof needs methods beyond the variance argument used here.

The notation

$$\overline{X}_n \xrightarrow{P} \mu$$

means that the probability of a noticeable difference between $\overline{X}_n$ and μ becomes small when the sample size becomes large.

The theorem does not say that the sample mean eventually becomes equal to μ. It says that substantial deviations become increasingly unlikely.

Why the result is plausible

Chebyshev's inequality gives

$$P\left(|\overline{X}_n - \mu| \ge \varepsilon\right) \le \frac{\mathrm{Var}(\overline{X}_n)}{\varepsilon^2}.$$

Using

$$\mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n},$$

we obtain

$$P\left(|\overline{X}_n - \mu| \ge \varepsilon\right) \le \frac{\sigma^2}{n\varepsilon^2}.$$

The right-hand side tends to zero as n increases.

This short argument shows the essential mechanism. The sampling distribution of the mean becomes concentrated because its variance decreases like $1/n$.

Remark

The law of large numbers is not a statement that random errors cancel perfectly. It is a probability statement saying that large deviations of the average become less likely as the sample size grows.

Sample proportions as a special case

For Bernoulli observations,

$$X_i = \begin{cases} 1, & \text{success}, \\ 0, & \text{failure}, \end{cases} \qquad P(X_i = 1) = p,$$

the sample proportion is

$$\widehat{p}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

Because $\mathbb{E}[X_i] = p$, the law of large numbers gives

$$\widehat{p}_n \xrightarrow{P} p.$$

This explains why relative frequencies are used to estimate probabilities. In a long sequence of independent repetitions, the observed proportion of successes becomes increasingly stable around the underlying success probability.

For example, if a mechanism has success probability $p = 0.4$, small samples may produce proportions such as

$$0.2, \qquad 0.5, \qquad 0.7.$$

With larger samples, proportions far from 0.4 become less common.

What the law does and does not provide

The law of large numbers provides the first foundation for estimation. If

$$\overline{X}_n \xrightarrow{P} \mu,$$

then a large-sample mean is a reasonable estimator of μ. Similarly, $\widehat{p}_n$ is a reasonable estimator of p.

But the law does not answer several important questions:

- How large must n be in a particular problem?
- What is the approximate probability of a specified estimation error?
- What is the shape of the remaining random fluctuations?
- How should uncertainty be reported after one sample is observed?

The law tells us that the estimator becomes close to the parameter. It does not yet describe the detailed pattern of the difference

$$\overline{X}_n - \mu.$$

That is the role of the central limit theorem.

Summary

The law of large numbers explains why averages and proportions can be used to learn about a probability model. As the sample size grows, their sampling distributions become concentrated around the corresponding population parameters. This produces stability, but it does not yet quantify the full shape of the remaining uncertainty.

10.3 The central limit theorem and the shape of fluctuations

The law of large numbers says that

$$\overline{X}_n$$

becomes close to

$$\mu$$

when the sample size is large. It explains stability, but it does not describe the remaining error

$$\overline{X}_n - \mu.$$

To use an observed sample mean for inference, we need more information. We need to know the typical size of this error and the approximate shape of its sampling distribution.

The central limit theorem provides this second regularity.

The scale of the fluctuations

Suppose

$$X_1, X_2, \ldots, X_n$$

are independent and identically distributed, with

$$\mathbb{E}[X_i] = \mu \qquad \text{and} \qquad \operatorname{Var}(X_i) = \sigma^2 < \infty.$$

Then

$$\mathbb{E}[\overline{X}_n] = \mu$$

and

$$\operatorname{SD}(\overline{X}_n) = \frac{\sigma}{\sqrt{n}}.$$

Therefore the natural scale of the difference

$$\overline{X}_n - \mu$$

is not 1, and not $1/n$, but

$$\frac{1}{\sqrt{n}}.$$

The standardized fluctuation is

$$Z_n = \frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}}.$$

This quantity measures how many standard deviations the sample mean lies above or below the population mean.

The central limit theorem

Under the assumptions above, the distribution of Z_n approaches the standard normal distribution as n increases:

$$Z_n = \frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1).$$

Theorem

For independent identically distributed observations with finite mean μ and finite positive variance σ^2,

$$\frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1).$$

This statement is called the **central limit theorem**.

The notation

$$\xrightarrow{d}$$

means convergence in distribution. For this course, the essential interpretation is that the standardized sampling distribution becomes approximately normal for large n.

Equivalently, the unstandardized sample mean is approximately distributed as

$$\overline{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right).$$

This is an approximation, not an exact identity in general.

What the theorem adds to the law of large numbers

The two limit theorems answer different questions.

The law of large numbers says

$$\overline{X}_n \approx \mu$$

for large n. It identifies the value around which the statistic stabilizes.

The central limit theorem says

$$\overline{X}_n - \mu$$

usually has size approximately

$$\frac{\sigma}{\sqrt{n}},$$

and that after dividing by this scale, the resulting distribution is approximately standard normal.

Thus:

$$\text{law of large numbers} \quad \longrightarrow \quad \text{location of the statistic,}$$

while

$$\text{central limit theorem} \quad \longrightarrow \quad \text{shape and scale of its fluctuations.}$$

Summary

The law of large numbers explains why the sample mean approaches μ. The central limit theorem explains how the sample mean continues to fluctuate around μ when the sample size is large.

The population need not be normal

A common misunderstanding is that the central limit theorem requires the individual observations to have a normal distribution. It does not.

The population distribution may be discrete, skewed, or otherwise non-normal. The theorem concerns the sampling distribution of the standardized mean, not the distribution of each individual observation.

For example, suppose

$$X_i \sim \text{Bernoulli}(p).$$

Each X_i takes only the values 0 and 1, so its distribution is not continuous and is certainly not normal. Nevertheless, the sample proportion

$$\widehat{p}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$$

is a sample mean. Since

$$\mathbb{E}[X_i] = p \qquad \text{and} \qquad \text{Var}(X_i) = p(1-p),$$

the central limit theorem gives

$$\frac{\widehat{p}_n - p}{\sqrt{p(1-p)/n}} \approx N(0,1)$$

for sufficiently large n.

Equivalently,

$$\widehat{p}_n \approx N\left(p, \frac{p(1-p)}{n}\right).$$

A numerical interpretation

Suppose a Bernoulli model has

$$p = 0.4$$

and the sample size is

$$n = 400.$$

Then

$$\mathbb{E}[\widehat{p}] = 0.4$$

and

$$\mathrm{SD}(\widehat{p}) = \sqrt{\frac{0.4(0.6)}{400}} \approx 0.0245.$$

The central limit theorem suggests that the standardized quantity

$$\frac{\widehat{p} - 0.4}{0.0245}$$

is approximately standard normal.

A sample proportion such as

$$\widehat{p} = 0.42$$

is less than one standard deviation above 0.4, so it is not surprising under the model. A value such as

$$\widehat{p} = 0.50$$

is more than four standard deviations above 0.4, so it would be highly unusual under the same model.

This is the beginning of statistical inference. The observed statistic is judged relative to the sampling distribution predicted by a proposed model.

When the approximation may be poor

The phrase "for large n" does not refer to one universal sample size. The quality of the approximation depends on the population distribution.

The normal approximation may be inaccurate when:

- the sample is small;
- the population distribution is extremely skewed;
- very rare and very large observations strongly affect the mean;
- the variance is infinite;
- the observations are strongly dependent.

For Bernoulli data, the approximation is usually more reliable when both

$$np \qquad \text{and} \qquad n(1-p)$$

are not too small. These quantities represent the expected numbers of successes and failures.

The central limit theorem is powerful because its conclusion is widely applicable, but every application still depends on assumptions about the sampling mechanism.

Remark

A large sample does not repair every defective model. Limit theorems describe the consequences of assumptions; they do not justify independence, identical distributions, or good data collection.

From approximation to inference

The approximation

$$\frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} \approx N(0,1)$$

allows probabilities about the statistic to be computed using the standard normal distribution.

It tells us which deviations from μ are ordinary and which are unusual. It also provides a way to construct ranges of parameter values that are compatible with an observed statistic.

To use this idea systematically, we need a name for the standard deviation of a sampling distribution. This quantity is called the standard error.

Summary

The central limit theorem provides a common approximate shape for many sampling distributions. After centring by the parameter and scaling by the standard deviation of the statistic, the remaining fluctuation is approximately standard normal. This regularity turns sampling distributions into practical tools for measuring uncertainty.

10.4 Standard errors and normal approximations

A statistic varies from sample to sample. The standard deviation of its sampling distribution measures the typical size of this variation.

Definition

The **standard error** of a statistic T is the standard deviation of its sampling distribution:

$$\mathrm{SE}(T) = \sqrt{\mathrm{Var}(T)}.$$

The term does not describe a computational mistake. It describes variation created by random sampling.

Standard deviation and standard error

The population standard deviation describes variation among individual observations. The standard error describes variation among statistics computed from repeated samples.

If individual observations have standard deviation σ, then

$$\mathrm{SD}(X_i) = \sigma, \qquad \mathrm{SE}(\overline{X}) = \frac{\sigma}{\sqrt{n}}.$$

A population may remain highly variable while its mean is estimated much more precisely.

Standard errors of common statistics

For independent observations with common variance σ^2,

$$\mathrm{SE}(\overline{X}) = \frac{\sigma}{\sqrt{n}}.$$

For a sample proportion based on independent Bernoulli observations,

$$\mathrm{SE}(\widehat{p}) = \sqrt{\frac{p(1-p)}{n}}.$$

Both formulas contain the scale $1/\sqrt{n}$. Multiplying the sample size by four divides the standard error by two.

Remark

Larger samples reduce random sampling variation, but they do not automatically remove bias, dependence, measurement error, or model misspecification.

Normal approximation in standard-error units

The central limit theorem suggests that, for sufficiently large samples,

$$\frac{\overline{X} - \mu}{\mathrm{SE}(\overline{X})} \approx N(0, 1).$$

Equivalently,

$$\overline{X} \approx \mu + \mathrm{SE}(\overline{X})Z, \qquad Z \sim N(0, 1).$$

The same form applies to a sample proportion:

$$\frac{\widehat{p} - p}{\mathrm{SE}(\widehat{p})} \approx N(0, 1).$$

A numerical difference should therefore be interpreted relative to its standard error.

Typical normal ranges

For $Z \sim N(0, 1)$,

$$P(-1.96 \leq Z \leq 1.96) \approx 0.95.$$

Consequently, when the approximation is appropriate,

$$P\left(\theta - 1.96\,\mathrm{SE}(T) \leq T \leq \theta + 1.96\,\mathrm{SE}(T)\right) \approx 0.95,$$

provided that the sampling distribution of T is centred at θ. This is a statement about the random statistic before the sample is observed.

Known and estimated standard errors

The theoretical standard error may contain unknown parameters. For the mean,

$$\mathrm{SE}(\overline{X}) = \frac{\sigma}{\sqrt{n}}.$$

The population standard deviation is commonly estimated using

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2,$$

so that

$$\widehat{\mathrm{SE}}(\overline{X}) = \frac{S}{\sqrt{n}}.$$

The denominator $n-1$ is used because, under independent identical sampling with finite variance,

$$\mathbb{E}[S^2] = \sigma^2.$$

For a proportion, the plug-in estimate is

$$\widehat{\mathrm{SE}}(\widehat{p}) = \sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}.$$

An estimated standard error is itself random. It varies from sample to sample and approximates the theoretical standard error only under the stated assumptions.

From standardized distances to inference

For a proposed parameter value θ_0,

$$Z_{\mathrm{obs}} = \frac{T_{\mathrm{obs}} - \theta_0}{\mathrm{SE}_{\theta_0}(T)}$$

measures the observed difference in standard-error units. Values near zero are ordinary under the proposed model; values far into a tail are unusual.

A standardized sampling statement may also be inverted to form a range of parameter values, provided that the dependence of the standard error on the parameter is handled correctly. This leads to confidence intervals.

Summary

A standard error is the standard deviation of a sampling distribution. It is the natural scale for comparing an observed statistic with a parameter value, and it connects limit theorems with confidence intervals and hypothesis tests.

10.5 Confidence intervals from sampling behaviour

Suppose a statistic T estimates a parameter θ, and assume that

$$\frac{T-\theta}{s_T} \approx N(0,1),$$

where s_T is known and does not depend on θ. Then

$$P\left(T - 1.96 s_T \leq \theta \leq T + 1.96 s_T\right) \approx 0.95.$$

The direct algebra is valid because s_T is fixed.

Definition

A **confidence interval** is produced by a procedure designed to contain the fixed parameter with a stated long-run frequency under repeated sampling.

After the sample is observed, the interval is numerical. The confidence level describes the long-run coverage of the procedure, not a probability distribution for the fixed parameter.

Mean

For independent observations with mean μ and variance σ^2, a large-sample interval with known σ is

$$\overline{X} \pm 1.96 \frac{\sigma}{\sqrt{n}}.$$

When σ is unknown, the large-sample plug-in version is

$$\overline{X} \pm 1.96 \frac{S}{\sqrt{n}}, \qquad S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \overline{X})^2.$$

Proportion

For a sample proportion, the theoretical standard error

$$\sqrt{\frac{p(1-p)}{n}}$$

depends on the unknown p. Replacing it by

$$\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}$$

gives the approximate interval

$$\widehat{p} \pm 1.96\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}.$$

This plug-in procedure is commonly called the **Wald interval**. It is not the literal algebraic inversion of the theoretical expression, and it can be inaccurate for small samples or values near the endpoints.

Interpretation

Larger samples usually reduce standard errors at the rate $1/\sqrt{n}$, but they do not remove bias, dependence, systematic measurement error, or an inappropriate model.

Summary

Direct inversion is straightforward when the standard error is known and does not depend on the unknown parameter. Estimated standard errors require an additional approximation that should be stated explicitly.

10.6 Statistical hypotheses and model compatibility

A confidence interval reports a range of parameter values compatible with an observed statistic. A hypothesis test begins with one proposed value and asks whether the observed statistic is compatible with it.

The null hypothesis has the form

$$H_0 : \theta = \theta_0.$$

It determines the sampling distribution used for comparison.

The alternative hypothesis

A test must also specify which departures from θ_0 are relevant. Common alternatives are

$$H_1 : \theta \neq \theta_0,$$

$$H_1 : \theta > \theta_0, \qquad \text{or} \qquad H_1 : \theta < \theta_0.$$

The alternative determines which values of a test statistic count as more extreme than the observed value. A two-sided alternative treats large deviations in either direction as evidence. A one-sided alternative uses only the direction specified in advance.

Remark

The direction of the alternative should be chosen from the scientific question, not after the data have been inspected.

The null distribution

Let T be a statistic whose distribution depends on θ. Under H_0, the model gives a particular sampling distribution for T. After observation, we calculate t_{obs} and locate it in that distribution. The reasoning is

$$H_0 \longrightarrow \text{null sampling distribution} \longrightarrow t_{\text{obs}} \longrightarrow \text{assessment of compatibility.}$$

An unusual observation is evidence against the null model, not a logical contradiction.

A standardized statistic

When a normal approximation is appropriate,

$$Z = \frac{T - \theta_0}{\mathrm{SE}_{H_0}(T)}$$

is approximately standard normal under H_0. The observed value is

$$z_{\text{obs}} = \frac{t_{\text{obs}} - \theta_0}{\mathrm{SE}_{H_0}(T)}.$$

The sign gives the direction of the difference, and the magnitude gives the distance from the null centre in standard-error units.

The p-value

Definition

The **p-value** is the probability, calculated under H_0, of obtaining a test statistic at least as extreme as the observed one in the direction or directions specified by H_1.

For $Z \sim N(0, 1)$, the three common forms are

$$P(|Z| \geq |z_{\text{obs}}|) \quad \text{for } H_1 : \theta \neq \theta_0,$$

$$P(Z \geq z_{\text{obs}}) \quad \text{for } H_1 : \theta > \theta_0,$$

and

$$P(Z \leq z_{\text{obs}}) \quad \text{for } H_1 : \theta < \theta_0.$$

A small p-value means that the observed statistic, or a value more extreme in the prespecified direction, would be unusual under H_0. A large p-value means that the observation is not unusual under H_0; it does not prove that the null hypothesis is true.

In particular,

$$p\text{-value} \neq P(H_0 \text{ is true} \mid \text{data}).$$

Example: a population proportion

For independent Bernoulli observations, consider

$$H_0 : p = p_0.$$

Under the null hypothesis,

$$\mathrm{SE}_{H_0}(\widehat{p}) = \sqrt{\frac{p_0(1-p_0)}{n}},$$

so

$$Z = \frac{\widehat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}.$$

For a two-sided question, the alternative is

$$H_1 : p \neq p_0$$

and the p-value uses both tails. For an upper-sided or lower-sided question, only the corresponding tail is used.

Decision rules and possible errors

A procedure may specify a significance level α. A conventional rule is

$$p\text{-value} \leq \alpha \quad \Longrightarrow \quad \text{reject } H_0.$$

Otherwise the correct statement is that the data do not provide sufficient evidence to reject H_0.

A type I error is rejection of a true null hypothesis. A type II error is failure to reject a false null hypothesis. The significance level controls the type I error probability under the null model. The type II error probability depends on the particular alternative and the sample size.

Statistical and practical importance

A large sample may make a small difference detectable. Statistical significance therefore does not imply practical importance. A complete analysis should report the estimated effect and its uncertainty, not only a threshold decision.

Relation to confidence intervals

For corresponding two-sided large-sample procedures, a value θ_0 is excluded by an approximate 95% confidence interval exactly when the corresponding level-0.05 test regards it as incompatible with the data. One-sided tests correspond to one-sided confidence bounds rather than ordinary two-sided intervals.

Dependence on the model

A small p-value can reflect an incorrect parameter value or failure of another assumption used to construct the null distribution. Dependence, biased sampling, measurement error, or an inappropriate probability model can all affect the conclusion.

Summary

A hypothesis test compares an observed statistic with the distribution predicted by H_0. The alternative hypothesis determines the relevant tail or tails, so it is part of the definition of extremeness and of the p-value.

10.7 Final checklist

The essential chain of statistical inference is

$$\text{model} \longrightarrow \text{sampling distribution} \longrightarrow \text{large-sample regularity} \longrightarrow \text{observed statistic} \longrightarrow \text{conclusion}.$$

The conclusion depends on the sampling mechanism and assumptions, not on the observed number alone.

Limit theorems

For independent identically distributed observations with mean μ and finite variance σ^2,

$$\overline{X}_n \xrightarrow{P} \mu.$$

This finite-variance law of large numbers explains stabilization. More general versions require only a finite expectation.

If $0 < \sigma^2 < \infty$, the central limit theorem gives

$$\frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1).$$

The law of large numbers identifies where the statistic stabilizes. The central limit theorem describes the scale and approximate shape of its remaining fluctuations.

Standard errors

The standard error is the standard deviation of a sampling distribution:

$$\mathrm{SE}(T) = \sqrt{\mathrm{Var}(T)}.$$

For a sample mean and a sample proportion,

$$\mathrm{SE}(\overline{X}) = \frac{\sigma}{\sqrt{n}}, \qquad \mathrm{SE}(\widehat{p}) = \sqrt{\frac{p(1-p)}{n}}.$$

Unknown quantities are often replaced by estimates. For the mean,

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

is used because $\mathbb{E}[S^2] = \sigma^2$ under independent identical sampling with finite variance.

Confidence intervals

A large-sample confidence interval often has the form

$$\text{estimate} \pm \text{critical value} \times \text{estimated standard error}.$$

The confidence level is the long-run coverage of the interval-producing procedure. Direct inversion is straightforward when the standard error is known and does not depend on the unknown parameter. A plug-in standard error creates an additional approximation.

Hypothesis tests

A test specifies both

$$H_0 : \theta = \theta_0$$

and an alternative such as

$$H_1 : \theta \neq \theta_0, \qquad H_1 : \theta > \theta_0, \qquad \text{or} \qquad H_1 : \theta < \theta_0.$$

The alternative determines the relevant tail or tails of the null distribution.

A standardized statistic often has the form

$$Z = \frac{T - \theta_0}{\mathrm{SE}_{H_0}(T)}.$$

The p-value is the probability under H_0 of a result at least as extreme as the observed result in the direction specified by H_1. It is not

$$P(H_0 \text{ is true} \mid \text{data}).$$

Failure to reject H_0 is not proof that it is true.

Typical tasks

You should be able to:

- distinguish the law of large numbers from the central limit theorem;
- calculate and interpret standard errors;
- explain why corrected sample variance uses $n - 1$;
- construct and interpret a basic large-sample confidence interval;
- identify when a standard error has been estimated rather than known;
- state both a null and an alternative hypothesis;
- select the correct tail or tails from the alternative;
- interpret a p-value as a probability about hypothetical data under the null model;
- distinguish statistical detectability from practical importance;
- identify the assumptions supporting an inferential conclusion.

Common mistakes

Avoid:

- claiming that a large-sample average becomes exactly equal to its mean;
- claiming that the central limit theorem makes the population normal;
- confusing population standard deviation with standard error;
- treating an estimated standard error as if it were known exactly;
- interpreting confidence level as posterior probability for a fixed parameter;
- defining a p-value without specifying the alternative direction;
- treating a large p-value as proof of the null hypothesis;
- treating statistical significance as practical importance;
- believing that a large sample repairs bias or an incorrect model.

Summary

Inference is possible because statistics have predictable sampling behaviour. Limit theorems provide stability and approximate shape, standard errors provide a scale, and interval and testing procedures connect these regularities with one observed sample while keeping assumptions explicit.

Chapter 11

AI-Assisted Visualisation and Simulation Projects

Modern AI coding assistants can generate small interactive applications quickly. For a probability course, this creates an opportunity to turn definitions and formulas into controlled computational experiments: sample spaces can be explored, probability laws can be changed with sliders, repeated samples can be generated, and long-run behaviour can be compared with exact theory.

The purpose of this chapter is not to replace mathematical work with generated software. Its purpose is to help students design visualisations and simulations that make the mathematical structure visible. Every project begins with a model, states its assumptions, identifies the quantities that should remain invariant, and requires the generated application to compare simulation with theory whenever possible.

The prompts below are deliberately more detailed than ordinary chat requests. They specify the learning objective, model, controls, diagrams, numerical checks, and expected behaviour of the application. A student may copy a prompt into an AI coding assistant, but should then inspect the code, test the mathematics, and be able to explain every displayed quantity. The projects are suggestions rather than rigid specifications: they may be simplified or extended as long as the mathematical objective and validation requirements are preserved.

11.1 How to Work with an AI Coding Assistant

An AI coding assistant responds better to a specification than to a request such as "make an animation about probability". A useful specification identifies:

1. the probability model and its assumptions;
2. the learning question;
3. the parameters controlled by the user;
4. the exact and simulated quantities to display;
5. test cases with known answers.

Generated software should be treated as a mathematical claim expressed through code. A convincing graph may still use an incorrect distribution, denominator, or independence assumption.

Reusable requirements

The following instructions may be appended to any project prompt in this chapter.

Example

Prompt fragment

> Build a single-page educational application using HTML, CSS, and vanilla JavaScript. It should open directly in a browser without a server. Use a responsive layout and SVG or Canvas for graphs.
>
> Include a title, learning objective, model assumptions, labelled controls, Reset and

Clear buttons, and a short explanation of every display. When simulation is used, provide controls for one repetition and many repetitions. Show theoretical values and simulation estimates side by side, and label them clearly.

Validate the input parameters and handle boundary cases. Before presenting the implementation, restate the model, formulas, and planned correctness checks. After the implementation, give mathematical test cases and the expected result of each test.

A disciplined sequence

1. Read the relevant chapter and identify one central mechanism.
2. Ask the assistant to restate the model before producing the application.
3. Verify the formulas and assumptions.
4. Begin with a minimal version containing one diagram and essential controls.
5. Test special cases whose answers are already known.
6. Add animation or further models only after the minimal version is correct.
7. Ask for an explanation detailed enough to modify the program independently.

A binomial application should be checked at $n = 1$. A conditional-probability application should be checked for disjoint, identical, and independent events. A confidence-interval simulator should be checked over many repetitions to see whether empirical coverage approaches the stated level under the assumed model.

Remark

A visualisation is not a proof. Monte Carlo computation can reveal patterns and test an implementation, but it does not replace a derivation or theorem. Every project should distinguish exact mathematics from numerical evidence.

11.2 Projects for Chapter 0: Introduction

The introductory chapter is about modelling discipline. The most useful project is therefore an interactive map of the objects that appear between a real situation and a probability calculation.

Project: model-construction map

The application should use several short scenarios, such as coin tosses, dice, cards, an urn, a waiting time, and a continuous measurement. For each scenario, the student should classify descriptions as

experiment, recording rule, outcome,
sample space, event, probability law,
random variable, data, statistic.

Example

Prompt to give an AI coding assistant

Build an interactive probability-model map. Include scenarios based on coins, dice, cards, urns, waiting times, and a continuous interval. Present cards describing the experiment, recording rule, elementary outcome, sample space, event, probability assignment, random variable, observed data, statistic, and possible conclusion.

Let the student arrange the cards in a logical order. Check the arrangement and explain every incorrect placement. Emphasize the difference between a theoretical probability law, one observed dataset, and a sampling distribution. Add controls that reveal one correct construction and load another scenario.

A useful extension is an assumption panel containing independence, equal likelihood, replacement, identical distribution, finite variance, and representative sampling. When an assumption is removed, the application should state which later conclusions may fail.

Summary

The project should make the modelling sequence visible:

$$\text{situation} \longrightarrow \text{model} \longrightarrow \text{calculation} \longrightarrow \text{interpretation}.$$

11.3 Projects for Chapter 1: Elements of Combinatorics

Combinatorics is especially suitable for interactive work because a program can display the objects being counted, not only the final formula. The central learning objective is to connect a recording rule with the correct class of mathematical objects.

Project: counting-model laboratory

The student should choose whether order matters, whether repetition is allowed, whether objects are distinguishable, and how many positions or selected objects are involved. For small parameters, the application should list every admissible object. For larger parameters, it should show only the count and explain which formula applies.

Example

Prompt to give an AI coding assistant

Create a combinatorics laboratory with controls for the number of available objects, the number selected, whether order matters, whether repetition is allowed, and whether repeated object types are distinguishable. Display the mathematical model selected by these choices, the corresponding counting formula, and the resulting number.

For small cases, enumerate all outcomes and show them as sequences, subsets, or multisets according to the recording rule. Provide a side-by-side comparison of two models that use the same numerical parameters but different recording rules. Include warnings when the user attempts to apply a formula to the wrong type of object.

The application should make visible the difference between

$$n^k, \qquad \frac{n!}{(n-k)!}, \qquad \binom{n}{k}, \qquad \binom{n+k-1}{k}.$$

It should not merely display these formulas; it should explain which mathematical objects each formula counts.

Project: overcounting microscope

This project should begin with labelled arrangements and then hide labels or identify arrangements that become equivalent. The display should group labelled outcomes into equivalence classes and show why division by factorials removes overcounting.

Example

Prompt to give an AI coding assistant

Build an overcounting visualiser for arrangements of a multiset. Let the user choose multiplicities, for example three objects of one type, two of another type, and one of a third type. First generate all arrangements as if every object had a private label. Then remove the private labels and group arrangements that become identical.

Show the size of every equivalence class, the total labelled count, the number of distinct visible arrangements, and the formula

$$\frac{n!}{n_1!n_2!\cdots n_r!}.$$

Allow one equivalence class to be opened so the student can inspect all labelled arrangements that collapse to the same visible arrangement.

A useful extension is a path-counting view of Pascal's triangle. Each entry should count lattice paths, and clicking an entry should show the recursion

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Summary

The program should always answer two questions:

What objects are being counted? Why is each object counted exactly once?

11.4 Projects for Chapter 2: Discovering Formalism

This chapter is less about numerical simulation and more about making abstract structure visible. Two projects are especially useful: an event-algebra playground and a finite sigma-algebra generator.

Project: event algebra playground

Use a small finite universe displayed as a grid of elementary outcomes. The user should define events A and B by clicking outcomes. The application should then display union, intersection, complements, differences, and symmetric difference.

Example

Prompt to give an AI coding assistant

Create an event-algebra playground on a finite sample space containing between eight and twenty elementary outcomes. Let the user select events A and B. Display $A \cup B$, $A \cap B$, A^c, B^c, $A \setminus B$, and the symmetric difference using consistent highlighting.

Next to every set operation, display the corresponding logical statement using words such as and, or, not, and but not. Add buttons that verify De Morgan's laws by comparing both sides outcome by outcome. Include a panel that writes the selected event in roster notation.

Project: sigma-algebra generated by a partition

A finite sigma-algebra can be visualised without advanced measure theory. Begin with a partition of a finite universe into blocks. The application should construct all unions of those blocks and explain why no other subsets belong to the generated sigma-algebra.

Example

Prompt to give an AI coding assistant

> Build an interactive finite sigma-algebra generator. Display a finite universe and let the user partition it into blocks. Generate the collection of all unions of complete blocks. Show that the empty set and the universe are included, that complements remain in the collection, and that unions of listed events remain in the collection.
>
> Allow the user to propose an additional subset. Explain whether it belongs to the generated sigma-algebra and, if it does not, identify the block that it splits. Keep the explanation elementary and connect the construction with the idea that events represent distinctions the model is able to observe.

A frequency panel may be added as motivation: repeated trials can produce relative frequencies for selected events. The program should state explicitly that frequency patterns motivate probability but do not prove the axioms.

Summary

The most useful visualisation in this chapter is structural:

$$\text{logical statement} \longleftrightarrow \text{event} \longleftrightarrow \text{set operation}.$$

11.5 Projects for Chapter 3: Sample Spaces and Elementary Probability

This chapter supports several interactive projects because the same experiment can be represented by lists, tables, trees, geometric regions, or symbolic notation.

Project: experiment-to-sample-space builder

Example

Prompt to give an AI coding assistant

> Create an experiment-to-sample-space builder with selectable models for repeated coin tosses, two dice, card selection, urn sampling, waiting for the first success, and a continuous uniform point on an interval.
>
> For every model, ask the user to choose the recording rule before displaying the sample space. Include controls for order, replacement, labels, and stopping rules when relevant. Show the same model as a list, table, tree, or symbolic set whenever those representations are meaningful. Let the user define an event and highlight all elementary outcomes in it.
>
> For finite uniform models, calculate probability by counting and display both the favourable count and the total count. For non-uniform discrete models, sum outcome probabilities. Do not use counting for the continuous interval model.

The student should be able to change one modelling decision and observe that the sample space, not merely the numerical answer, changes.

Project: continuous uniform geometry

Use a horizontal interval $[a, b]$ and a draggable event interval $[c, d]$. Display

$$P(c \leq X \leq d) = \frac{d - c}{b - a}.$$

The visualisation should show that a single point has length zero, although every realised observation is a point.

Example

Prompt to give an AI coding assistant

> Build a continuous-uniform probability visualiser. Let the user set the endpoints of the sample interval and drag the endpoints of an event interval. Shade the event region and show its length, total length, and exact probability.
>
> Add a discretisation control that replaces the interval by an increasingly fine grid. Compare counting on the grid with the limiting length ratio, and explain why the continuous model does not assign positive probability to individual points.

Summary

The application should reinforce the rule

define the outcome and event first, assign probability second.

11.6 Projects for Chapter 4: Conditional Probability, Independence, and Bayes' Theorem

Conditional probability is one of the strongest candidates for visualisation because conditioning changes the reference population. The denominator should visibly change when new information is imposed.

Project: conditioning changes the space

Example

Prompt to give an AI coding assistant

> Create an interactive finite probability space displayed as a grid of weighted elementary outcomes. Let the user define events A and B. Show $P(A)$, $P(B)$, $P(A \cap B)$, $P(A \mid B)$, and $P(B \mid A)$.
>
> When the user activates conditioning on B, visually fade all outcomes outside B, renormalise the remaining weights, and show that
>
> $$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$
>
> Include presets for disjoint events, nested events, independent events, and dependent events. Display whether the equality $P(A \cap B) = P(A)P(B)$ holds.

Project: Bayes diagnostic-test laboratory

This project should use prevalence, sensitivity, and specificity as adjustable parameters. A tree diagram and an icon array should display the same model.

Example

Prompt to give an AI coding assistant

> Build a Bayes theorem laboratory for a diagnostic test. Provide sliders for prevalence, sensitivity, specificity, and population size. Show a probability tree, an expected-count table, and an icon array. Calculate the positive predictive value and negative predictive value from exact probabilities.
>
> Highlight the difference between $P(\text{positive} \mid \text{condition})$ and $P(\text{condition} \mid \text{positive})$. Add a simulation mode that generates individuals and compares empirical proportions with the exact Bayes calculation. Include a button that changes only prevalence so the student can see why predictive values change even when the test characteristics remain fixed.

An extension may display a partition $B_1, \ldots, B_k$ and animate the law of total probability before reversing the reasoning with Bayes' theorem.

Summary

The key visual principle is that conditioning restricts the space and then rescales probability inside the restricted space.

11.7 Projects for Chapter 5: Discrete Random Variables and Probability Laws

This chapter is ideal for interactive graphics because a discrete law has several equivalent representations: a mechanism, a mapping from outcomes to numbers, a probability mass function, a cumulative distribution function, and numerical summaries such as expectation and variance.

Project: outcome-to-random-variable mapper

Example

Prompt to give an AI coding assistant

> Create an interactive application that begins with a finite sample space and lets the user define a random variable by assigning a numerical value to each elementary outcome. Group all outcomes that receive the same value and construct the support, probability mass function, and cumulative distribution function automatically.
>
> When the user clicks a value of the random variable, highlight its complete inverse image in the sample space. Display
>
> $$P(X = x), \qquad F_X(t) = P(X \le t), \qquad \mathbb{E}[X], \qquad \mathrm{Var}(X).$$
>
> Include preset mappings for the number of heads in coin tosses, the sum of two dice, a payoff, and an indicator variable. Verify that PMF values sum to one and that CDF jumps equal point masses.

Project: discrete-law explorer

The application should include Bernoulli, binomial, geometric, and Poisson laws. Parameters should be controlled by sliders, while support, PMF, CDF, mean, and variance update immediately.

Example

Prompt to give an AI coding assistant

> Build a discrete probability-law explorer for Bernoulli, binomial, geometric, and Poisson distributions. For each law, show the random mechanism, parameter restrictions, support, PMF, CDF, expectation, and variance.
>
> Use linked PMF and CDF plots. Clicking a PMF bar should highlight the corresponding jump in the CDF. Add a Monte Carlo panel with controls for one sample and many samples. Overlay empirical relative frequencies on the exact PMF and report the maximum absolute difference.
>
> For the binomial law, allow n and p to change and verify that $n = 1$ gives a Bernoulli law. For the geometric law, state clearly whether the support begins at zero or one and keep that convention consistent.

Possible extension: expectation as a balance point

Draw the PMF as masses placed on a number line and mark $\mathbb{E}[X]$ as the balancing location. The student should be able to move probability mass while preserving total mass one and observe how expectation and variance change.

Summary

A good discrete-law application links four views:

$$\text{mechanism} \longrightarrow \text{random variable} \longrightarrow \text{PMF and CDF} \longrightarrow \text{moments.}$$

11.8 Projects for Chapter 6: Continuous Random Variables and Probability Laws

Continuous laws are particularly well suited to linked density and CDF plots. The application should make clear that density is not point probability and that interval probability is area.

Project: density, CDF, and quantile explorer

Example

Prompt to give an AI coding assistant

> Create an interactive explorer for uniform, exponential, normal, gamma, and beta distributions. Provide suitable parameter controls and display the parameter restrictions. Draw the density and CDF in linked panels.
>
> Let the user drag interval endpoints a and b. Shade the density area over the interval and display
> $$P(a \le X \le b) = F_X(b) - F_X(a).$$
> Mark the mean, median, selected quantiles, and one standard deviation around the mean when defined. Add a probability control that moves a quantile marker to the value q_p satisfying $F_X(q_p) = p$.

> Use numerical integration when a closed-form CDF is not implemented. State the numerical method, display an estimated integration error, and verify that total density area is approximately one.

The application should explicitly show that changing the value of a density at one isolated point does not change interval probabilities.

Project: theoretical law and generated sample

Example

Prompt to give an AI coding assistant

> Extend the continuous-distribution explorer with random sampling. Generate samples from the selected distribution and display a histogram and empirical CDF beside the theoretical density and CDF. Provide controls for sample size, histogram bin width, and repeated resampling.
>
> Show the sample mean, sample variance, empirical median, and selected empirical quantiles next to their theoretical counterparts. Do not imply that the histogram is the density or that the empirical CDF is the theoretical CDF. Label every object precisely.

A useful cautionary mode may compare distributions with similar means but different shapes, or similar variances but different tail behaviour.

Summary

The central visual relation is

$$\text{density} \xrightarrow{\text{integration}} \text{CDF} \xrightarrow{\text{differences}} \text{interval probabilities.}$$

11.9 Projects for Chapter 7: Empirical Distributions and Descriptive Statistics

This chapter naturally leads to a data laboratory in which one dataset can be viewed through a frequency table, histogram, empirical CDF, numerical summaries, and robust summaries.

Project: interactive descriptive-statistics laboratory

Example

Prompt to give an AI coding assistant

> Create an interactive descriptive-statistics laboratory. Let the user enter, paste, edit, or generate a numerical dataset. Display the raw observations, sorted observations, frequency table, histogram, empirical CDF, boxplot, mean, median, empirical variance, corrected sample variance, standard deviation, quartiles, interquartile range, and outlier fences.
>
> Provide a histogram bin-width slider and explain that changing bins changes the picture but not the data. Let the user drag one observation or add an extreme observation. Update all summaries immediately and highlight which statistics are sensitive to the change.

Display both

$$v_n = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

and

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

with a short explanation of their different roles.

Project: empirical versus theoretical distribution

Example

Prompt to give an AI coding assistant

Add a panel that generates data from a selected theoretical distribution. Overlay the histogram with the theoretical density or PMF, and overlay the empirical CDF with the theoretical CDF. Let the user change the sample size and resample repeatedly.

Display numerical discrepancies such as the difference between empirical and theoretical means, variances, and selected CDF values. Emphasize that one sample may differ substantially from the model even when it was generated from that model.

Suggested investigation

Students may record how the mean and median react to one moving outlier, how histogram shape changes with bin width, and how the empirical CDF changes one jump at a time when an observation is added.

Summary

The project should keep three levels separate:

observed values, empirical distribution, theoretical probability law.

11.10 Projects for Chapter 8: Sampling and Distributions of Statistics

This chapter is one of the richest sources of simulation projects. The essential animation should show a complete random sample being transformed into one statistic, followed by repetition of that process.

Project: repeated-sampling laboratory

Example

Prompt to give an AI coding assistant

Create a repeated-sampling laboratory. Let the user select a population law, its parameters, the sample size n, and a statistic. Include at least the sample mean, sample proportion when appropriate, sample median, empirical variance, and corrected sample variance.

Use three linked panels. The first panel shows the population law. The second shows one newly generated sample and calculates the selected statistic. The third accumulates the statistic over repeated samples and displays its sampling distribution as a histogram or discrete PMF.

Provide controls for one sample, ten samples, one thousand samples, continuous animation, pause, and clear. Display the empirical mean and standard deviation of the simulated statistics. When theory is available, display the theoretical centre and standard error and overlay them on the simulation.

The application must distinguish the histogram of observations inside one sample from the histogram of a statistic across many samples.

Project: exact small-sample enumeration

For small finite populations or mechanisms, the sampling distribution can be found exactly rather than simulated.

Example

Prompt to give an AI coding assistant

Build an exact sampling-distribution constructor for small Bernoulli samples and small samples of die outcomes. Enumerate all ordered samples for a user-selected small n, calculate the chosen statistic for every sample, and group samples that produce the same statistic value.

Show each statistic value, its complete preimage of samples, and its exact probability. Then run a Monte Carlo simulation beside the exact distribution and display the discrepancy. Include sample mean, number of successes, sample proportion, maximum, and range as selectable statistics.

A strong extension is a transition animation:

$$(X_1, \ldots, X_n) \longrightarrow T = g(X_1, \ldots, X_n) \longrightarrow \text{one point in the sampling distribution.}$$

Summary

The student should be able to identify what is random at each level:

observations, complete sample, statistic, sampling distribution.

11.11 Projects for Chapter 9: Limit Theorems and Statistical Inference

This chapter supports several complementary simulations. No single animation should try to represent the law of large numbers, the central limit theorem, confidence intervals, and hypothesis tests at once. Separate applications make the distinct logical roles clearer.

Project: law of large numbers and central limit theorem

Example

Prompt to give an AI coding assistant

Create a split-screen application comparing the law of large numbers with the central limit theorem. Let the user select Bernoulli, uniform, exponential, or another finite-variance population law.

On the left, generate one long sequence and plot the running mean against the sample size, together with the population mean. On the right, for a selected n, generate many independent samples and plot the histogram of

$$\frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}}.$$

Overlay the standard normal density. Let the user change n, the number of repetitions, and the population parameters.

Explain that the left panel concerns convergence of the average, while the right panel concerns the shape and scale of fluctuations. Do not describe either panel as a proof.

An optional cautionary example may use a distribution without a finite mean or variance, provided the application clearly states which theorem assumption fails.

Project: confidence-interval coverage

Example

Prompt to give an AI coding assistant

Create a confidence-interval coverage simulator. Include a population-mean mode with known standard deviation and a population-proportion mode using the Wald interval. Let the user set the true parameter, sample size, confidence level, and number of repetitions.

For each repetition, draw the interval on a horizontal line and mark whether it contains the fixed true parameter. Keep a running coverage count and compare it with the nominal confidence level. Display interval width and show how it changes with sample size.

Explain that the parameter remains fixed while the intervals vary from sample to sample. In the proportion mode, warn that the Wald interval may have poor coverage for small samples or probabilities near zero or one.

Project: hypothesis test and p-value simulator

Example

Prompt to give an AI coding assistant

Build a hypothesis-testing simulator for a Bernoulli proportion. Let the user specify $H_0 : p = p_0$, choose a two-sided, upper-sided, or lower-sided alternative, select n, and enter or simulate an observed number of successes.

Display the null sampling distribution, mark the observed statistic, shade the tail or tails defined by the alternative, and calculate the exact binomial p-value together with a normal approximation when appropriate. Explain every set of outcomes included in the p-value.

Add a repeated-experiment mode in which the true probability may equal or differ from p_0. Estimate the rejection rate. When the null is true, interpret this as an empirical type I error rate; when it is false, interpret it as empirical power.

A further extension may compare confidence intervals and corresponding two-sided tests, showing when a proposed parameter value is excluded by the interval and rejected by the test.

Summary

The projects should preserve the logical separation:

$$\begin{aligned} \text{LLN}: &\quad \text{stability}, \\ \text{CLT}: &\quad \text{fluctuation shape}, \\ \text{confidence interval}: &\quad \text{coverage procedure}, \\ \text{hypothesis test}: &\quad \text{compatibility with a proposed model}. \end{aligned}$$

11.12 Final Checklist for an AI-Assisted Project

A visually attractive application is not automatically a correct mathematical application. Before submitting or relying on a generated project, check the following points.

Mathematical model

1. Is the experiment and recording rule stated?
2. Is the sample space or population law identified?
3. Are all parameters and assumptions visible?
4. Is independence used only when it is part of the model?
5. Are exact formulas distinguished from numerical approximations?
6. Are theoretical distributions distinguished from empirical and sampling distributions?

Simulation

1. Can one repetition be inspected before many repetitions are run?
2. Is the simulated object clearly identified?
3. Is the number of repetitions displayed?
4. Can the simulation be cleared and reproduced?
5. Are Monte Carlo estimates compared with exact values when exact values exist?
6. Does the program avoid presenting numerical agreement as proof?

Visual communication

1. Are axes, units, events, and parameters labelled?
2. Does every histogram state what values are being counted?
3. Are density, probability mass, frequency, and relative frequency kept distinct?
4. Do controls prevent invalid parameter combinations?
5. Does the application explain what changes and what remains fixed?

Validation

Test boundary and special cases. Examples include:

$$P(A \mid A) = 1, \qquad P(A \mid B) = 0 \text{ for disjoint } A, B,$$

$$\sum_x p_X(x) = 1, \qquad \lim_{t\to\infty} F_X(t) = 1,$$

$$n = 1 \text{ for a binomial model}, \qquad \mathbb{E}[S^2] = \sigma^2 \text{ under the stated sampling assumptions.}$$

For inference projects, check whether empirical coverage or type I error approaches the nominal value only when the assumptions and implementation are appropriate.

Student explanation

A student should be able to answer:

- What mathematical question does the application investigate?
- Which formulas are implemented?
- Which output is exact and which is simulated?
- Which assumptions are required?
- Which test cases were used?
- What did the AI assistant generate, and what did the student verify or modify?
- What can the experiment suggest, and what still requires proof?

Theorem

AI can reduce the cost of building an experiment, but it does not remove the need for mathematical judgement. The educational value of the project lies in specifying the model, checking the implementation, interpreting the output, and explaining its limitations.